NETWORK RECEIVER
R-N402/R-N402D
SERVICE MANUAL

IMPORTANT NOTICE

This manual has been provided for the use of authorized Yamaha Retailers and their service personnel.
It has been assumed that basic service procedures inherent to the industry, and more specifically Yamaha Products, are already known and understood by the users, and have therefore not been restated.

WARNING: Failure to follow appropriate service and safety procedures when servicing this product may result in personal injury, destruction of expensive components, and failure of the product to perform as specified. For these reasons, we advise all Yamaha product owners that any service required should be performed by an authorized Yamaha Retailer or the appointed service representative.

IMPORTANT: The presentation or sale of this manual to any individual or firm does not constitute authorization, certification or recognition of any applicable technical capabilities, or establish a principle-agent relationship of any form.

The data provided is believed to be accurate and applicable to the unit(s) indicated on the cover. The research, engineering, and service departments of Yamaha are continually striving to improve Yamaha products. Modifications are, therefore, inevitable and specifications are subject to change without notice or obligation to retrofit. Should any discrepancy appear to exist, please contact the distributor's Service Division.

WARNING: Static discharges can destroy expensive components. Discharge any static electricity your body may have accumulated by grounding yourself to the ground buss in the unit (heavy gauge black wires connect to this buss).

IMPORTANT: Turn the unit OFF during disassembly and part replacement. Recheck all work before you apply power to the unit.

■ CONTENTS

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YAMAHA

YAMAHA CORPORATION
P.O.Box 1, Hamamatsu, Japan
'16.10

■ TO SERVICE PERSONNEL

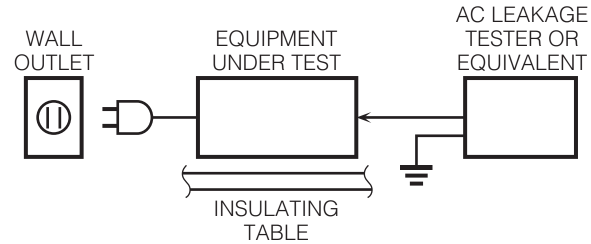
1. Critical Components Information

   Components having special characteristics are marked ⚠ and must be replaced with parts having specifications equal to those originally installed.

2. Leakage Current Measurement (For 120V Models Only)

   When service has been completed, it is imperative to verify that all exposed conductive surfaces are properly insulated from supply circuits.

- Meter impedance should be equivalent to 1500 ohms shunted by 0.15 μF.



- Leakage current must not exceed 0.5mA.
- Be sure to test for leakage with the AC plug in both polarities.

For U model
"CAUTION"
"F5401: FOR CONTINUED PROTECTION AGAINST RISK OF FIRE, REPLACE ONLY WITH SAME TYPE 2A, 250V FUSE."
"F5402: FOR CONTINUED PROTECTION AGAINST RISK OF FIRE, REPLACE ONLY WITH SAME TYPE 8A, 125V FUSE."

CALIFORNIA PROPOSITION 65 WARNING

This product contains chemicals known to the State of California to cause cancer, or birth defects or other reproductive harm.

DO NOT PLACE SOLDER, ELECTRICAL/ELECTRONIC OR PLASTIC COMPONENTS IN YOUR MOUTH FOR ANY REASON WHATSOEVER!

Avoid prolonged, unprotected contact between solder and your skin! When soldering, do not inhale solder fumes or expose eyes to solder/flux vapor!

If you come in contact with solder or components located inside the enclosure of this product, wash your hands before handling food.

About lead free solder

All of the P.C.B.s installed in this unit and solder joints are soldered using the lead free solder.

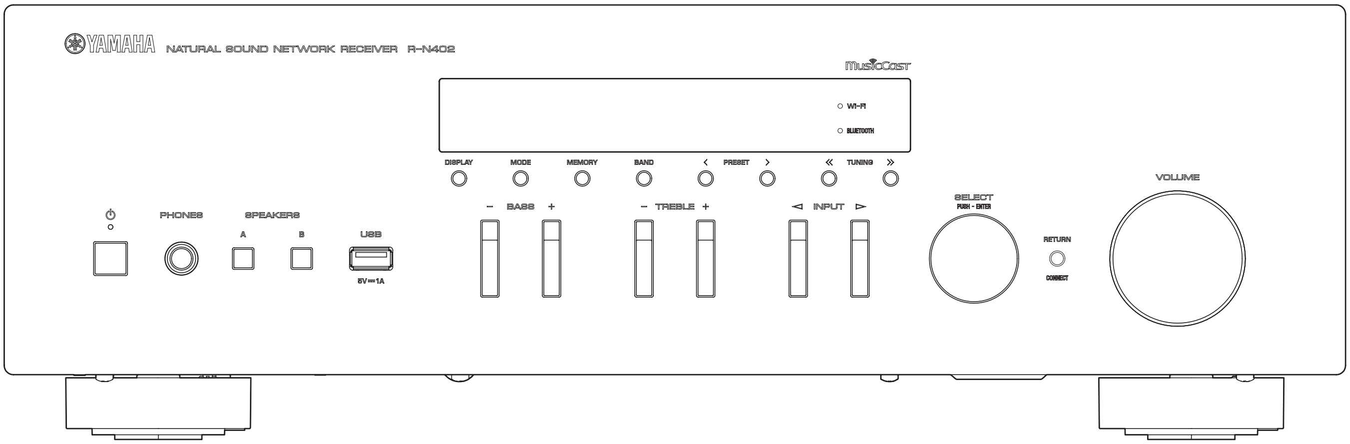
Among some types of lead free solder currently available, it is recommended to use one of the following types for the repair work.

- Sn + Ag + Cu (tin + silver + copper)
- Sn + Cu (tin + copper)
- Sn + Zn + Bi (tin + zinc + bismuth)

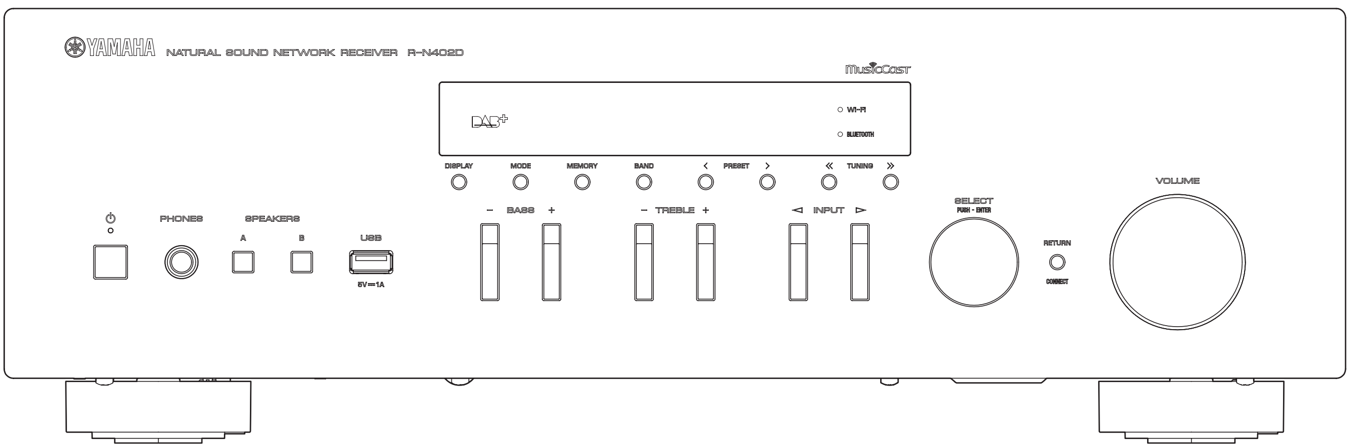
Caution:
As the melting point temperature of the lead free solder is about 30° C to 40° C (50° F to 70° F) higher than that of the lead solder, be sure to use a soldering iron suitable to each solder.

■ FRONT PANELS

R-N402

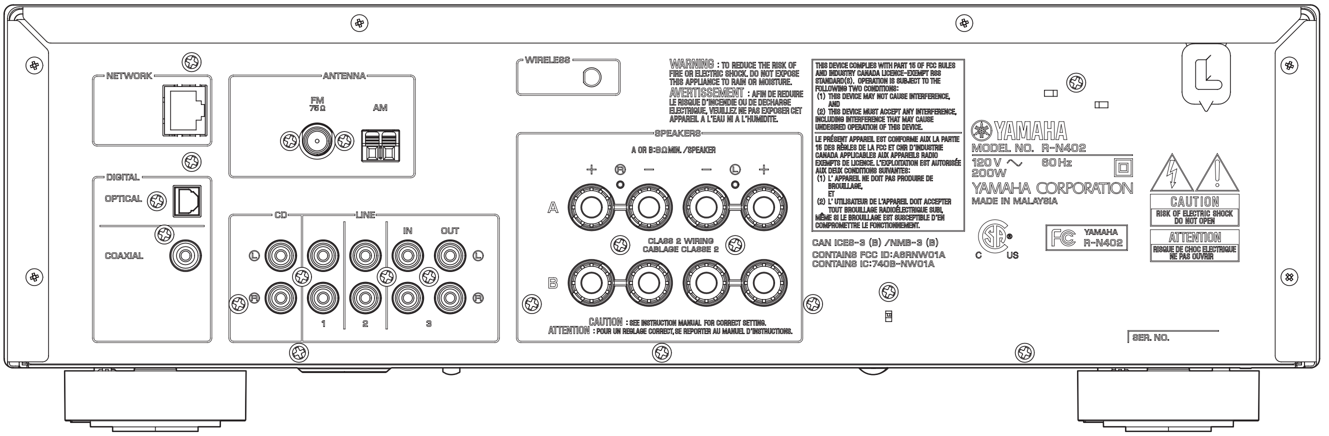


R-N402D

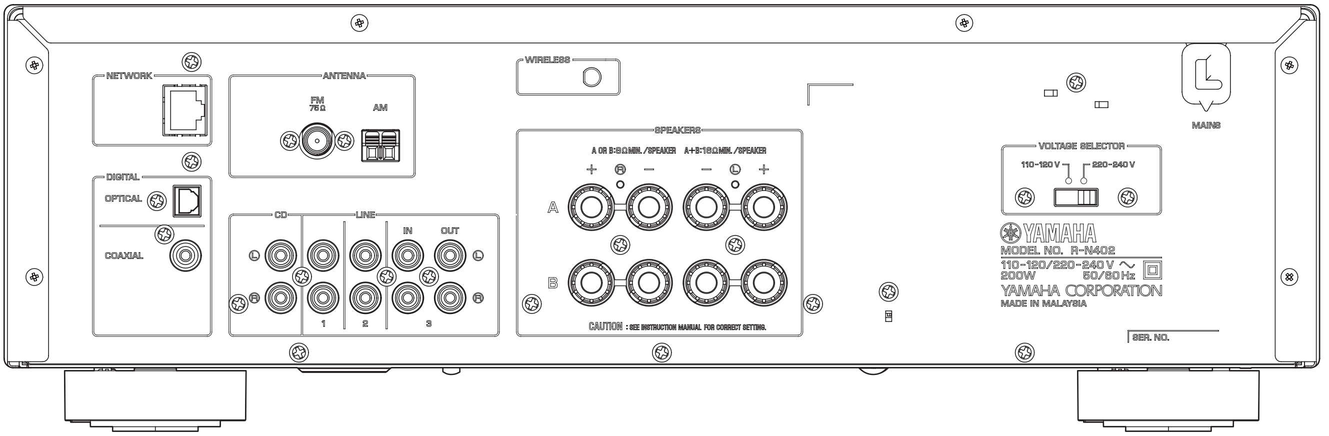


■ REAR PANELS

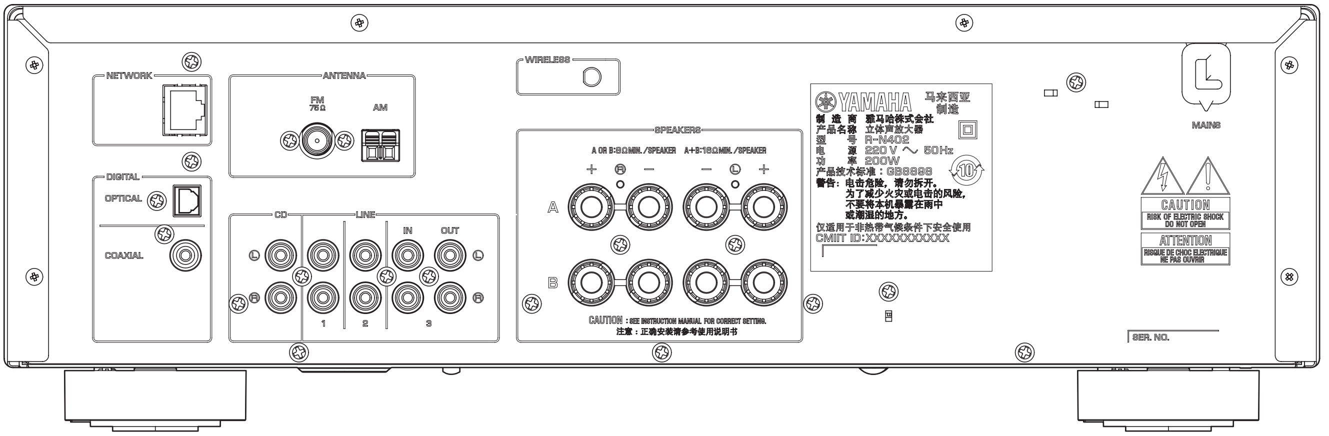
R-N402 (U model)



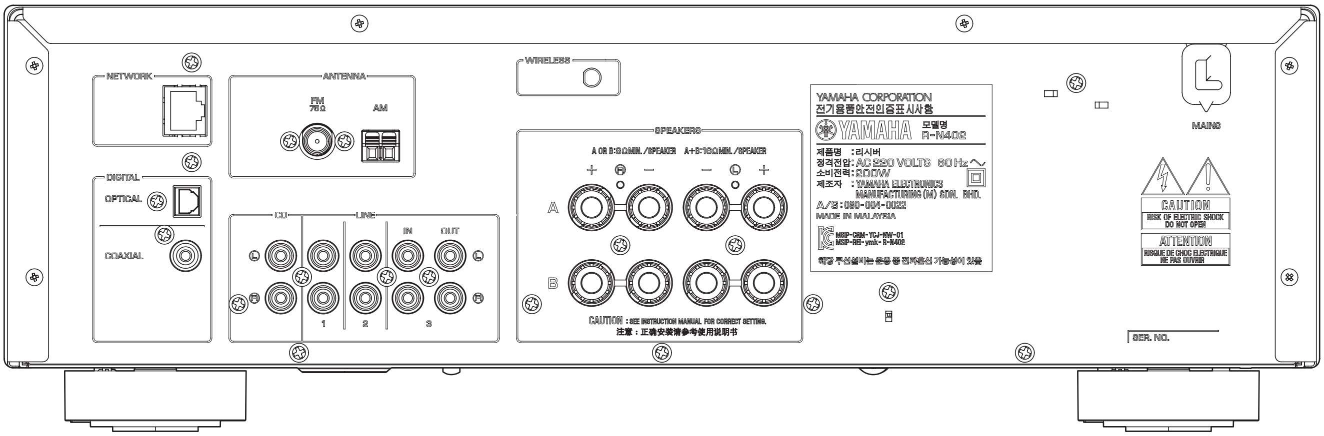
R-N402 (R model)



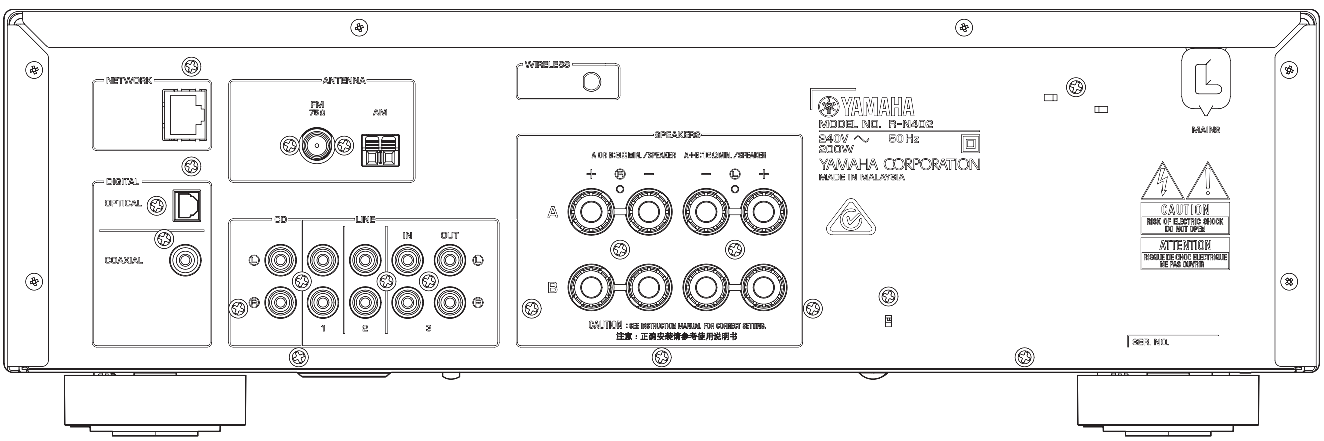
R-N402 (T model)



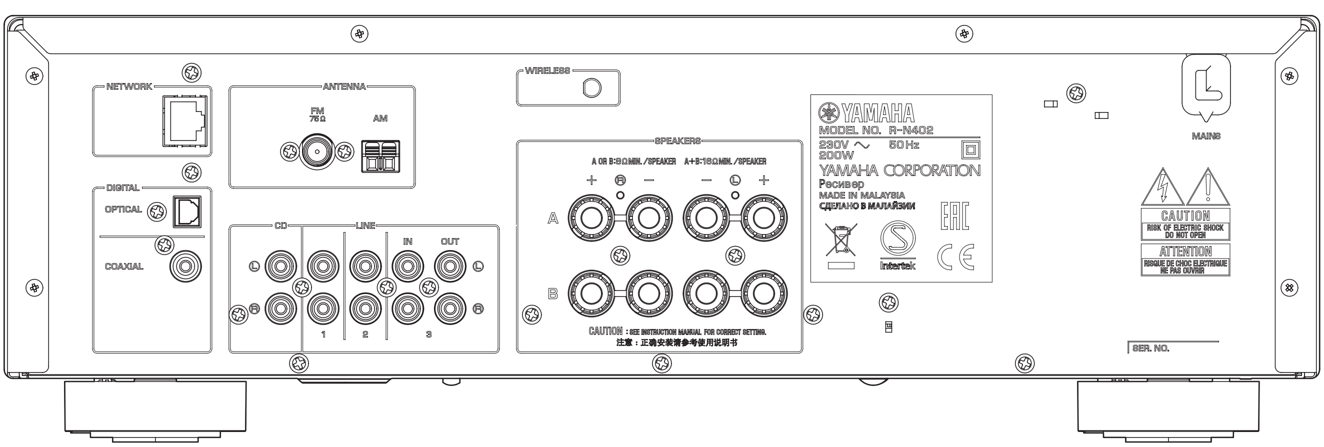
R-N402 (K model)



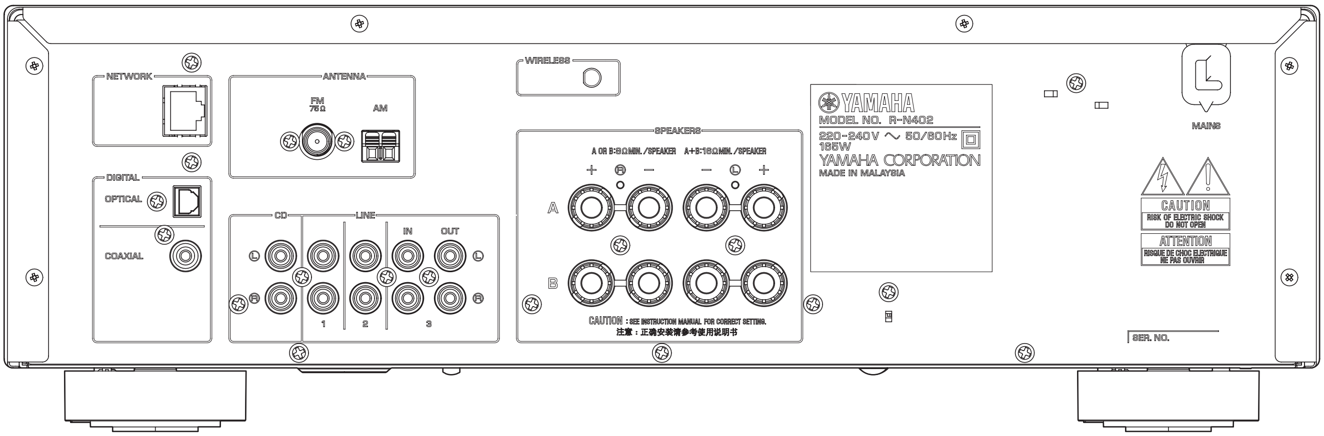
R-N402 (A model)



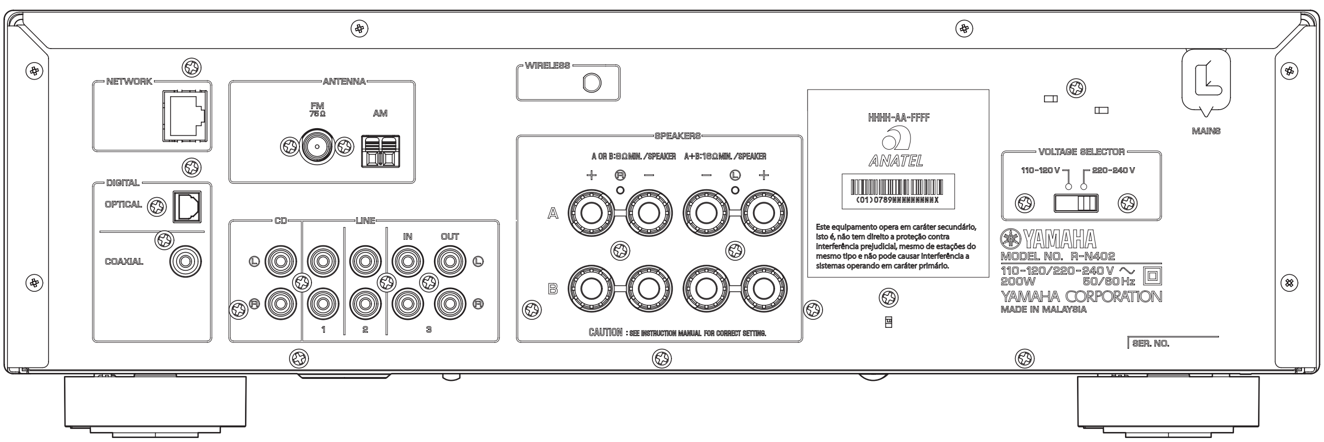
R-N402 (B, G models)



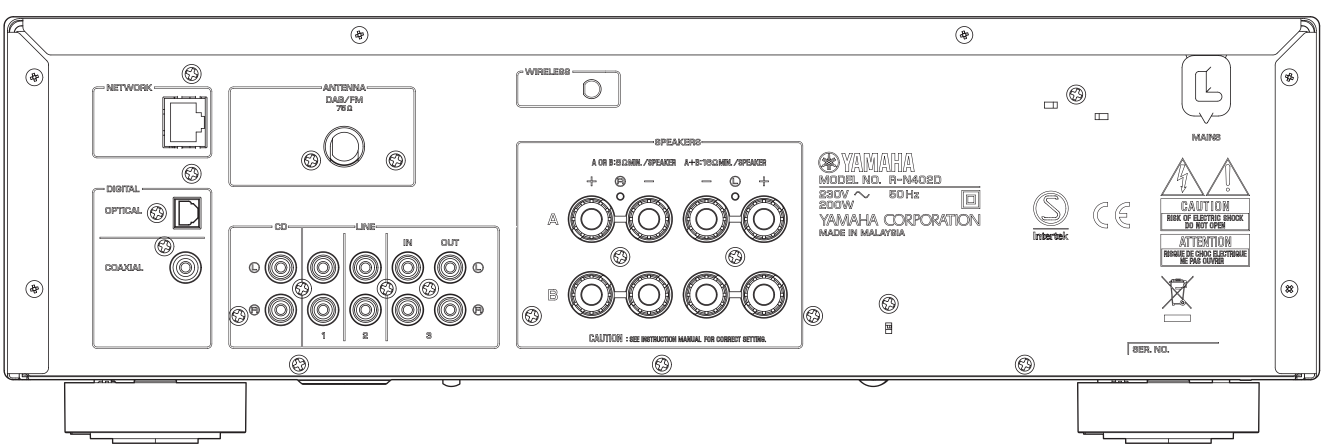
R-N402 (L model)



R-N402 (S model)

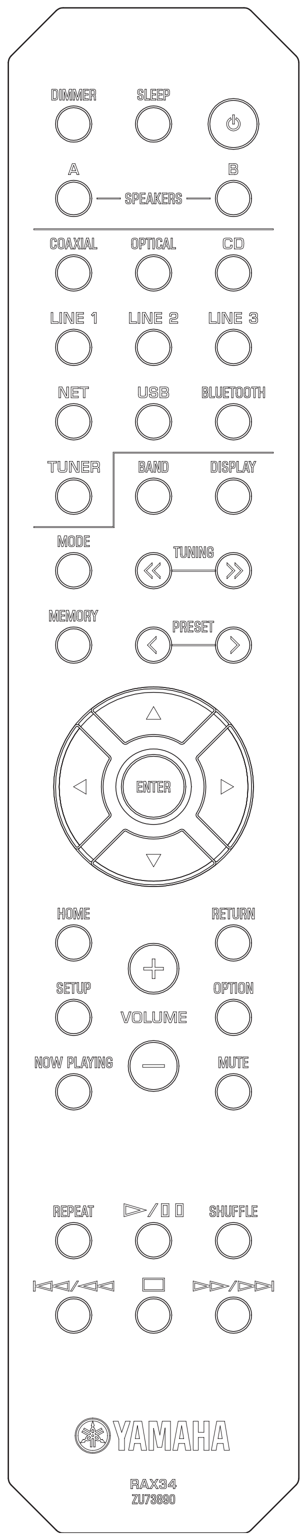


R-N402D (B, G models)



■ REMOTE CONTROL PANEL

RAX34



■ SPECIFICATIONS

■ Audio Section

Minimum RMS Output Power (Power Amp. Section)
(40 Hz to 20 kHz, 0.2 % THD, 8 ohms / L/R drive)
U, R, T, K, A, B, G, S models .... 100 W + 100 W
L model .... 85 W + 85 W

Dynamic Power Per Channel (IHF)
8 / 6 / 4 / 2 ohms .... 125 / 150 / 165 / 180 W

Maximum Power Per Channel (1 kHz, 0.7 % THD, 4 ohms)
[B, G models]
.... 115 W

IEC Power (1 kHz, 0.2 % THD, 8 ohms)
[B, G models]
.... 110 W

Damping Factor (1 kHz, 8 ohms)
SPEAKER-A ....120 or more

Maximum Effective Output Power (JEITA)
(1 kHz, 10 % THD, 8 ohms)
R model ....140 W
L model...125 W

Input Sensitivity/Input Impedance (1 kHz, 100 W/8 ohms)
CD etc..500 mV / 47 k-ohms

Maximum Input Signal (1 kHz, 0.5% THD)
CD etc..2.2 V or more

Output Level/Output Impedance (1 kHz, 500 mV, CD etc. input)
LINE OUT...500 mV / 2.2 k-ohms
8 ohms load, Headphone jack ....470 mV / 470 ohms

Frequency Response (CD etc.)
20 Hz to 20 kHz ....0 ± 0.5 dB
10 Hz to 100 kHz ....0 ± 3.0 dB

Total Harmonic Distortion (20 Hz to 20 kHz, 50 W/8 ohms)
CD etc. to SP OUT... 0.2 % or less

Signal to Noise Ratio (IHF-A)
CD etc., Input shorted 500 mV .... 100 dB or more

Residual Noise (IHF-A)
.... 70 μ V

Channel Separation
(CD etc. Input 5.1 k-ohms shorted)
1 kHz.. 65 dB or more
10 kHz.. 50 dB or more

Tone Control Characteristics
Bass
Boost/Cut (50 Hz).. ±10 dB
Treble
Boost/Cut (20 kHz).. ±10 dB

Optical Jack, Coaxial Jack Support fs
................................32 kHz / 44.1 kHz / 48 kHz / 88.2 kHz / 96 kHz / 176.4 kHz / 192 kHz

■ FM Section

Tuning Range

U, R, L, S models .... 87.5 to 107.9 MHz
R, T, K, A, B, G, L, S models .... 87.50 to 108.00 MHz

50 dB Quieting Sensitivity (IHF-A) (1 kHz, 100 % MOD.)

Mono .... 3 μ V (20.8 dBf)

Signal to Noise Ratio (IHF-A)

Mono / Stereo .... 71 dB / 70 dB

Harmonic Distortion (1 kHz)

Mono / Stereo .... 0.4 % / 0.4 %

Antenna Input

.... 75 ohms unbalanced

■ AM Section

Tuning Range

U, R, L, S models .... 530 to 1,710 kHz
R, T, K, A, B, G, L, S models .... 531 to 1,611 kHz

Antenna

.... Loop antenna

■ DAB Section (R-N402D)

[B, G models]

Tuning Range

.... 174 to 240 MHz (Band III)

Support Audio Format

.... MPEG 1 Layer II / MPEG 4 HE ACC v2 (ACC+)

Antenna

.... 75 ohms unbalanced

■ General

Power Supply

U model .... AC 120 V, 60 Hz
R, S models .... AC 110–120/220–240 V, 50/60 Hz
T model .... AC 220 V, 50 Hz
K model .... AC 220 V, 60 Hz
A model .... AC 240 V, 50 Hz
B, G models .... AC 230 V, 50 Hz
L model .... AC 220–240 V, 50/60 Hz

Power Consumption

U, R, T, K, A, B, G, L, S models .... 200 W
L model .... 165 W

Standby Power Consumption

Network Standby (Wired, Wi-Fi, Wireless Direct, Bluetooth)
ALL OFF .... 0.1 W
Network Standby ON
Wired ON .... 1.7 W
Wi-Fi (Wireless) ON .... 1.8 W
Wireless Direct ON .... 1.8 W
Bluetooth ON .... 1.8 W

Dimensions (W x H x D)

.... 435 x 141 x 340 mm (17-1/8" x 5-1/2" x 13-3/8")

Reference Dimensions (W x H x D) with attaching wireless antenna

.... 435 x 200 x 340 mm (17-1/8" x 7-7/8" x 13-3/8")

Weight

.... 7.3 kg (16.1 lbs.)

Finish

[R-N402]
U, R, A, G, L, S models .... Black color
R, T, K, A, B, G, L models .... Silver color

[R-N402D]
B, G models .... Black / Silver color

Accessories

Remote control .... x 1
Battery (R6, AA, UM-3) .... x 2
FM antenna (1.4 m) (R-N402) .... x 1
AM antenna (1.2 m) (R-N402) .... x 1
DAB/FM antenna (1.6 m) (R-N402D) .... x 1

*** Specifications are subject to change without notice.**

U .... U.S.A. model
R .... General model
T .... Chinese model
K .... Korean model
A .... Australian model
B .... British model
G .... European model
L .... Singapore model
S .... Brazilian model

Supports iOS 7 or later for setup using Wireless Accessory Configuration.

"Made for iPod," "Made for iPhone," and "Made for iPad" mean that an electronic accessory has been designed to connect specifically to iPod, iPhone, or iPad, respectively, and has been certified by the developer to meet Apple performance standards.

Apple is not responsible for the operation of this device or its compliance with safety and regulatory standards.

Please note that the use of this accessory with iPod, iPhone, or iPad may affect wireless performance.

Made for.

iPhone 6s Plus, iPhone 6s, iPhone 6 Plus, iPhone 6, iPhone 5s, iPhone 5c, iPhone 5, iPhone 4s
iPad Pro, iPad mini 4, iPad Air 2, iPad mini 3, iPad Air, iPad mini 2, iPad mini, iPad (3rd and 4th generation), iPad 2
iPod touch (5th and 6th generation)
(as of June 2016)

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Bluetooth protocol stack (Blue SDK)

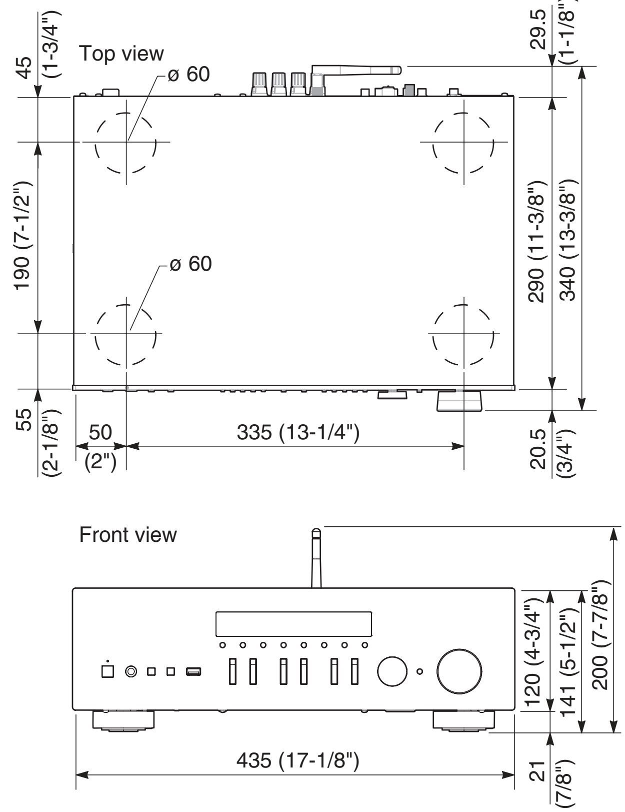
Explanations regarding GPL

This product utilizes GPL/LGPL open-source software in some sections. You have the right to obtain, duplicate, modify, and redistribute this open-source code only. For information on GPL/LGPL open source software, how to obtain it, and the GPL/LGPL license, refer to the Yamaha Corporation website
(http://download.yamaha.com/sourcecodes/musiccast/).

MusicCast is a trademark or registered trademark of Yamaha Corporation.

(For R-N402D)
The unit supports DAB/DAB+ tuning.

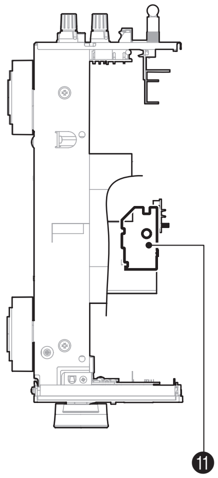
- **DIMENSIONS**



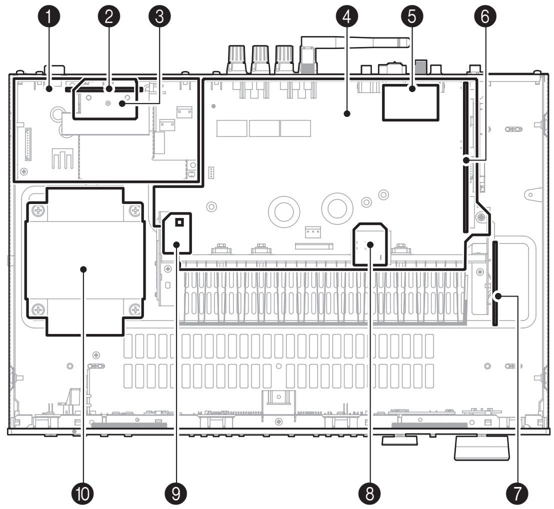
Unit: mm (inch)

■ INTERNAL VIEW

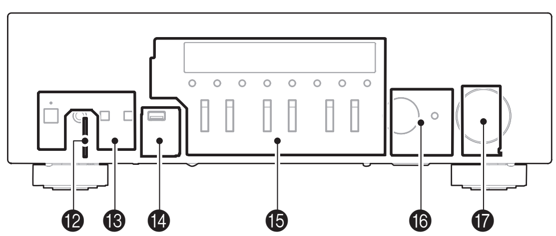
Side view



Top view



Front view



❶ MAIN (8) P.C.B.
❷ SWITCH P.C.B. (R, S models)
❸ MAIN (12) P.C.B.
❹ MAIN (1) P.C.B.
❺ TUNER Module (R-N402)
DAB P.C.B. + DAB Module
(R-N402D)
❻ DIGITAL (1) P.C.B.
❼ MAIN (9) P.C.B.
❽ MAIN (6) P.C.B.
❾ MAIN (11) P.C.B.
❿ POWER TRANSFORMER
⓫ MAIN (10) P.C.B.
⓬ MAIN (5) P.C.B.
⓭ MAIN (3) P.C.B.
⓮ DIGITAL (2) P.C.B.
⓯ MAIN (2) P.C.B.
⓰ MAIN (7) P.C.B.
⓱ MAIN (4) P.C.B.

■ SERVICE PRECAUTIONS

Safety measures

- Some internal parts in this product contain high voltages and are dangerous.
  Be sure to take safety measures during servicing, such as wearing insulating gloves.
- Note that the capacitors indicated below are dangerous even after the power is turned off because an electric charge remains and a high voltage continues to exist there.

  Before starting any repair work, connect a discharging resistor (5 k-ohms/10 W) to the terminals of each capacitor indicated below to discharge electricity.

  The time required for discharging is about 30 seconds per each.

  C140 and C143 on MAIN (1) P.C.B.

  For details, refer to “PRINTED CIRCUIT BOARDS”.

■ DISASSEMBLY PROCEDURES

(Remove parts in the order as numbered.)
Disconnect the power cable from the AC outlet.

1. Removal of Top Cover

a. Remove screw (①), 4 screws (②) and 6 screws (③). (Fig. 1)
b. Remove the top cover. (Fig. 1)

2. Removal of Front Panel Unit and Sub-Chassis Unit

a. Remove the knobs. (Fig. 1)
b. Remove 5 screws (④). (Fig. 1)
c. Release 6 hooks (Ⓐ) and then remove the front panel unit forward. (Fig. 1)
d. Remove screw (⑤) and 6 screws (⑥). (Fig. 1)
e. Remove CB101, CB201, CB301 (DIGITAL (1)), CB301 (MAIN (1)) and CB503. (Fig. 1)
f. Release 2 hooks (Ⓑ) and then remove the sub-chassis unit. (Fig. 1)

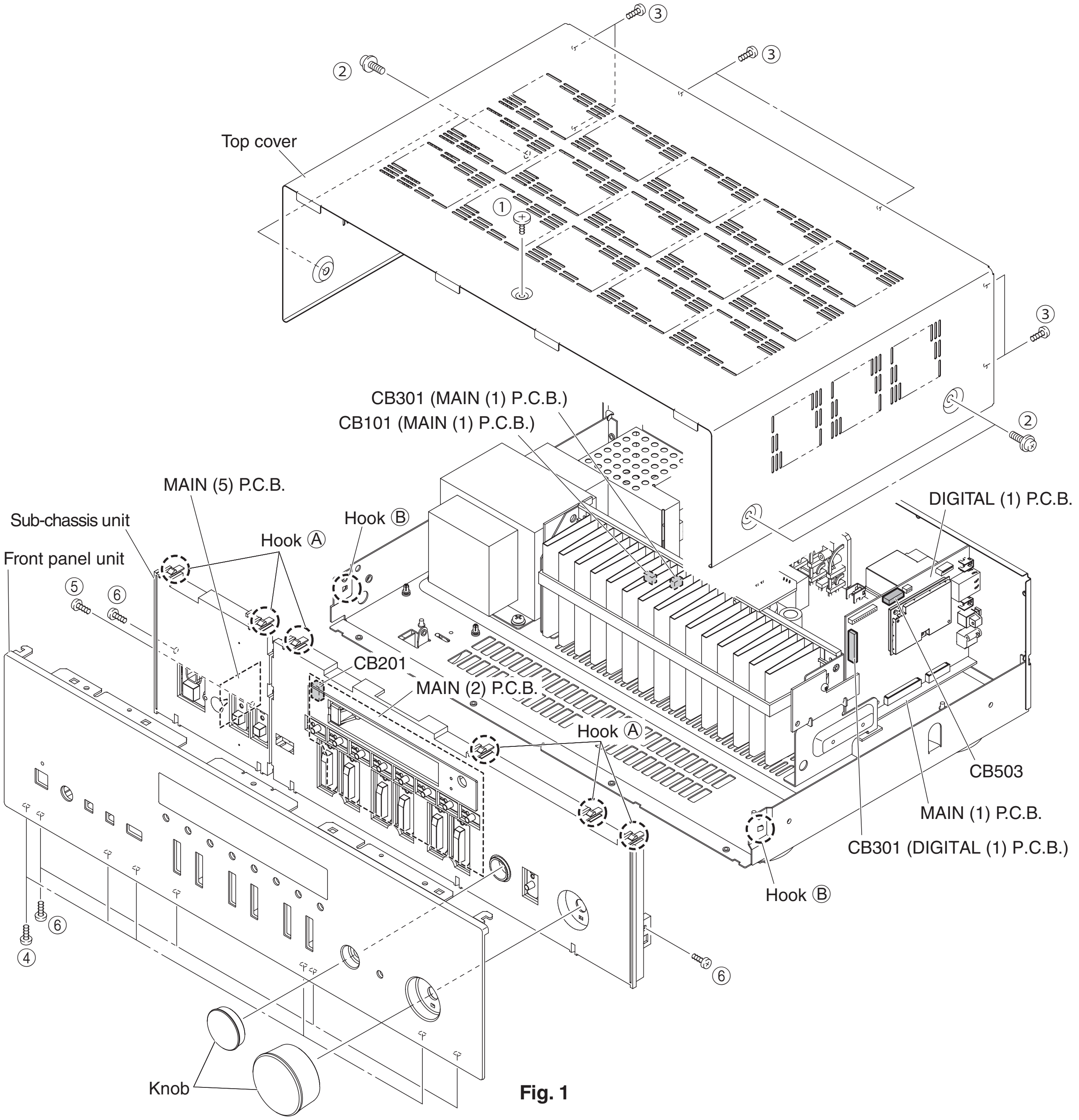


Fig. 1

3. Removal of Network Module

a. Remove the Wireless LAN Antenna Connector by using MHF Connector Remover. (Fig. 2)

b. Remove a screw (⑦). (Fig. 2)

c. Remove the Network Module forward. (Fig. 2)
The network module is directly connected to the DIGITAL (1) P.C.B. with the board-to-board connectors.

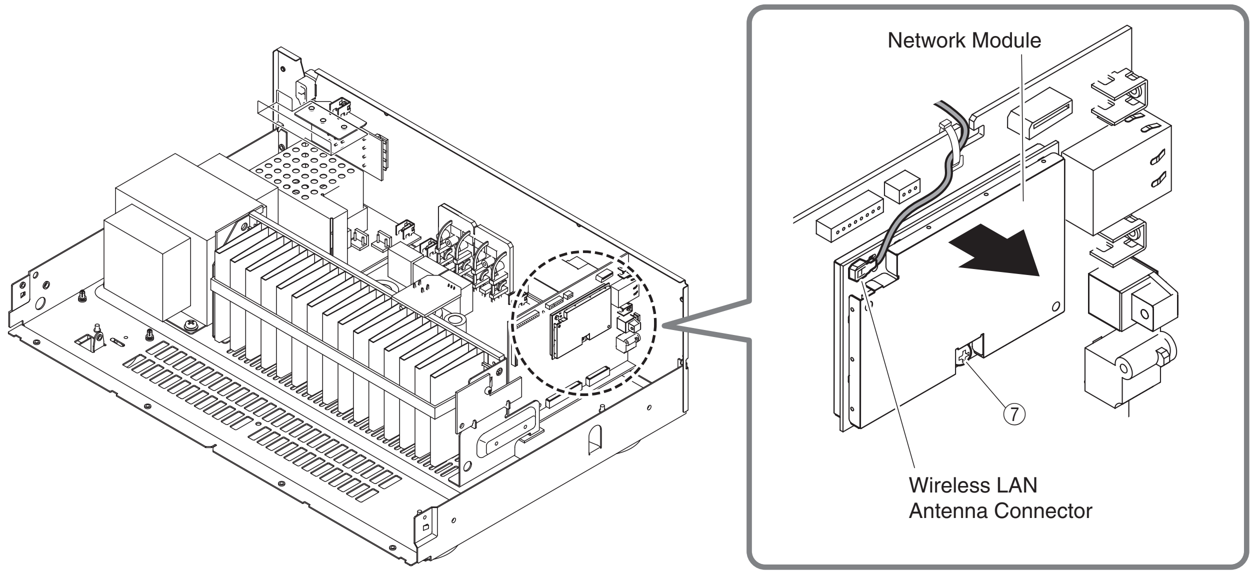


Fig. 2

CAUTION !

- To remove the Wireless LAN Antenna Connector, use the special MHF Connector Remover. Hook the tip of this tool on the cover of the Wireless LAN Antenna Connector and pull it straight in the direction of the engaging axis of the Wireless LAN Antenna Connector. (Fig. 3)

  Special removing tool
  ZK708100: MHF Connector Remover

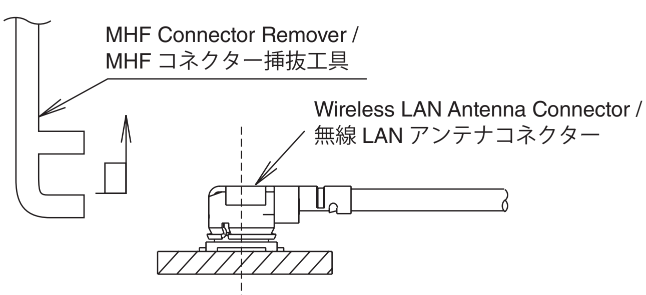


Fig. 3

- The Wireless LAN Antenna Connector should only be plugged back to the Network Module after it has been installed back onto the DIGITAL (1) P.C.B.
- When plugging the Wireless LAN Antenna Connector back to the Network Module, make sure to hold it and insert it vertically using the MHF Connector Remover. Make sure not to insert the Wireless LAN Antenna Connector at a sharp angle as it may break.
- The Wireless LAN Antenna Connector can be inserted and removed up to 5 times only.

4. Removal of AMP Unit

a. Remove 3 screws (⑧). (Fig. 5)
b. Remove screw (⑨) and 3 screws (⑩). (Fig. 4)
c. Unlock the locking card spacer. (Fig. 4)
d. Remove CB3 (R, S models), CB103, CB541 and CB544. (Fig. 4)
e. Remove the AMP unit. (Fig. 4)

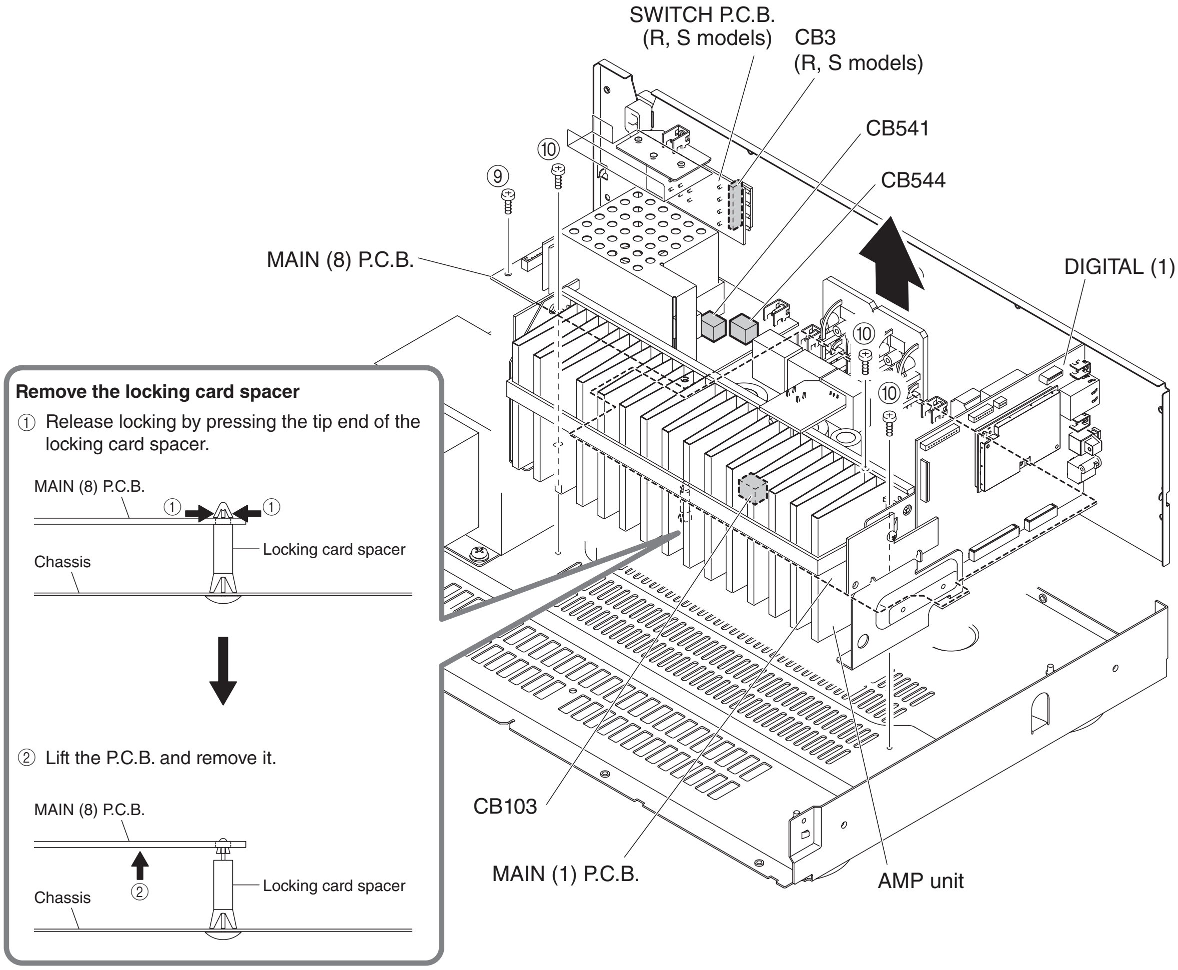


Fig. 4

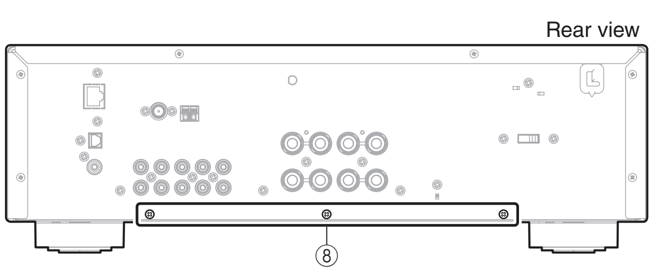


Fig. 5

When checking the MAIN (1) P.C.B.:

- Place the P.C.B.s (with rear panel) upright. (Fig. 6)
- Connect the heatsink and rear panel to the chassis with a ground lead or the like. (Fig. 6)
- Reconnect all cables (connectors) that have been disconnected.
- When connecting the flexible flat cable, be careful with polarity.

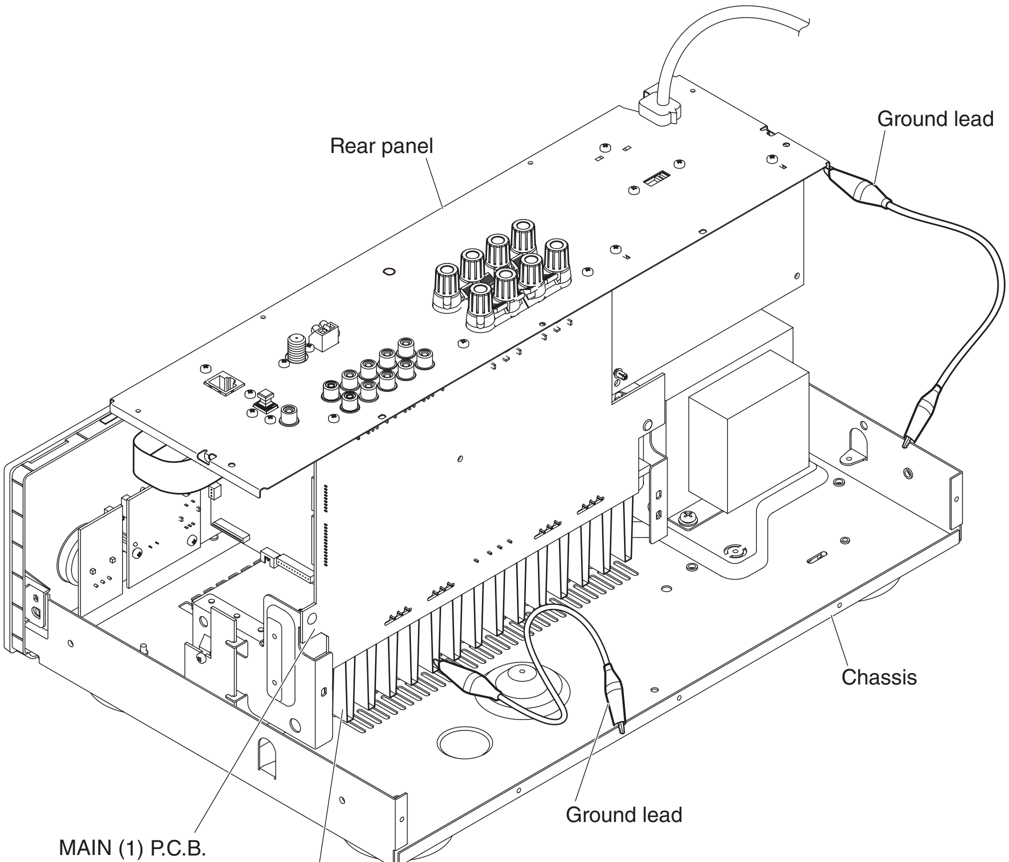


Fig. 6

R-N402/R-N402D

■ UPDATING FIRMWARE

When the following parts are replaced, the firmware must be updated to the latest version.

DIGITAL P.C.B.
Network module

● Confirmation of firmware version

Before and after updating the firmware, check the firmware version by using the self-diagnostic function menu.

Start up the self-diagnostic function and select “S2. ROM VERSION/CHECKSUM” menu.

Using the sub-menu, have the firmware version displayed, and note them down.

(For details, refer to “SELF-DIAGNOSTIC FUNCTION”)

* When the firmware version is different from written one after updating, perform the updating procedure again from the beginning.

● Initializing the back-up IC (EEPROM: IC301 on DIGITAL P.C.B.)

After updating the firmware, the back-up IC MUST be initialized by the following procedure store the setting information (soundfield parameters, system memory and tuner presetting, etc.) properly.be initialized by the following procedure store the setting information (soundfield parameters, system memory and tuner presetting, etc.) properly.

Start up the self-diagnostic function and select “S1. FACTORY PRESET” menu.
(For details, refer to “SELF-DIAGNOSTIC FUNCTION”)

Select “PRESET RSRV”, press the “⏻” (Power) key to turn off the power once and turn on the power again. Then the back-up IC is initialized.

● Required Tools

- USB storage device
- Latest firmware

Download the latest firmware from the specified download source to the PC.

And then, copy the firmware to the root folder of the USB storage device.

If the firmware is copied to a subfolder of the USB storage device, the update will not proceed.

● Operation Procedures

* Disconnect the power cable of this unit from the AC outlet.

1. Insert the USB storage device to the USB jack. (Fig. 1)
2. While pressing the “DISPLAY” key, connect the power cable to the AC outlet. (Fig. 1)

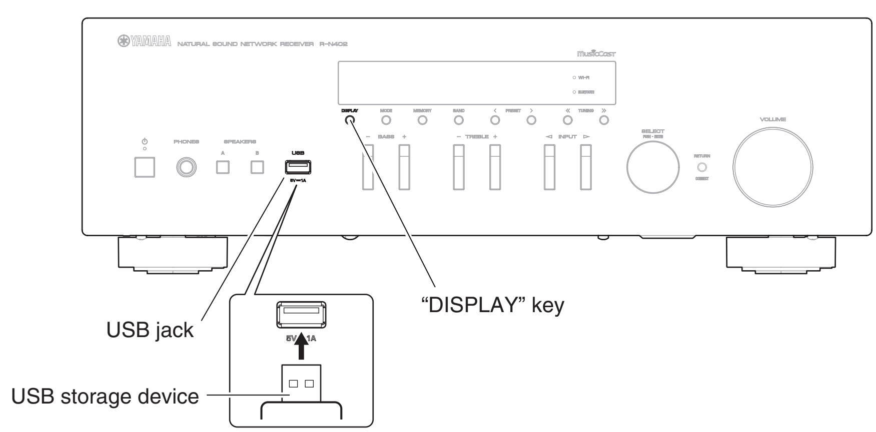


Fig.1

3. The USB UPDATE mode is activated and “USB UPDATE” is displayed. Writing of the firmware starts automatically. (Fig. 2)

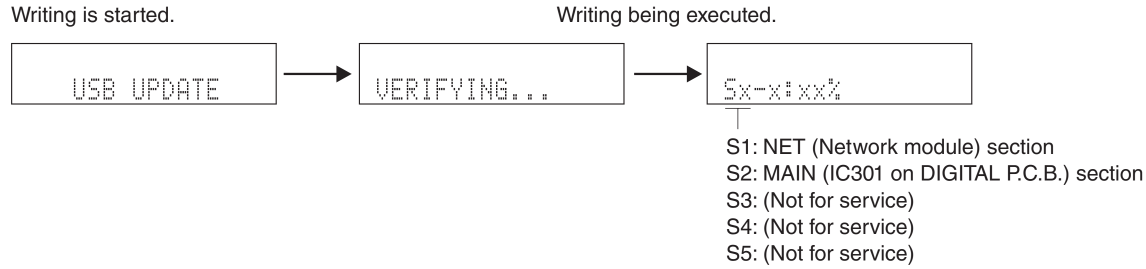


Fig. 2

* If error message is displayed during writing of the firmware, refer to “List of Error Messages” to determine the cause and perform the updating procedure again from the beginning.

4. When writing of the firmware is completed, “UPDATE SUCCESS”, “PLEASE...” and “POWER OFF!” are displayed repeatedly. (Fig. 3)

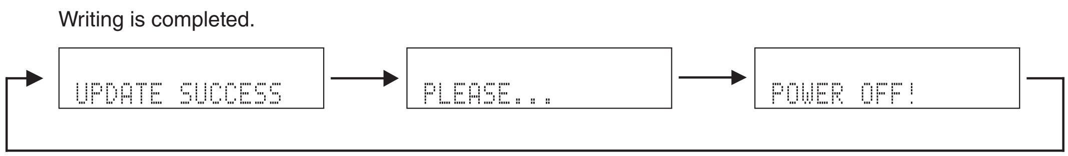


Fig. 3

5. Press the “⏻” (Power) key to turn off the power. (Fig. 4)
6. Remove the USB storage device from the USB jack. (Fig. 4)
7. Start up the self-diagnostic function and check that the firmware version is the same as written ones. (For details, refer to “Confirmation of firmware version”)

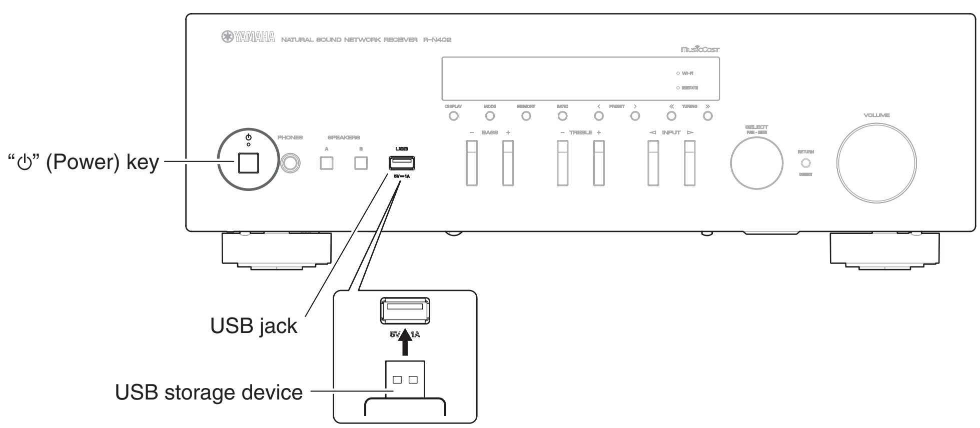
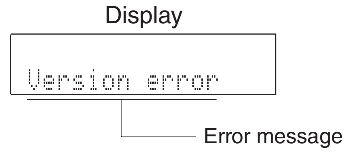


Fig.4

R-N402/R-N402D

List of Error Messages



Error Message	Cause	Remedy
Access denied	Access to the PC is denied.	Configure the sharing settings and select the unit as a device to which music contents are shared.
Access error	The unit cannot access the USB device.	Turn off the unit and reconnect your USB device. If the problem persists, try another USB device.
	There is a problem with the signal path from the network to the unit.	Make sure your router and modem are turned on. Check the connection between the unit and your router (or hub).
Check SP Wires	The speaker cables short circuit.	Twist the bare wires of the cables firmly and connect to the unit and speakers properly.
No content	There are no playable files in the selected folder.	Select a folder that contains files supported by the unit.
Please wait	The unit is preparing for connecting to the network.	Wait until the message disappears. If the message stays more than 3 minutes, turn off the unit and turn it on again.
Unable to play	The unit cannot play back the songs stored on the PC for some reason.	Check if the format of files you are trying to play is supported by the unit. For information about the formats supported by the unit, see “Playing back music stored on media servers (PCs/NAS)”. If the unit supports the file format, but still cannot play back any files, the network may be overloaded with heavy traffic.
Version error	Firmware update via the internet failed.	Update the firmware again.

■ SELF-DIAGNOSTIC FUNCTION

This unit has self-diagnostic functions that are intended for inspection, measurement and location of faulty point.
Each item has a main menu, each of which has sub-menu items.
Listed in the table below are main menu items and sub-menu items.
Note: Some of the menu items listed below may not apply to the models covered in this service manual.

No.	Main menu	No.	Sub-menu
A: Audio system			
A1	AUDIO SET	1	NORMAL
		2	PRE MUTE
		3	VOL MUTE
A2	DIR PLL	1	DIR PLL
D: Display system			
D1	FL CHECK	1	INITIAL DISPLAY
		2	ALL SEGMENT OFF
		3	ALL SEGMENT ON
		4	CHECK PATTERN 1
		5	CHECK PATTERN 2
U: Universal system			
U1	USB	1	USB FRONT 1 TRACK
		2	USB_VBUS HIGH POWER (Not for service)
N: Network system			
N1	NETWORK	1	IP ADDRESS CHECK
		2	MAC ADDRESS CHECK
		3	LINE NOISE 100 MDI (Not for service)
		4	LINE NOISE 100 MDIX (Not for service)
		5	LINE NOISE 10 MDI (Not for service)
		6	LINE NOISE 10 MDIX (Not for service)
		7	LINK CHECK
		8	EXT TEST
		10	PING (Not for service)
N2	WiFi	1	WiFi ON JIG01 (Not for service)
		2	WiFi ON JIG02 (Not for service)
		3	WiFi ON JIG03 (Not for service)
		4	WiFi ON JIG04 (Not for service)
		5	WiFi ON JIG05 (Not for service)
		6	WiFi ON JIG06 (Not for service)
		7	WiFi ON JIG07 (Not for service)
		8	WiFi ON JIG08 (Not for service)
		9	WiFi ON JIG09 (Not for service)
		10	WiFi ON JIG10 (Not for service)
		11	WiFi OFF (Not for service)
		12	WiFi MAC ADDRESS
		13	WiFi RF TEST (Not for service)
N3	BLUETOOTH	1	Bluetooth MODULE VERSION (Not for service)
		2	Bluetooth ADDRESS
		3	PLAYBACK
N4	NETWORK AUDIO (Not for service)	1	LOOP BACK
		2	NETWORK AUDIO PLAYBACK
N5	NET P.C.B. CONNECT CHECK (Not for service)	1	SPI
C: Communication system			
C1	ACCESS CHECK	1	ALL
		2	BUS DIR
		3	EEPROM
		4	TUNER

No.	Main menu	No.	Sub-menu
C2	NETWORK IC CHECK	1	ALL
		2	NET RAM (Not for service)
		3	PHY (Ethernet PHYceiver) TEST (Not for service)
		4	APL (Apple) ID CHECK
		5	CLK GEN
		6	PMIC
		7	WL MOD (Not for service)
		8	NET EEPROM
R: DAB			
R1	DAB	1	INVALID ITEM (Not for service)
		2	INVALID ITEM (Not for service)
P: Power supply and protection system			
P1	AD DATA CHECK	1	DC
		2	PS
		3	TM
		4	OUT LEVEL
		5	USB-VBUS
		6	KEY
P2	PROTECTION HISTORY	1	1. HISTORY 1
		2	1. POWER PORT
		3	1. LAST INPUT
		4	1. LAST VOLUME LEVEL
		5	2. HISTORY 2
		6	2. POWER PORT
		7	2. LAST INPUT
		8	2. LAST VOLUME LEVEL
		9	3. HISTORY 3
		10	3. POWER PORT
		11	3. LAST INPUT
		12	3. LAST VOLUME LEVEL
		13	4. HISTORY 4
		14	4. POWER PORT
		15	4. LAST INPUT
		16	4. LAST VOLUME LEVEL
T: Troubleshooting Information			
T1	TROUBLE SHOOTING INFORMATION	1	OPERATING TIME
		2	POWER-RELAY ON
S: System and version system			
S1	FACTORY PRESET	1	PRESET INHIBIT
		2	PRESET RESERVED
S2	ROM VERSION/CHECKSUM	1	SYSTEM VERSION
		2	MICROPROCESSOR VERSION
		3	MICROPROCESSOR CHECKSUM
		4	NETWORK VERSION
		5	NETWORK CHECKSUM
S3	SOFT SWITCH (Not for service)	1	SWITCH MODE
S4	SYSTEM INFORMATION	1	MODEL/DESTINATION
		2	VERIFY (Not for service)

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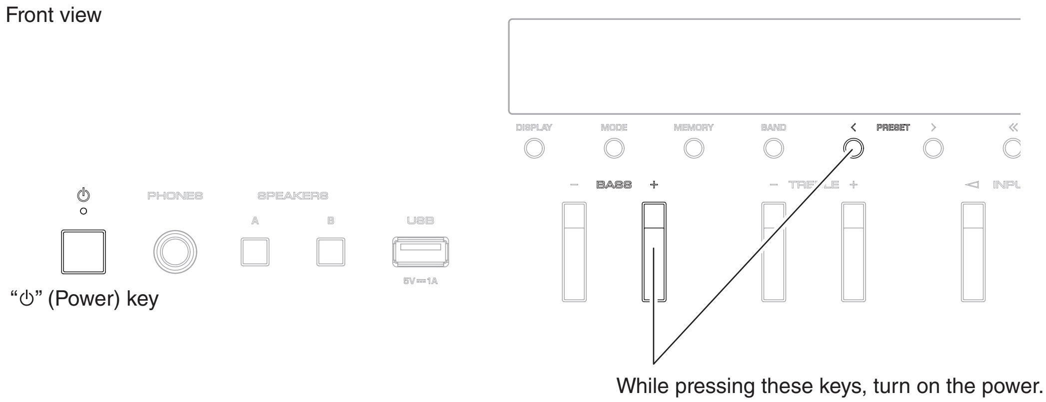
Starting Self-Diagnostic Function

While pressing the “PRESET <” and “BASS +” keys, press the “⏻” (Power) key to turn on the power, and release those 2 keys.

The self-diagnostic function mode is activated.

Keys of this unit



Starting Self-Diagnostic Function in the protection cancel mode

If the protection function works and causes hindrance to troubleshooting, cancel the protection function by the procedure below, and it will be possible to enter the self-diagnostic function mode. (The protection functions other than the excess current detect function will be disabled.)

While pressing the “PRESET <” and “BASS +” keys, press the “⏻” (Power) key to turn on the power and keep pressing those 2 keys for 3 seconds or longer.

The self-diagnostic function mode is activated with the protection functions disabled.

In this mode, the “SLEEP” segment of the FL display flashes to indicate that the mode is self-diagnostic function mode with the protection functions disabled.

CAUTION:
Using this unit with the protection function disabled may cause further damage to this unit. Use special care for this point when using this mode.

Canceling Self-Diagnostic Function

1. Before canceling self-diagnostic function, execute setting for “S1. FACTORY PRESET” menu. (Memory initialization inhibited or Memory initialized).
   * In order to keep the user memory preserved, be sure to select PRESET INHIBIT (Memory initialization inhibited).
2. Press the “⏻” (Power) key to turn off the power.

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● Display provided when Self-Diagnostic Function started

The display is as described below depending on the situation when the power to this unit is turned off.

1. When the power is turned off by usual operation:

"NO PROTECT" is displayed. Then "A1-1. NORMAL" is displayed in a few seconds.

Opening message	After a few seconds	Main menu display
NO PROTECT	⇨	A1-1 NORMAL

2. When the protection function worked to turn off the power:

The information of protection function which worked at that time is displayed. Then "A1-1. NORMAL" is displayed in a few seconds.

Note: At that time if you restart the self-diagnostic function after turning off the power once, "NO PROTECT" will be displayed. That is because that situation is equal to "1. When the power is turned off by usual operation:". However history of the protection function is stored in memory as backup data. For details, refer to "P2. PROTECTION HISTORY" menu.

2-1. When there is a history of protection function due to excess current.

```
I PROTECT
```

Cause: An excessive current flowed through the power amplifier.

Supplementary information: As over current of the power amplifier is detected, the abnormal channel can be identified by checking the current detect transistor.

Turning on the power without correcting the abnormality will cause the protection function to work immediately and the power supply will instantly be shut off.

Notes:
- **Applying the power to this unit without correcting the abnormality can be dangerous and cause additional circuit damage. To avoid this, if "I PROTECT" protection function works 1 time, the power will not turn on even when the "⏻" (Power) key is pressed. In order to turn on the power again, start up the self-diagnostic function.**
- **The output transistors in each amplifier channel should be checked for damage before applying power to this unit.**
- **Amplifier current should be monitored by measuring DC voltage across the emitter resistors for each channel.**

2-2. When the protection function worked due to abnormal DC output.

DC PRT:xxxH

H: Displayed when the voltage is HIGHER than upper limit
L: Displayed when the voltage is LOWER than lower limit

xxx: A/D conversion value of voltage at the moment when the protection function worked (Reference voltage: 3.3 V=255)

Cause: DC output of the power amplifier is abnormal.

Supplementary information: The protection function worked due to a DC voltage appearing at the speaker terminal. A cause could be a defect in the amplifier.

Turning on the power without correcting the abnormality will cause the protection function to work in 5 seconds and the power supply will be shut off.

2-3. When the protection function worked due to abnormal voltage in the power supply section.

PSxPRT:xxxL

H: Displayed when the voltage is HIGHER than upper limit
L: Displayed when the voltage is LOWER than lower limit

xxx: A/D conversion value of voltage at the moment when the protection function worked (Reference voltage: 3.3 V = 255)

PS1/PS2/PS3

Cause: The voltage in the power supply section is abnormal.

Supplementary information: The protection function worked due to a defect or overload in the power supply.

Turning on the power without correcting the abnormality will cause the protection function to work in 1 seconds and the power supply will be shut off.

Notes:
- **Applying the power to this unit without correcting the abnormality can be dangerous and cause additional circuit damage. To avoid this, if "PS" and "DC" protection function works 3 times consecutively, the power will not turn on even when the "⏻" (Power) key is pressed. In order to turn on the power again, start up the self-diagnostic function.**
- **The output transistors in each amplifier channel should be checked for damage before applying power to this unit.**
- **Amplifier current should be monitored by measuring DC voltage across the emitter resistors for each channel.**

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2-4. When the protection function worked due to excessive heatsink temperature.

TMPxPRT:xxxL

H: Displayed when the voltage is HIGHER than upper limit
L: Displayed when the voltage is LOWER than lower limit

xxx: A/D conversion value of voltage at the moment when the protection function worked (Reference voltage: 3.3 V=255)

TMP1/TMP2/TMP3

Cause: The temperature of the heatsink is excessive

Supplementary information: The protection function worked due to the temperature limit being exceeded. Causes could be poor ventilation or a defect related to the thermal sensor.

Turning on the power without correcting the abnormality will cause the protection function to work in 1 seconds and the power supply will be shut off.

● History of protection function

When the protection function has worked, its history is stored in memory as backup data.

Even if no abnormality is noted while servicing the unit, an abnormality which has occurred previously can be defined as long as the backup data has been stored.

For details, refer to “P2. PROTECTION HISTORY” menu.

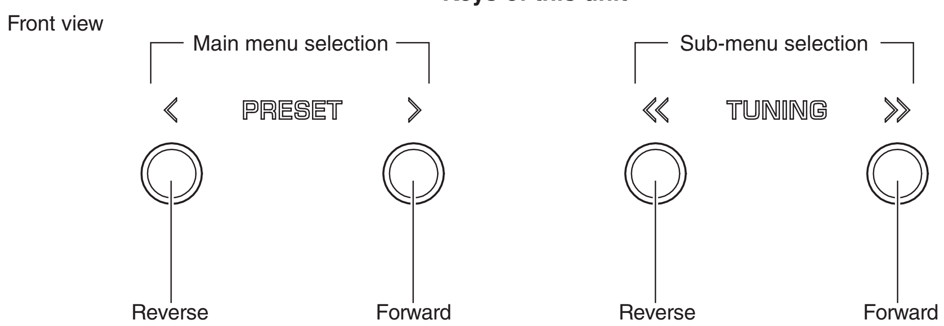
● Operation procedure of Main menu and Sub-menu

Each item has a main menu, each of which has sub-menu items.

Main menu selection
Select the main menu using “PRESET > ” (forward) and “PRESET < ” (reverse) keys.

Sub-menu selection
Select the sub-menu using “TUNING >> ” (forward) and “TUNING << ” (reverse) keys.

Keys of this unit



● Functions in Self-Diagnostic Function mode

In addition to the self-diagnostic function menu items, functions listed below are available.

- Power ON/OFF
- Master volume
- Muting
- Input selection

* Functions related to the tuner and the set menu are not available.

● Initial settings when Self-Diagnostic Function started

The following initial settings are used when self-diagnostic function is started.
When self-diagnostic function is canceled, these settings are restored to those before starting self-diagnostic function.

- Master volume: 60 (-20 dB)
- Input: LINE1
- Speaker setting: SPEAKERS A ON
SPEAKERS B ON

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● Details of Self-Diagnostic Function menu

A1. AUDIO SET

This menu is used to check audio signal route.

A1-1. NORMAL

The audio signal input to LINE 1 jack is output.

```
A1-1
NORMAL
```

A1-2. PRE MUTE

The audio signal output is muted.

MUTE control port: MAIN_N_MUTE (FRONT L/R, SURROUND L/R channels, 58 pin on IC302)
REC_N_MUTE (REC L/R channel, 60 pin on IC302)

```
A1-2
PRE MUTE
```

A1-3. VOL MUTE

The sound signal output is muted on Volume IC.

```
A1-3
VOL MUTE
```

A2. DIR PLL (Phase Lock Loop)

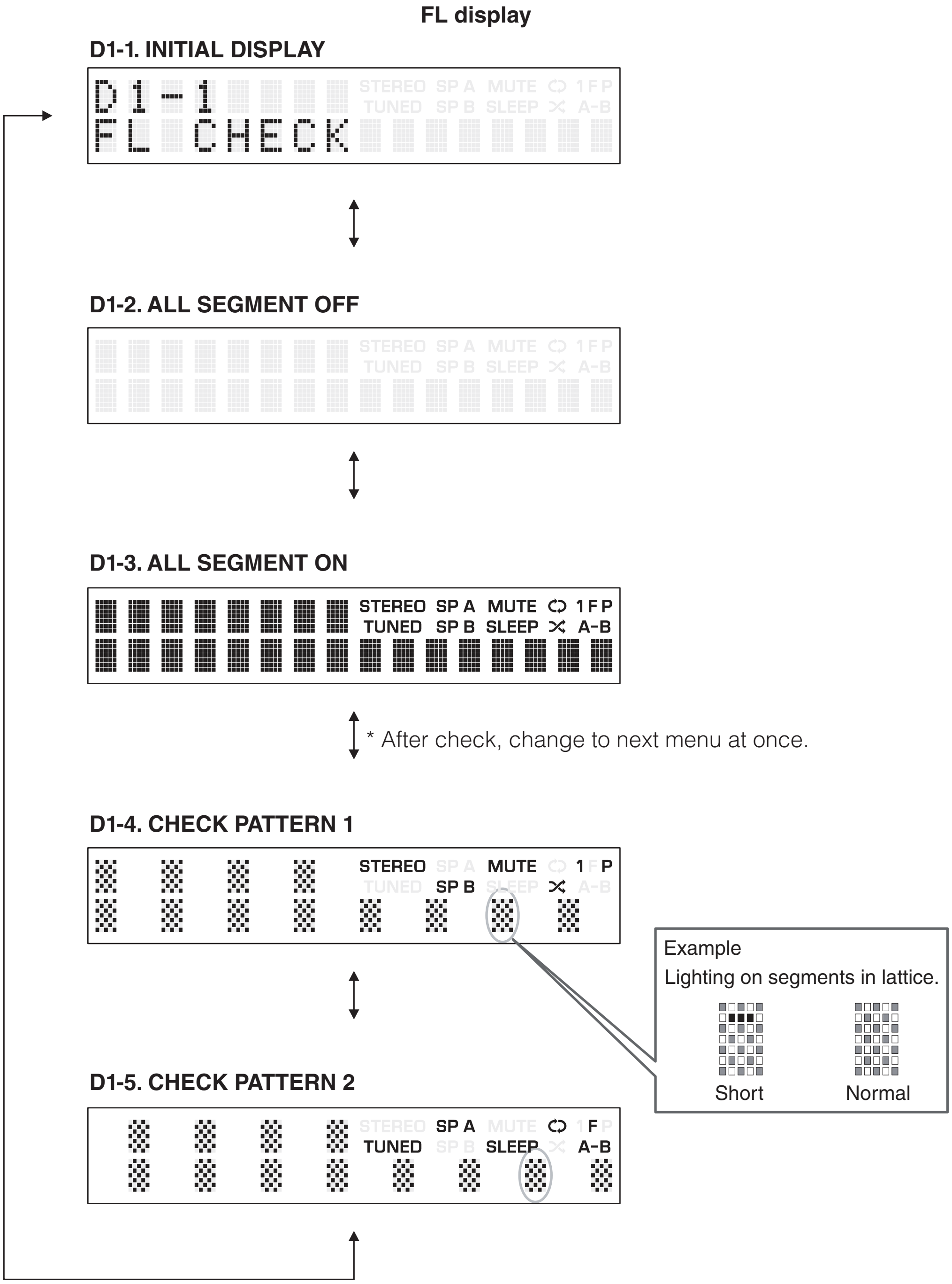
This menu is used to check the route of digital audio signal input to OPTICAL/COAXIAL jack.

```
A2-1
DIR PLL:---
```

Lock: Lock
Unlock: Unlock
---: When the analog input is selected for the input source

D1. FL CHECK

This menu is used to check operation of the FL display.



Segment conditions of the FL tube is checked by turning ON and OFF all segments.

Next, a short between segments next to each other is checked by turning ON and OFF all segments alternately (in lattice).

(In the above example, the segments in the second row from the top are shorted.)

U1. USB

This menu is used to check the audio signal route from USB storage device.

U1-1. USB FRONT 1 TRACK

The 1st music file stored in the USB storage device connected to the USB jack is played.

* Copy 2 or more music files from PC to the root folder of the USB storage device in advance.

```
U1-1
USB_F 1 TRACK
```

U1-2. USB_VBUS HIGH POWER

Not for service.

```
U1-2
USB_VBUS_HPWR
```

N1. NETWORK

This menu is used to check functions related to NETWORK.

Connect between LAN port of broadband router and NETWORK jack of this unit with a network cable.

* When the network condition varies while sub-menu is displayed (e.g., the network is deactivated once), the correct result will not be displayed.

  In that case, once turn off the power to this unit, then start up the self-diagnostic function again and select this menu.

N1-1. IP ADDRESS CHECK

This menu is used to check that IP address can be obtained.

```
N1-1
IP AD CHK:OK
```

OK: Connected (IP address obtained)
NG: No traffic / Disconnected

N1-2. MAC ADDRESS CHECK

This menu is used to check that MAC address is written.

```
N1-2
MAC AD CHK:OK
```

OK: Normal
NG: Unwritten

N1-3. LINE NOISE 100 MDI

Not for service.

```
N1-3
LN MDI 100
```

N1-4. LINE NOISE 100 MDIX

Not for service.

```
N1-4
LN MDIX 100
```

N1-5. LINE NOISE 10 MDI

Not for service.

```
N1-5
LN MDI 10
```

N1-6. LINE NOISE 10 MDIX

Not for service.

```
N1-6
LN MDIX 10
```

N1-7. LINK CHECK

This menu is used to check that the broadband router is connected correctly.

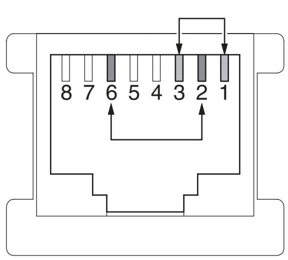
```
N1-7
LINK CHK:OK
```

OK: Connected
NG: No traffic / Disconnected

N1-8. EXT TEST

Transmission/reception of the NETWORK port is checked.
With the power turned off, short the pins of the NETWORK jack as shown in the figure below.
Start up the self-diagnostic function and select this menu.
Transmission/reception test is executed and its result is displayed.

Note) Be sure to return the shorted pins to their original condition after executing this test.



NETWORK jack

```
N1-8
EXT TEST:OK
```

OK: Normal
NG: Abnormal
--: Checking

N1-10. PING

Not for service.

```
N1-10
PING:NG
```

N2. WiFi

This menu is used to set functions related to wireless LAN adaptor.

N2-1. to N2-10. WiFi ON JIG01 to JIG10

Not for service.

```
N2-1                      N2-10
WIFI ON JIG01  ........   WIFI ON JIG10
```

N2-11. WiFi OFF

Not for service.

```
N2-11
WIFI OFF
```

N2-12. WiFi MAC ADDRESS

MAC address of the wireless LAN adaptor is displayed.

```
N2-12
68C90B 0DF98B
```

N2-13. WiFi RF TEST

Not for service.

```
N2-13
WIFI RF TEST
```

N3. BLUETOOTH

This menu is used to display Bluetooth module version and Bluetooth address.

N3-1. Bluetooth MODULE VERSION

Not available.

* Selection of menu is available.

```
N3-1
BT VERSION
```

N3-2. Bluetooth ADDRESS

Address of the Bluetooth is displayed.

```
N3-2
68C90B 0DF98A
```

N3-3. PLAYBACK

Connect (by pairing) with the device usable with Bluetooth and reproduce the music file stored in the device.

```
N3-3
PLAYBACK
```

N4. NETWORK AUDIO

Not for service.

```
N4-1                    N4-2
LOOPB0:NG  ..........   N APB0:OK
```

N5. NET P.C.B. CONNECT CHECK

Not for service

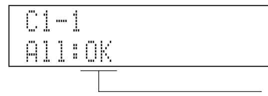
```
N5-1
SPI:OK
```

C1. ACCESS CHECK

This menu is used to check the communication and bus line connection between devices on DIGITAL P.C.B.

C1-1. ALL

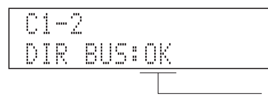
The total detection result of sub-menus from C1-2 to C1-4 is displayed.



OK: No error detected
NG: An error is detected
--: Checking

C1-2. BUS DIR

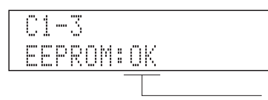
Communication and bus line connection between microprocessor (IC302) and DIR (IC508) are checked.



OK: No error detected
NG: An error is detected
--: Checking

C1-3. EEPROM

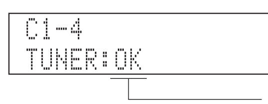
EEPROM (IC301)'s reading/writing is checked.



OK: No error detected
NG: An error is detected
--: Checking

C1-4. TUNER

The BAND TUNER I2C (Inter integrated circuit) bus line connection is checked.



OK: No error detected
NG: An error is detected
--: Checking

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C2. NETWORK IC CHECK

This menu is used to check the communication and bus line connection between devices related to network.

C2-1. ALL

The total detection result of sub-menus from C2-2 to C2-8 is displayed.

```
C2-1
ALL:OK
```

OK: No error detected
NG: An error is detected
--: Checking

C2-2. NET RAM

Not for service.

```
C2-2
NET RAM:OK
```

OK: No error detected
NG: An error is detected
--: Checking

C2-3. PHY (Ethernet PHYceiver) TEST

Not for service.

```
C2-3
PHY TEST:OK
```

OK: No error detected
NG: An error is detected
--: Checking

C2-4. APL (Apple) ID CHECK

Apple authentication IC (IC511) device ID is checked.

```
C2-4
APL ID:OK
```

OK: No error detected
NG: An error is detected
--: Checking

C2-5. CLK GEN

Communication between NET CPU and the clock generator IC is checked.

```
C2-5
CLK GEN:NG
```

C2-6. PMIC

Communication between NET CPU and PMIC is checked.

```
C2-6
PMIC:NG
```

C2-7. WL MOD

Not for service.

```
C2-7
WL MOD:NG
```

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C2-8. NET EEPROM

Communication between NET CPU and EEPROM is checked.

```
C2-8
NET EEPROM:NG
```

R1 DAB

Not for service.

```
R1-1          R1-2
N/A   ....... DAB:N/A
```

P1. AD DATA CHECK

This menu is used to display the A/D conversion value of the microprocessor which detects panel keys and protection functions by using the sub-menu.

When "P1-6. KEY" sub-menu is selected, keys become inoperable due to detection of the values of all keys. However, it is possible to advance to the next menu by turning the VOLUME knob.

* Numeric values in the figure are given as reference only.

P1-1. DC

Power amplifier DC (DC voltage) output is detected.

The voltage at 120 pin (DC_PRT) of IC302 is displayed.

Normal value: 27 to 136
(Reference voltage: 3.3 V=255)

* If DC becomes out of the normal value range, the protection function works to turn off the power.

```
P1-1
DC: 088
```

P1-2. PS

Power supply voltage (PS) protection detection.

The voltage at 126 pin (PS1_PRT)/107 pin (PS2_PRT)/106 pin (PS3_PRT) of IC302 are displayed.

Voltage detects
PS1: FLDC, VP, AC, +5T, 8.44V
PS2: +3.3N
PS3: +5.5V

Normal value
PS1: 153 to 194
PS2: 103 to 129
PS3: 139 to 165
(Reference voltage: 3.3 V=255)

* If PS1, PS2 or PS3 becomes out of the normal value range, the protection function works to turn off the power.

```
P1-2
PS:184/117/156
```

PS3
PS2
PS1

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P1-3. TM

Temperature of the heatsink (THM) is detected.
The voltage at 122 pin (THM1), 110 pin (THM3) of IC302 is displayed.

Normal value
THM1: 15 to 134
THM2: Not for service
THM3: 15 to 134
(Reference voltage: 3.3 V=255)
* If THM1 or THM3 become out of the normal value range, the protection function works to turn off the power.

```
P1-3
TM:071/000/223
```

THM3
THM2 (Not for service)
THM1

P1-4. OUTPUT LEVEL

Output level of speaker output is detected.
The voltage at 121 pin (AMP_OLV) of IC302 is displayed.
(Reference voltage: 3.3 V=255)

```
P1-4
OUTLVL: 255
```

P1-5. USB-VBUS

Power supply voltage of USB jack is detected.
The voltage at 109 pin (USB_VBUS_PRT) of IC302 is detected.
(Reference voltage: 3.3 V=255)

```
P1-5
USB-VBUS:199
```

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P1-6. KEY

Panel key is detected.

When the A/D conversion value of the panel key becomes out of the specified range, normal operation will not be available.

In that case, check the constant of voltage dividing resistor, solder condition, etc. Refer to table.

* When "P1-6. KEY" menu is selected, keys become inoperable due to detection of the values of all keys. However, it is possible to advance to the next menu by turning the VOLUME knob.

(Reference voltage: 3.3 V=255)

```
P1-6
KY:255/255 OFF
```

Key name
KEY1
KEY0

Display		KEY0
A/D value	Key name	
000 – 011	INP+	INPUT ►
012 – 034	INP-	INPUT ◄
035 – 059	TRE+	TREBLE +
060 – 083	TRE-	TREBLE –
084 – 109	BASS+	BASS +
110 – 135	BASS-	BASS –
136 – 161	RTN	RETURN
162 – 186	ENTER	SELECT PUSH – ENTER (jog dial)
255	OFF	—

Display		KEY1
A/D value	Key name	
000 – 011	TUN+	TUNING »
012 – 034	TUN-	TUNING «
035 – 059	PST+	PRESET >
060 – 083	PST-	PRESET <
084 – 109	BAND	BAND
110 – 135	MEM	MEMORY
136 – 161	MODE	MODE
162 – 186	DISP	DISPLAY
187 – 213	SPA	SPEAKER A
214 – 241	SPB	SPEAKER B
255	OFF	—

P2. PROTECTION HISTORY

This menu is used to display the history of protection function.
In the history 1 to 4, the setting information for operation of each protection function will be stored.
All history of protection function and setting information will be erased by pressing the “RETURN” key.

* Numeric values in the figure are given as reference only.

```
P2-1
1.PS1    000L
```

P2-1. HISTORY 1

H: Displayed when the voltage is HIGHER than upper limit.

L: Displayed when the voltage is LOWER than lower limit.

xxx: A/D conversion value of voltage at the moment when the protection function worked. (Reference voltage: 3.3 V=255)

```
P2-2
1.P-PORT:----
```

P2-2. POWER PORT 1

Not for service.

```
P2-3
1.INPUT: 0x09
```

P2-3. LAST INPUT 1

The input source which was selected last is displayed in hexadecimal notation.

```
P2-4
1.VOL.:
```

P2-4. LAST VOLUME LEVEL 1

The volume level which was selected last is displayed.
-80.0 to +16.5 dB

```
P2-5
2.I Prt
```

P2-5. HISTORY 2

```
P2-6
2.P-PORT:----
```

P2-6. POWER PORT 2

```
P2-7
2.INPUT: 0x0D
```

P2-7. LAST INPUT 2

```
P2-8
2.VOL.:
```

P2-8. LAST VOLUME LEVEL 2

```
P2-9
3.Dc
```

P2-9. HISTORY 3

```
P2-10
3.P-PORT:----
```

P2-10. POWER PORT 3

```
P2-11
3.INPUT: 0x05
```

P2-11. LAST INPUT 3

```
P2-12
3.VOL.:
```

P2-12. LAST VOLUME LEVEL 3

```
P2-13
4.Tmpx
```

P2-13. HISTORY 4

```
P2-14
4.P-PORT:----
```

P2-14. POWER PORT 4

```
P2-15
4.INPUT: 0x06
```

P2-15. LAST INPUT 4

```
P2-16
4.VOL.:
```

P2-16. LAST VOLUME LEVEL 4

R-N402/R-N402D

T1. TROUBLE SHOOTING INFORMATION

This menu is used to display the operating time and operation frequency of this unit.

* The operating time and operation frequency during the self-diagnostic function mode will not be stored.

T1-1. OPERATING TIME

The operating time of this unit is displayed.

The operating time will be erased by pressing the “RETURN” key.

```
T1-1
0000D 00H 00M
```

Minute (0M to 59M)
Hour (0H to 23H)
Day (0D to 9999D)

T1-2. POWER-RELAY ON

The operation frequency of the power relay (RY541) is displayed in hexadecimal notation.

The operation frequency will be erased by pressing the “RETURN” key.

```
T1-2
PR-ON Num:0000
```

Operation frequency
0 to FFFF (up to 65,535 times)

T1-3. NRC (Net Restart Counter)

Not for service.

```
T1-3
NRC:      0
```

S1. FACTORY PRESET

This menu is used to reserve/inhibit initialization of the back-up IC (EEPROM: IC301 on DIGITAL P.C.B.).

```
S1-1
PRESET INHI
```

S1-1. PRESET INHIBIT (Initialization inhibited)

Initialization of the back-up IC is not executed. Select this sub-menu to protect the values set by the user.

```
S1-2
PRESET RSRV
```

S1-2. PRESET RESERVED (Initialization reserved)

Initialization of the back-up IC is reserved. (Actual initialization is executed when the power is turned on next.) To reset to the original factory settings or to reset the backup IC, select this sub-menu and press the “⏻” (Power) key to turn off the power.

CAUTION: Before setting to the PRESET RESERVED, write down the existing preset memory content of the tuner. (This is because setting to the PRESET RESERVED will cause the user memory content to be erased.)

S2. ROM VERSION/CHECKSUM

The firmware version and checksum values are displayed.

The checksum is obtained by adding the data at every 8-bit and expressing the result as a hexadecimal notation.

* Numeric values in the figure are given as reference only.

```
S2-1
System:  1.02
```

S2-1. SYSTEM VERSION

The firmware version is displayed.

```
S2-2
Ver:  0023
```

S2-2. MICROPROCESSOR VERSION

The firmware version of MICROPROCESSOR (IC302 on DIGITAL P.C.B.) is displayed.

```
S2-3
Sum:  88E8
```

S2-3. MICROPROCESSOR CHECKSUM

The checksum value of MICROPROCESSOR (IC302 on DIGITAL P.C.B.) is displayed.

```
S2-4
N:0011
```

S2-4. NETWORK VERSION

The firmware version of Network module is displayed.

```
S2-5
N-Sum:D360D26E
```

S2-5. NETWORK CHECKSUM

The checksum value of Network module is displayed.

S3. SOFT SWITCH

Not for service.

```
S3-1
   SW   :PCB
```

S4. SYSTEM INFORMATION

This menu is used to display the model name and destination.

S4-1. MODEL/DESTINATION

The model name and destination are displayed.

```
S4-1
N4 ---  G 055
```

Not for service.

Destination

U / R (R, S) / K / A / G (B, G, F) / L

Not for service.

Model name

N4 (R-N402)

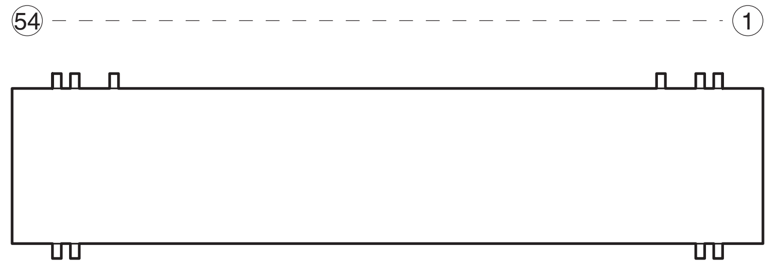
S4-2. VERIFY

Not for service.

```
S4-2
Verify 255  00
```

■ DISPLAY DATA

● V2001 : 016ST106INK (MAIN (2) P.C.B.)



● PIN CONNECTION

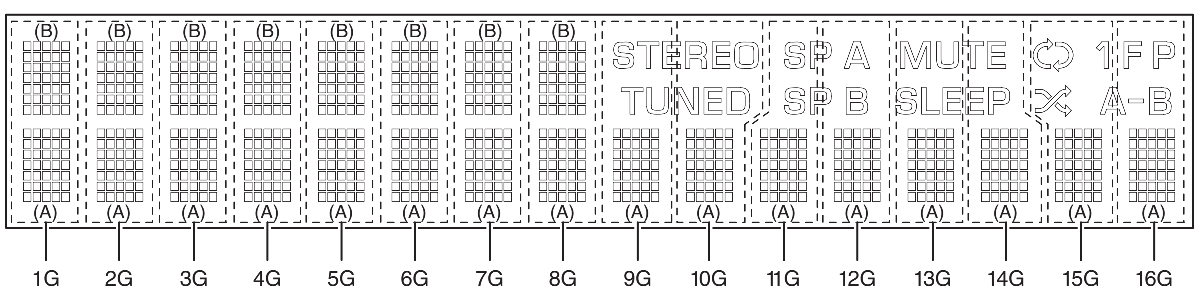
Pin No.	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
Connection	NX (11G)	NX (11G)	NX (12G)	NX (12G)	NX	NX (13G)	NX (13G)	NX	NX (14G)	NX (14G)	NX (15G)	NX	NX (15G)	NX (16G)	NX (16G)	NP	NP	F–	F–

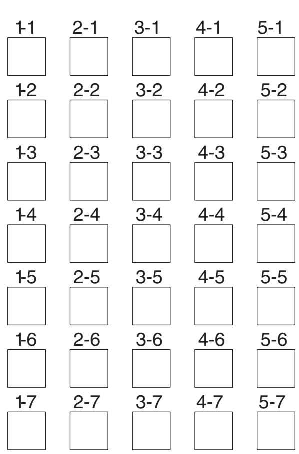
Pin No.	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20
Connection	$\overline{\text{RESET}}$	NX (5G)	NX (5G)	$\overline{\text{CS}}$	NX (6G)	NX (6G)	NX (7G)	CP	NX (7G)	NX (8G)	NX (8G)	DA	NX (9G)	NX (9G)	TSA	NX (10G)	TSB	NX (10G)

Pin No.	54	53	52	51	50	49	48	47	46	45	44	43	42	41	40	39	38
Connection	F+	F+	NP	NP	LGND	PGND	VH	NX (1G)	NX (1G)	VDD	NX (2G)	NX (2G)	NX (3G)	NX (3G)	OSC	NX (4G)	NX (4G)

Note: 1) F+, F- ..... Filament 2) NP ..... No pin 3) DL ..... Datum line 4) LGND ..... Logic GND pin
5) PGND ..... Power GND pin 6) VH ..... High voltage supply pin 7) VDD ..... Logic voltage supply pin 8) CP ..... Shift register clock
9) DA ..... Serial data input 10) TSA, B ..... Test pin 11) $\overline{\text{CS}}$ ..... Chip select input pin 12) OSC ..... Pin for self-oscillation
13) $\overline{\text{RESET}}$ ..... Reset input 14) NX ..... No extended pin

● GRID ASSIGNMENT





1G – 16G

● ANODE CONNECTION

	1G – 8G	9G – 14G	15G, 16G
D0B	1-1B	–	–
D1B	2-1B	–	–
D2B	3-1B	–	–
D3B	4-1B	–	–
D4B	5-1B	–	–
D5B	1-2B	–	–
D6B	2-2B	–	–
D7B	3-2B	–	–
D8B	4-2B	–	–
D9B	5-2B	–	–
D10B	1-3B	–	–
D11B	2-3B	–	–
D12B	3-3B	–	–
D13B	4-3B	–	–
D14B	5-3B	–	–
D15B	1-4B	–	–
D16B	2-4B	–	–
D17B	3-4B	–	–
D18B	4-4B	–	–
D19B	5-4B	–	–
D20B	1-5B	–	–
D21B	2-5B	–	–
D22B	3-5B	–	–
D23B	4-5B	–	–
D24B	5-5B	–	–
D25B	1-6B	–	–
D26B	2-6B	–	–
D27B	3-6B	–	–
D28B	4-6B	–	–
D29B	5-6B	–	–
D30B	1-7B	–	–
D31B	2-7B	–	–
D32B	3-7B	–	–
D33B	4-7B	–	P
D34B	5-7B	–	A-B
D0A	1-1A	1-1A	1-1A
D1A	2-1A	2-1A	2-1A

	1G – 8G	9G, 10G	11G, 12G	13G, 14G	15G, 16G
D2A	3-1A	3-1A	3-1A	3-1A	3-1A
D3A	4-1A	4-1A	4-1A	4-1A	4-1A
D4A	5-1A	5-1A	5-1A	5-1A	5-1A
D5A	1-2A	1-2A	1-2A	1-2A	1-2A
D6A	2-2A	2-2A	2-2A	2-2A	2-2A
D7A	3-2A	3-2A	3-2A	3-2A	3-2A
D8A	4-2A	4-2A	4-2A	4-2A	4-2A
D9A	5-2A	5-2A	5-2A	5-2A	5-2A
D10A	1-3A	1-3A	1-3A	1-3A	1-3A
D11A	2-3A	2-3A	2-3A	2-3A	2-3A
D12A	3-3A	3-3A	3-3A	3-3A	3-3A
D13A	4-3A	4-3A	4-3A	4-3A	4-3A
D14A	5-3A	5-3A	5-3A	5-3A	5-3A
D15A	1-4A	1-4A	1-4A	1-4A	1-4A
D16A	2-4A	2-4A	2-4A	2-4A	2-4A
D17A	3-4A	3-4A	3-4A	3-4A	3-4A
D18A	4-4A	4-4A	4-4A	4-4A	4-4A
D19A	5-4A	5-4A	5-4A	5-4A	5-4A
D20A	1-5A	1-5A	1-5A	1-5A	1-5A
D21A	2-5A	2-5A	2-5A	2-5A	2-5A
D22A	3-5A	3-5A	3-5A	3-5A	3-5A
D23A	4-5A	4-5A	4-5A	4-5A	4-5A
D24A	5-5A	5-5A	5-5A	5-5A	5-5A
D25A	1-6A	1-6A	1-6A	1-6A	1-6A
D26A	2-6A	2-6A	2-6A	2-6A	2-6A
D27A	3-6A	3-6A	3-6A	3-6A	3-6A
D28A	4-6A	4-6A	4-6A	4-6A	4-6A
D29A	5-6A	5-6A	5-6A	5-6A	5-6A
D30A	1-7A	1-7A	1-7A	1-7A	1-7A
D31A	2-7A	2-7A	2-7A	2-7A	2-7A
D32A	3-7A	3-7A	3-7A	3-7A	3-7A
D33A	4-7A	4-7A	4-7A	4-7A	4-7A
D34A	5-7A	5-7A	5-7A	5-7A	5-7A
AD1	–	TUNED	SP B	SLEEP	
AD2	–	STEREO	SP A	MUTE	
AD3	–	–	–	–	1
AD4	–	–	–	–	F

R-N402/R-N402D

	1G	2G	3G	4G	5G	6G	7G	8G	9G	10G	11G	12G	13G	14G	15G	16G
D0B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D1B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D2B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D3B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D4B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D5B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D6B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D7B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D8B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D9B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D10B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D11B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D12B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D13B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D14B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D15B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D16B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D17B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D18B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D19B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D20B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D21B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D22B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D23B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D24B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D25B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D26B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D27B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D28B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D29B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D30B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D31B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D32B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	–	–
D33B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	T20	
D34B	T1	T2	T3	T4	T5	T6	T7	T8	–	–	–	–	–	–	T20	
D0A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D1A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16

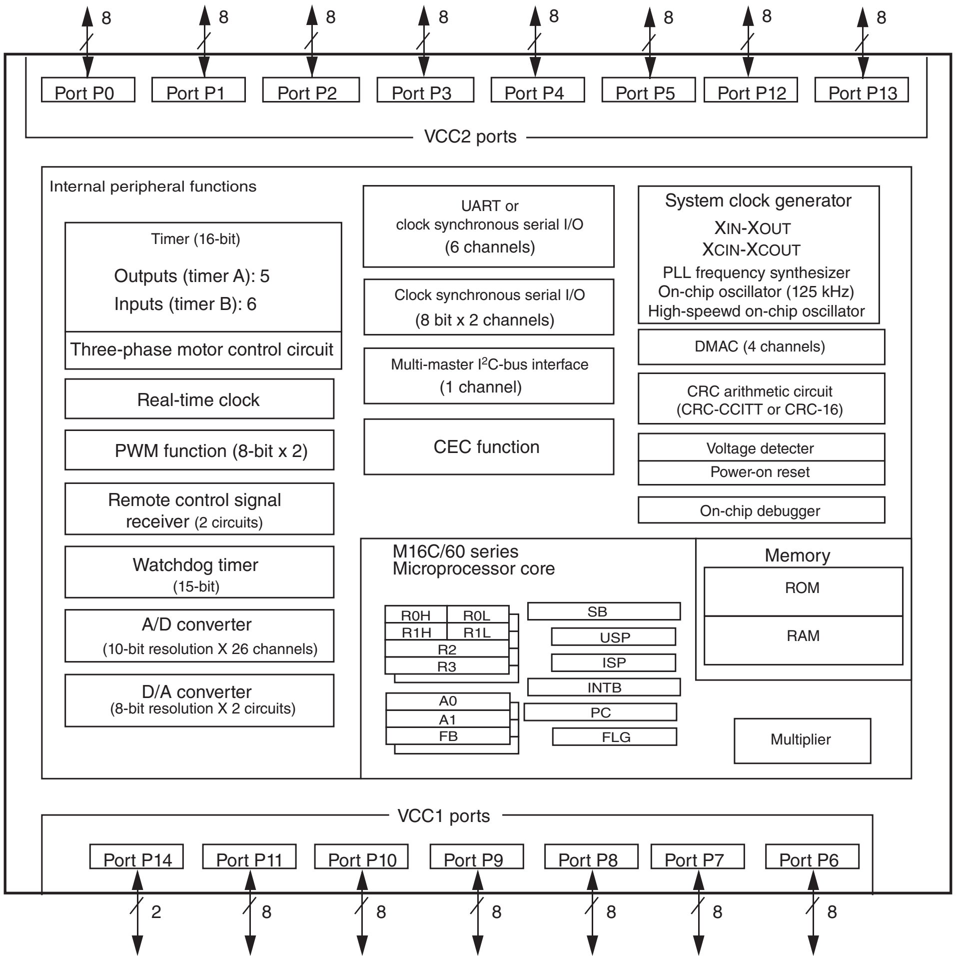
R-N402/R-N402D

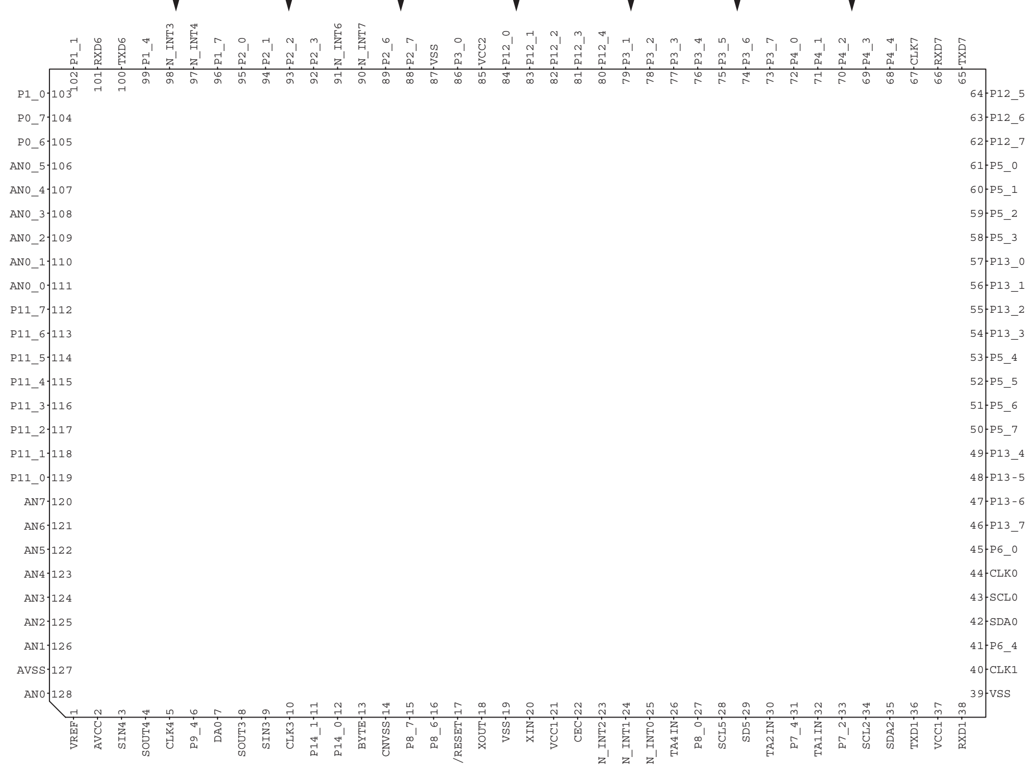
	1G	2G	3G	4G	5G	6G	7G	8G	9G	10G	11G	12G	13G	14G	15G	16G
D2A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D3A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D4A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D5A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D6A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D7A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D8A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D9A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D10A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D11A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D12A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D13A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D14A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D15A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D16A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D17A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D18A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D19A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D20A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D21A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D22A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D23A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D24A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D25A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D26A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D27A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D28A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D29A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D30A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D31A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D32A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D33A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
D34A	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16
AD1	–	–	–	–	–	–	–	–	T17		T18		T19		T20	
AD2	–	–	–	–	–	–	–	–	T17		T18		T19		T20	
AD3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	T20	
AD4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	T20	

■ IC DATA

IC302: R5F3651TNFC (DIGITAL P.C.B.)
Main microprocessor

*** No replacement part available.**





Pin No.	Function Name (P.C.B.)	I/O				Detail of Function
		FULL ON	POWER OFF	MCU SLEEP	AC OFF	
1	VREF	MCU	MCU	MCU	MCU	AD reference voltage
2	AVCC	MCU	MCU	MCU	MCU	CPU power supply
3	EEP_MISO	SI	SI	O	SI	EEPROM synchronized data input
4	EEP_MOSI	SO	SO	O	SO	EEPROM synchronized data output
5	EEP_SCK	SO	SO	O	SO	EEPROM synchronized clock output
6	(no_use)	O	O	O	O	
7	(no_use)	O	O	O	O	
8	DIR_MOSI	SO	O	O	O	DIR synchronized data output
9	DIR_MISO	SI	I-	I-	I-	DIR synchronized data input
10	DIR_SCK	SO	O	O	O	DIR synchronized clock output
11	DAC_N_RST	O	O	O	O	DAC reset signal
12	TUN_N_RST	O	O	O	O	Tuner reset signal
13	BYTE	MCU	MCU	MCU	MCU	External data bus width select input / Single chip mode : L (16bit)
14	E8a	MCU	MCU	MCU	MCU	Processor modes select / L: Single chip mode
15	PCM/DSD_SEL	O	O	O	O	PCM / DSD audio signal selector L: PCM / H: DSD
16	DIAG_FCT	O	O	O	O	Diag OK : Output High / Diag NG : Output Low (default)
17	N_RST	MCU	MCU	MCU	MCU	CPU reset signal
18	XOUT	MCU	MCU	MCU	MCU	Oscillator circuit output
19	VSS	MCU	MCU	MCU	MCU	CPU ground
20	XIN	MCU	MCU	MCU	MCU	Oscillator circuit input
21	VCC1	MCU	MCU	MCU	MCU	CPU power supply
22	(no_use)	O	O	O	O	
23	DIR_N_INT	IRQ	I-	I-	I-	DIR emphasis detect signal
24	(no_use)	O	O	O	O	
25	DIR_N_INT0	IRQ	I-	I-	I-	DIR error interrupt signal
26	DAC_ZERO_DET	I	O	O	O	DAC data ZERO detect signal
27	(no_use)					
28	FL_SCK	O	O	O	O	FL driver system clock
29	FL_MOSI	SO	SO	O	O	FL driver synchronized data output
30	TUN_TUNED	TMR	O	O	O	
31	DIR_N_RST	O	O	O	O	DIR reset signal
32	ACPWR_DET	TMR	I+	I+	I+	AC power detect / L: Power down
33	DIR_N_CS	I	I	I	I	DIR chip select signal
34	(no_use)					
35	(no_use)					
36	E8A_TXD	SO	SO	I+	I+	CPU debug
37	VCC1	MCU	MCU	MCU	MCU	CPU power supply
38	E8A_RXD	SI	SI	I+	I+	CPU debug
39	VSS	MCU	MCU	MCU	MCU	CPU ground
40	E8A_SCK	SI	SI	I+	I+	CPU debug
41	E8A_BUSY	I-	I-	I-	I-	CPU debug
42	TUN_SDA	SI	O	O	O	Tuner I2C data
43	TUN_SCL	SO	O	O	O	Tuner I2C clock
44	(no_use)	O	O	O	O	
45	(no_use)	O	O	O	O	
46	VOL_MOSI	SO	O	O	O	Electronic volume IC synchronized data output
47	VOL_SCK	O	O	O	O	Electronic volume IC synchronized system clock
48	I_PRT	I-	I-	I-	I-	Overcurrent protection detection
49	(no_use)	O	O	O	O	
50	(no_use)	O	O	O	O	
51	(no_use)	O	O	O	O	

Pin No.	Function Name (P.C.B.)	I/O				Detail of Function
		FULL ON	POWER OFF	MCU SLEEP	AC OFF	
52	E8A_N_EPM	I-	I-	I-	I-	CPU debug
53	HP_N_DET	I+	I+	I+	I+	Headphone detect port
54	HPRY	O	O	O	O	Headphone relay control
55	SPRY_B	O	O	O	O	Speaker relay control (SP B)
56	(no_use)	O	O	O	O	
57	SPRY_A	O	O	O	O	Speaker relay control (SP A)
58	MAIN_N_MUTE	O	O	O	O	Mute control (MAIN AMP)
59	(no_use)	O	O	O	O	
60	REC_N_MUTE	O	O	O	O	Mute control (LINE 3 REC)
61	E8A_N_CE	I+	I+	I+	I+	CPU debug
62	NCPU_MODE	I-	I-	I-	I-	+3.3V (H) : Standard mode / 0V (L) : MAC address write access
63	(no_use)	O	O	O	O	
64	PRY	O	O	O	O	Power relay control / H: On
65	NCPU_RXD	SO	SO	SO	SO	NCPU UART output
66	NCPU_TXD	SI	SI	SI	SI	NCPU UART input
67	(no_use)	O	O	O	O	
68	(no_use)	O	O	O	O	
69	(no_use)	O	O	O	O	
70	(no_use)	O	O	O	O	
71	N_FCT	I	I	I	I	FCT detect / H: NORMAL mode L: FCT mode
72	(no_use)	O	O	O	O	
73	(no_use)	O	O	O	O	
74	VOL_RB	I+	I+	I+	I+	Volume rotary encoder B
75	VOL_RA	I+	I+	I+	I+	Volume rotary encoder A
76	FLD_N_CS	O	O	O	O	FL driver chip select
77	FLD_N_RST	O	O	O	O	FL driver reset
78	(no_use)	O	O	O	O	
79	STBY_LED	O	O	O	O	Stanby LED / H: LED light
80	BT_LED	O	O	O	O	Bluetooth LED / H: LED light
81	WIFI_LED	O	O	O	O	WiFi LED / H: LED light
82	+3.3S_PON	O	O	O	O	Power on signal for LDO +3.3S
83	NCPU_PHOLD	O	O	O	O	NW-01 internal DC-DC power on signal
84	(no use)	O	O	O	O	
85	VCC2	MCU	MCU	MCU	MCU	CPU power supply
86	EEP_N_CS	O	O	O	O	EEPROM chip select
87	VSS	MCU	MCU	MCU	MCU	CPU ground
88	MODE	I+	I+	I+	I+	+3.3V：Normal mode / 0V : enable MAC address write
89	VBUS_PON	O	O	O	O	VBUS power supply power on signal
90	REM_IN	IRQ	IRQ	IRQ	I	Remote control pulse input
91	PWR_N_DET	IRQ	IRQ	IRQ	I	Power switch detetct / L: Standby key ON
92	NCPU_PON	O	O	O	O	5NET LDO power on
93	3.3D_PON	O	O	O	O	3.3D LDO power on for DIR & OPT
94	NCPU_N_RST	O	O	O	O	NW-01 reset signal
95	NCPU_VBUSDRV	I	I	I	I	NW-01 USB VBUS signal
96	NET_RF	I+	I+	I+	I+	Network selector encoder B
97	NCPU_N_INT	IRQ	IRQ	IRQ	IRQ	NW-01 interrupt signal
98	NCPU_MUTE0	O	O	O	O	NW-01 mute signal (0)
99	NET_RE	I+	I+	I+	I+	Network selector encoder A
100	NCPU_SPI_MOSI	SO	O	O	O	NW-01 SPI data output
101	NCPU_SPI_MISO	SI	O	O	O	NW-01 SPI data input
102	NCPU_SPI_SCK	O	O	O	O	NW-01 SPI clock output

Pin No.	Function Name (P.C.B.)	I/O				Detail of Function
		FULL ON	POWER OFF	MCU SLEEP	AC OFF	
103	NCPU_SPI_N_CS	O	O	O	O	NW-01 SPI chip select
104	DAC_N_CS	O	O	O	O	DAC chip select
105	NCPU_ADT_MUTE	O	O	O	O	Network audio distribution mute signal
106	PS3_PRT	AD				PS protection 3 DIR & DAC power supply monitor
107	PS2_PRT	AD				PS Protection 2 NW-01 power supply monitor
108	NCPU_MUTE1	O	O	O	O	NW-01 mute signal (1)
109	USB_VBUS_PRT	AD				USB Protection USB VBUS power supply monitor
110	THM3_PRT	AD	I	I	I	Thermal sensor under diode bridge
111	DEST	AD				AD reserve
112	(no_use)	O	O	O	O	
113	DIR_SD0	I	I	O	I	Data audio signal check in PCB digital (DIAG)
114	DIR_WCK	I	I	O	I	Word clock audio signal check in PCB digital (DIAG)
115	(no_use)	O	O	O	O	
116	MODEL	I	I	I	I	For Model recognization R-N402 : H / R-N402D : L
117	(no_use)	O	O	O	O	
118	(no_use)	O	O	O	O	
119	(no_use)	O	O	O	O	
120	DC_PRT	AD	I	I	I	Power amplifier DC detect
121	(no_use)					
122	THM	AD	I	I	I	Thermal sensor
123	(no_use)	AD				
124	KEY1	AD	I	I	I	KEY AD value uptake 2
125	KEY0	AD	I	I	I	KEY AD value uptake 1
126	PS1_PRT	AD	I	I	I	PPS protection 1 MAIN power supply monitor
127	AVSS	MCU	MCU	MCU	MCU	CPU ground
128	FCT_OUTPUT	AD	O	O	O	Audio signal output check in FCT

FULL ON : During normal operation
POWER OFF : During transition from normal operation to standby state
MCU SLEEP : Standby state
AC OFF : During transition from normal operation to AC off

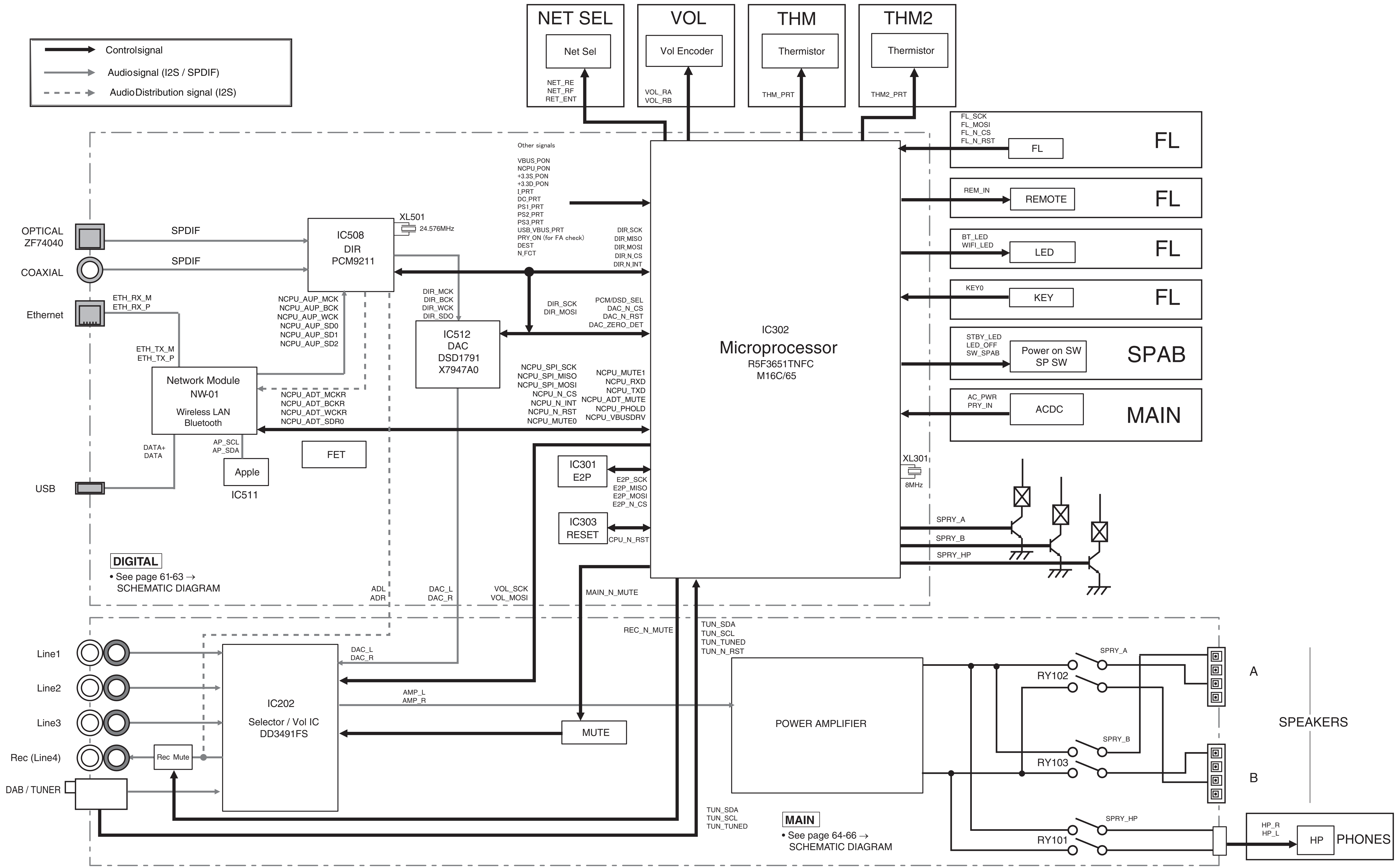
Key detection for A/D port
Key input (A/D) pull-up resistance 10 k-ohms

	0 Ω	1 k Ω	1.3 k Ω	1.6 k Ω	2.2 k Ω	3.3 k Ω	4.7 k Ω	7.5 k Ω
Detected voltage value at 125 pin	0.000 – 0.150 V	0.150 – 0.459 V	0.459 – 0.771 V	0.771 – 1.088 V	1.088 – 1.425 V	1.425 – 1.765 V	1.765 – 2.093 V	2.093 – 2.424 V
A/D value (3.3 V = 255)	000 – 012	012 – 035	035 – 060	060 – 084	084 – 110	110 – 136	136 – 162	162 – 187
KEY0	INPUT ►	INPUT ◄	TREBLE +	TREBLE –	BASS +	BASS –	RETURN	SELECT PUSH – ENTER (jog dial)

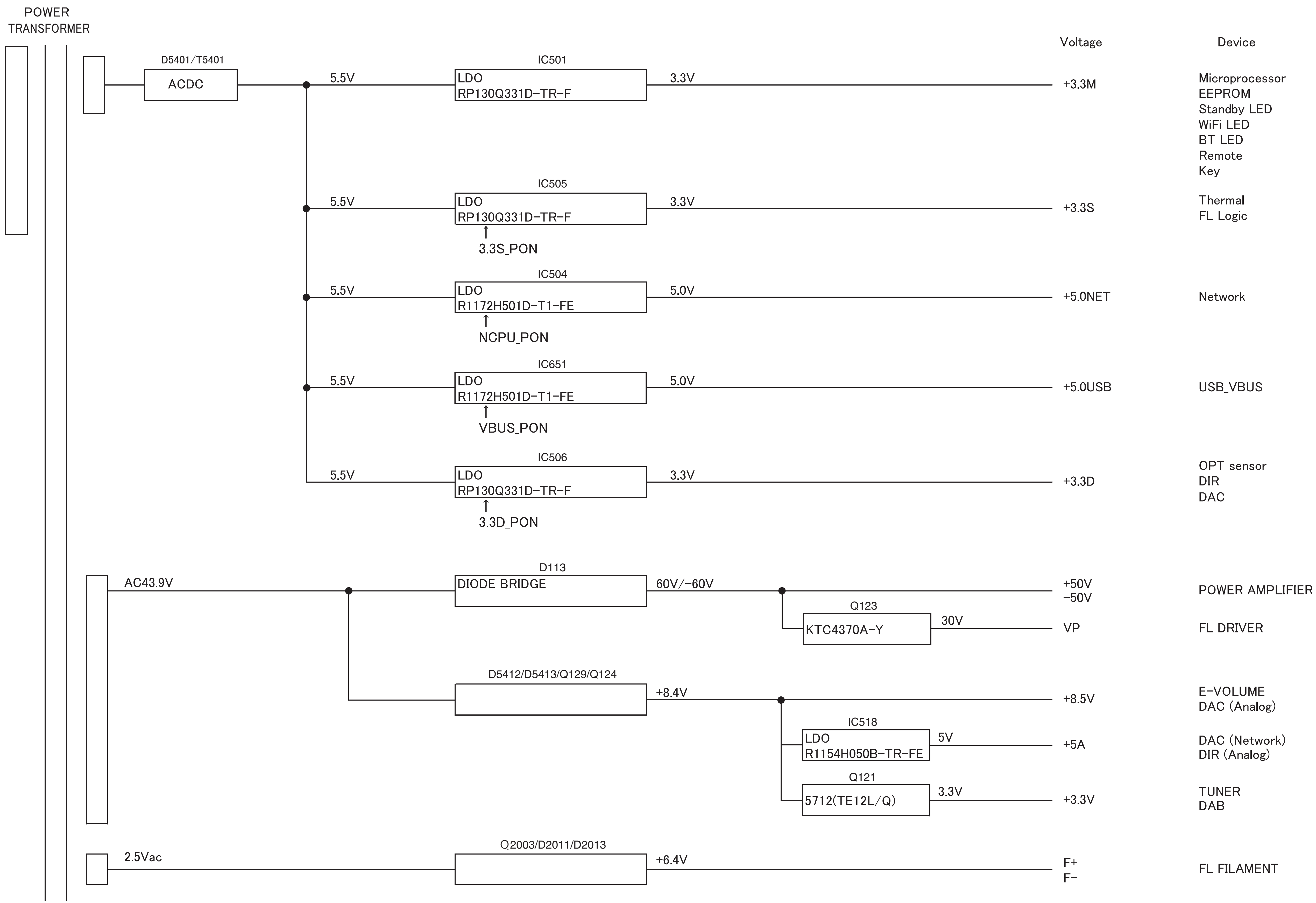
	0 Ω	1 k Ω	1.3 k Ω	1.6 k Ω	2.2 k Ω	3.3 k Ω	4.7 k Ω	7.5 k Ω	15 k Ω	47 k Ω
Detected voltage value at 124 pin	0.000 – 0.150 V	0.150 – 0.459 V	0.459 – 0.771 V	0.771 – 1.088 V	1.088 – 1.425 V	1.425 – 1.765 V	1.765 – 2.093 V	2.093 – 2.424 V	2.424 – 2.770 V	2.770 – 3.124 V
A/D value (3.3 V = 255)	000 – 012	012 – 035	035 – 060	060 – 084	084 – 110	110 – 136	136 – 162	162 – 187	187 – 214	214 – 241
KEY1	TUNING »	TUNING «	PRESET ›	PRESET ‹	FM/AM	MEMORY	FM MODE	DISPLAY	SPEAKER A	SPEAKER B

■ BLOCK DIAGRAMS

Audio Section Block Diagram

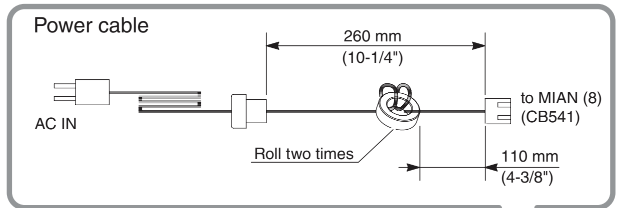


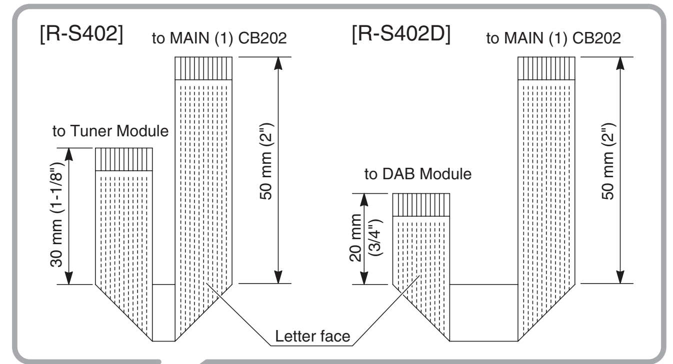
Power Supply Section Block Diagram

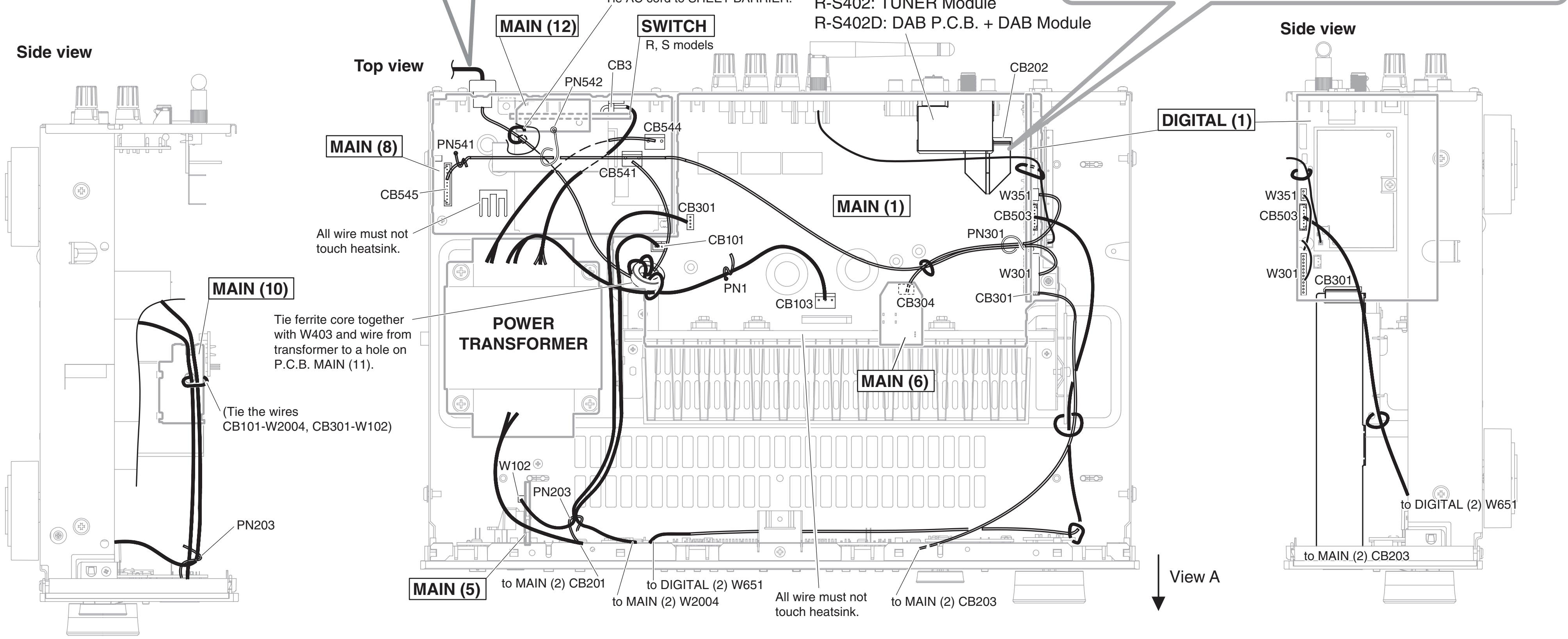


■ WIRING DIAGRAM

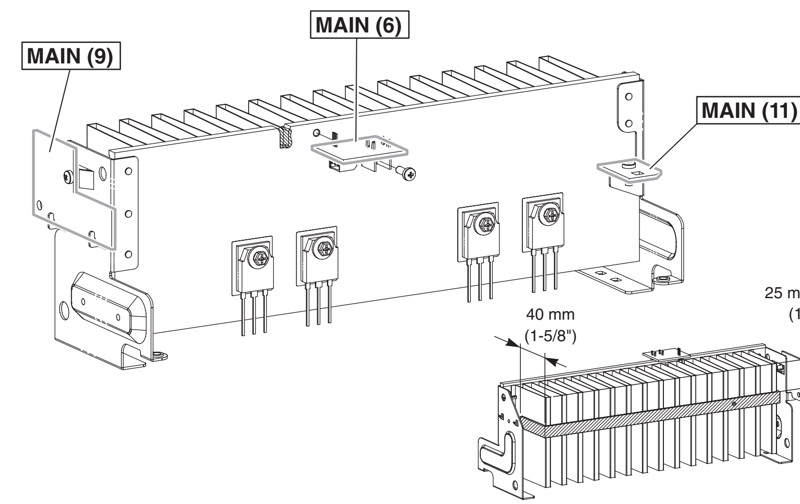
• OVERALL ASSEMBLY



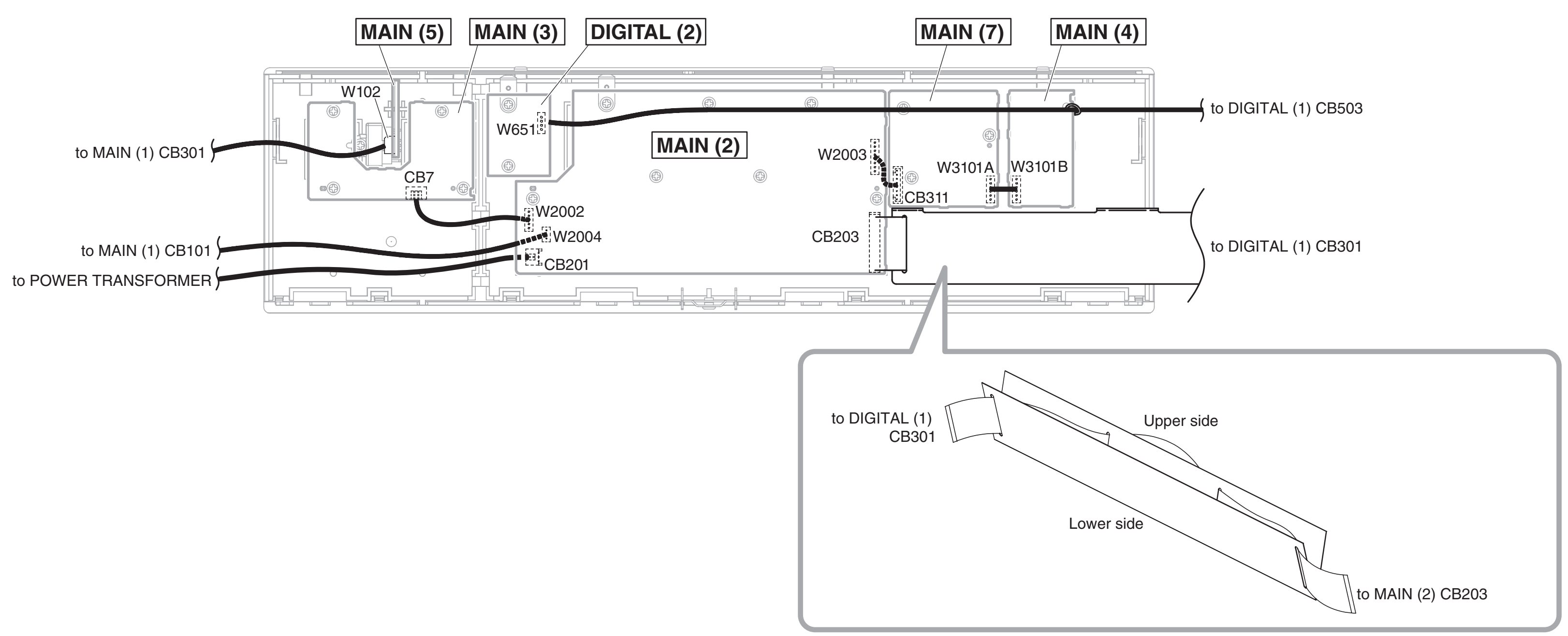




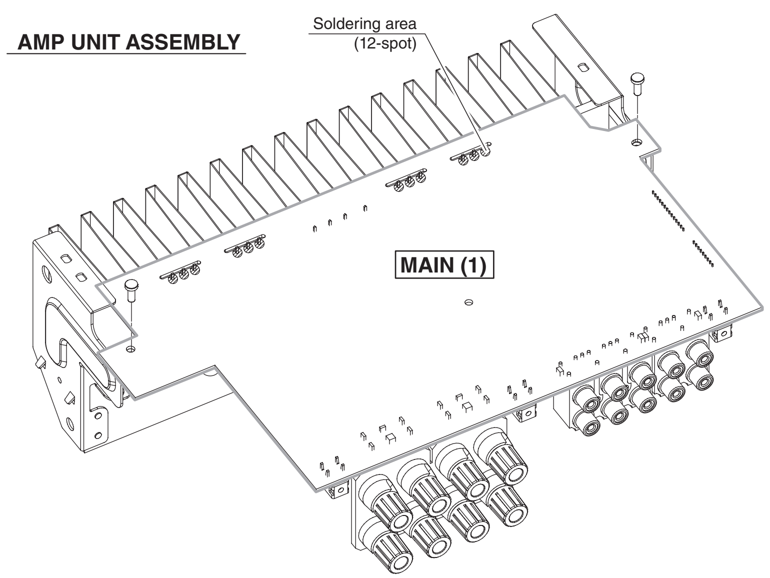
HEATSINK ASSEMBLY



Front panel (View A)

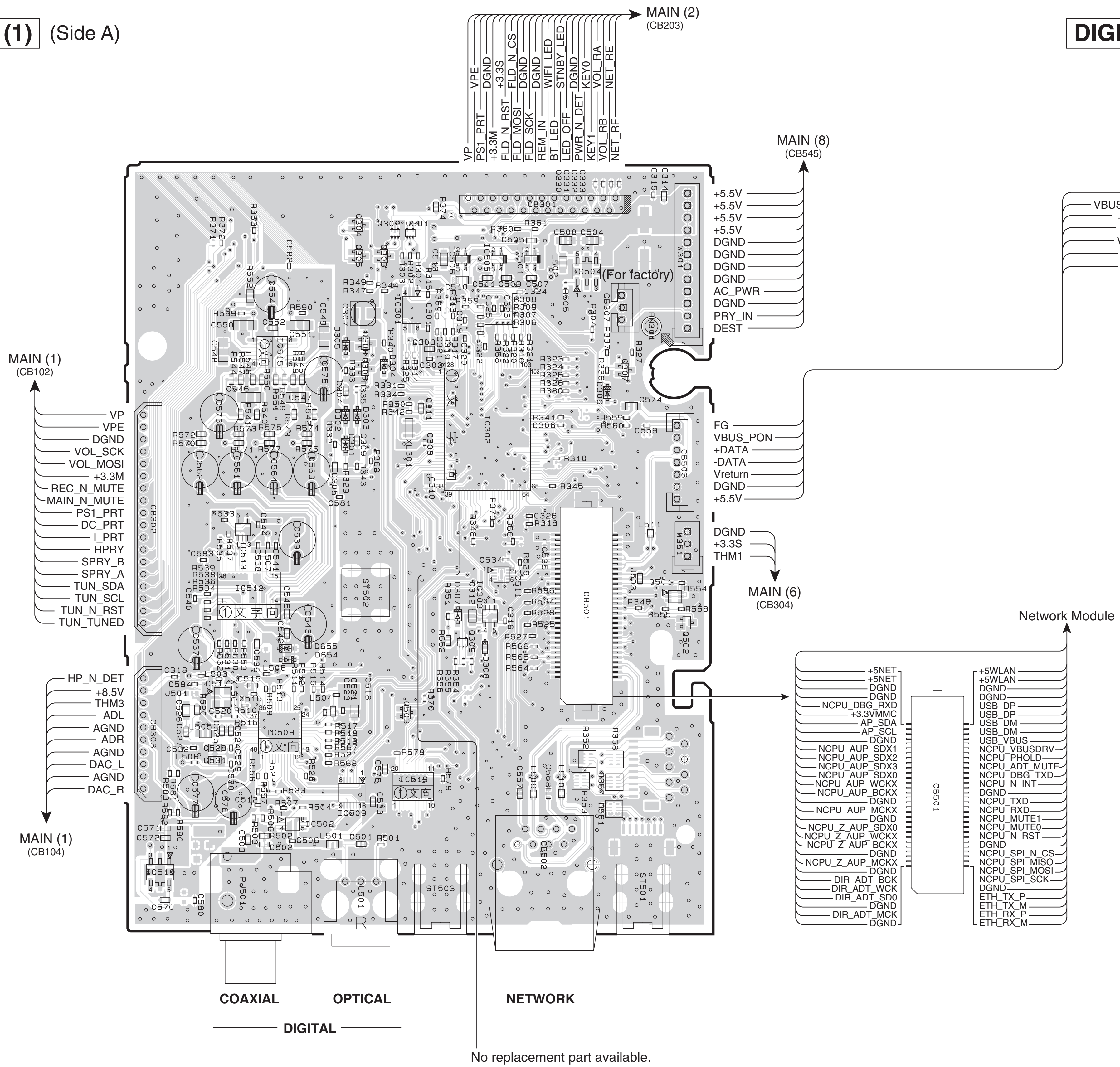


AMP UNIT ASSEMBLY

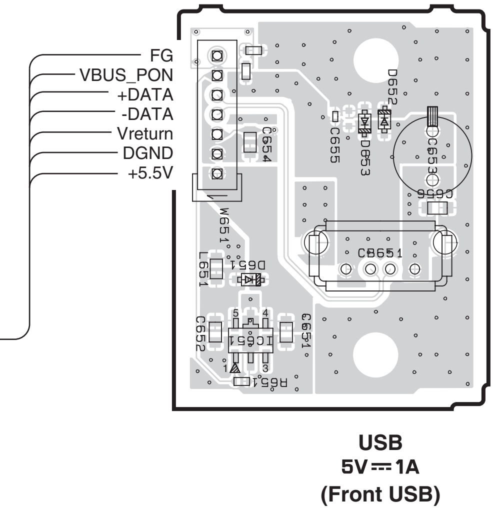


■ PRINTED CIRCUIT BOARDS

DIGITAL (1) (Side A)



DIGITAL (2) (Side A)

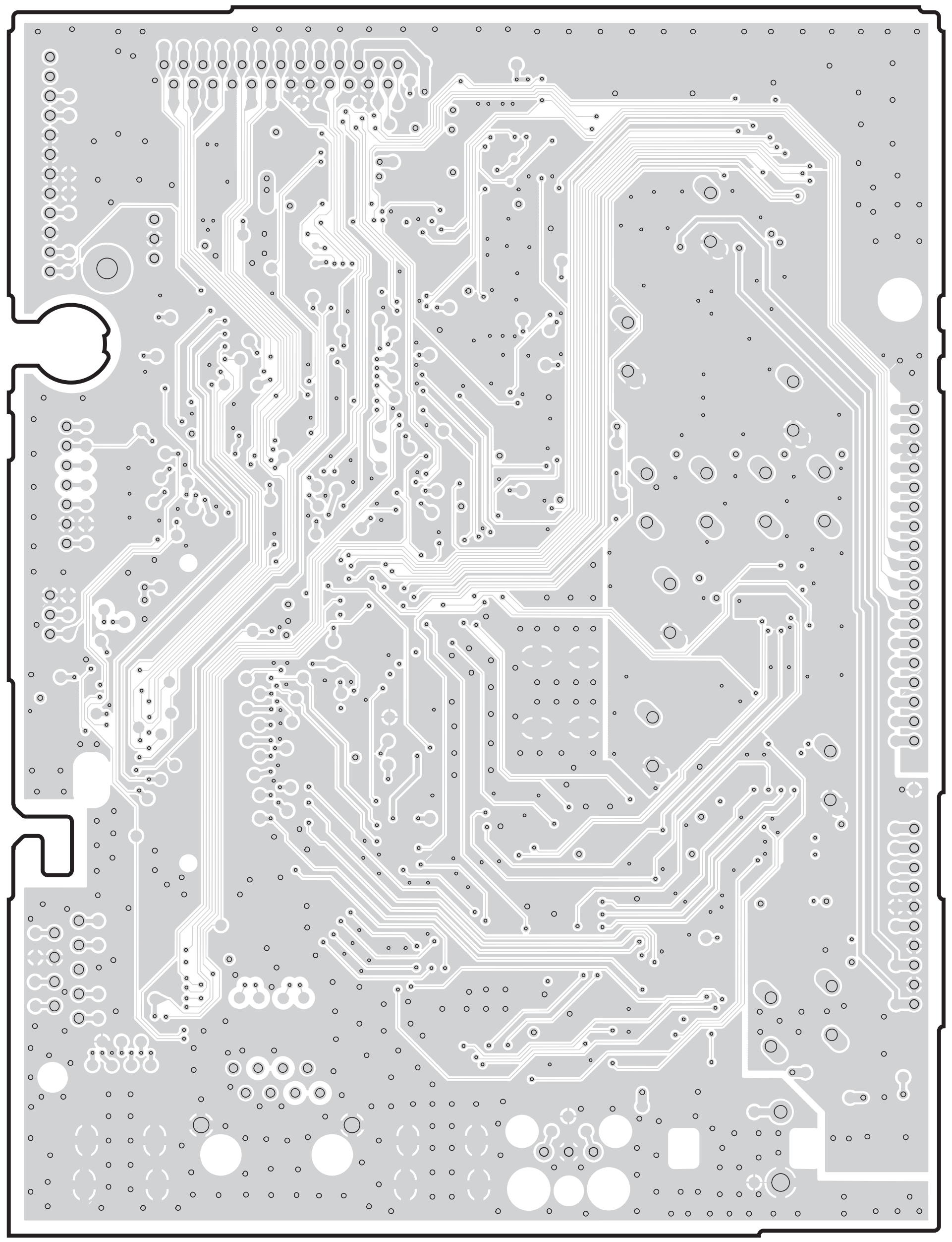


• Semiconductor Location

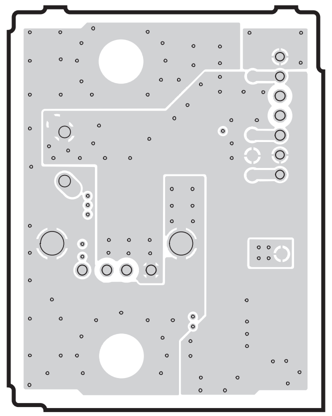
Ref no.	Location
D301	D4
D302	D4
D303	D4
D304	D3
D305	D3
D306	E3
D307	D5
D308	D5
D651	H3
D652	I3
D653	H3
D654	C5
D655	C5
IC301	D3
IC302	D4
IC303	D5
IC501	E3
IC502	C6
IC504	E3
IC505	D3
IC506	D3
IC508	C5
IC509	D6
IC511	D4
IC512	C4
IC513	C4
IC515	C3
IC518	C6
IC519	D5
IC651	H3
Q301	D3
Q302	D3
Q303	D3
Q304	D2
Q305	D3
Q306	D3
Q307	E3
Q308	D3
Q309	D5
Q501	E4
Q502	E5
Q503	D5

1 2 3 4 5 6 7

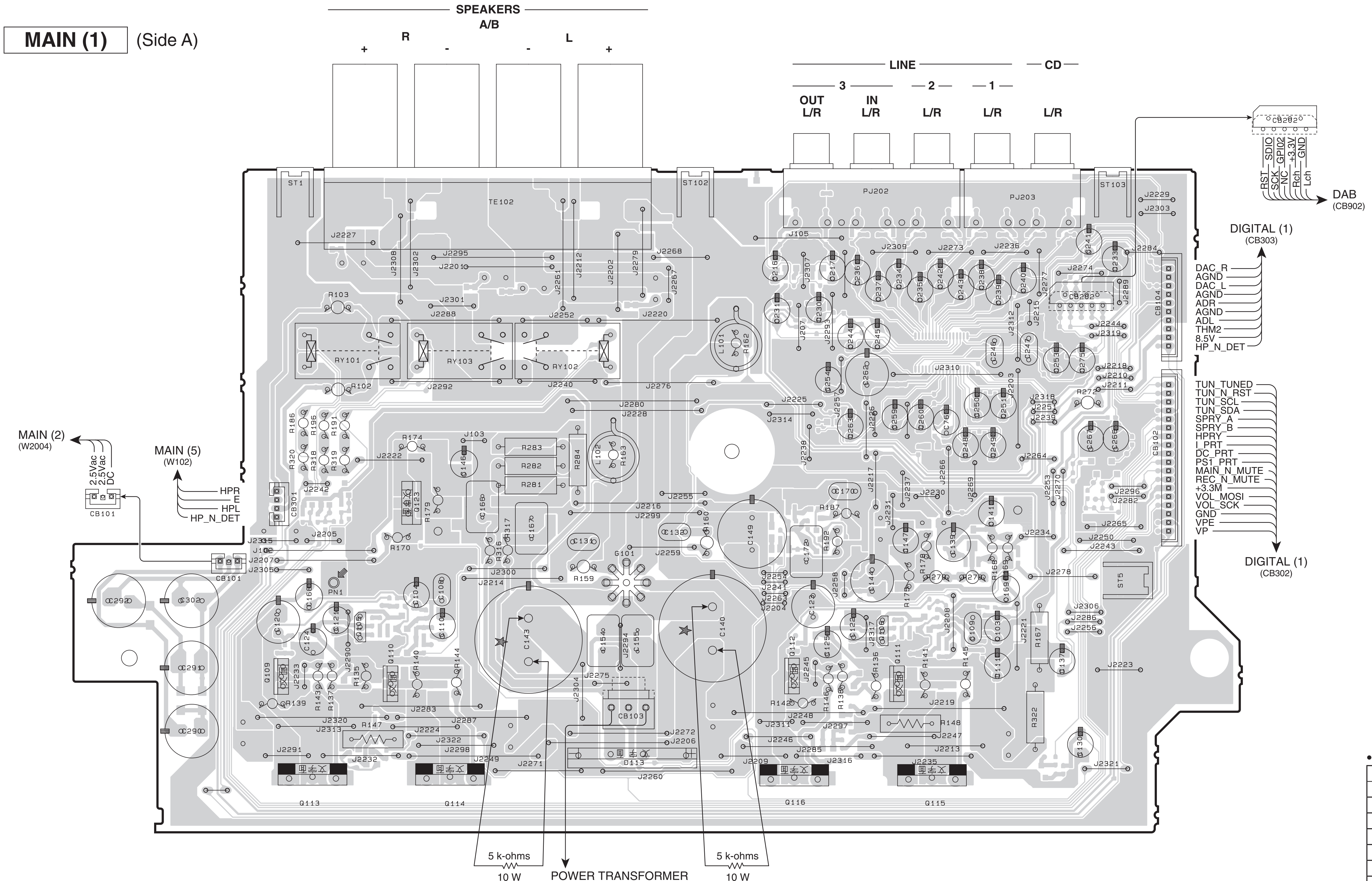
DIGITAL (1) (Side B)



DIGITAL (2) (Side B)



MAIN (1) (Side A)



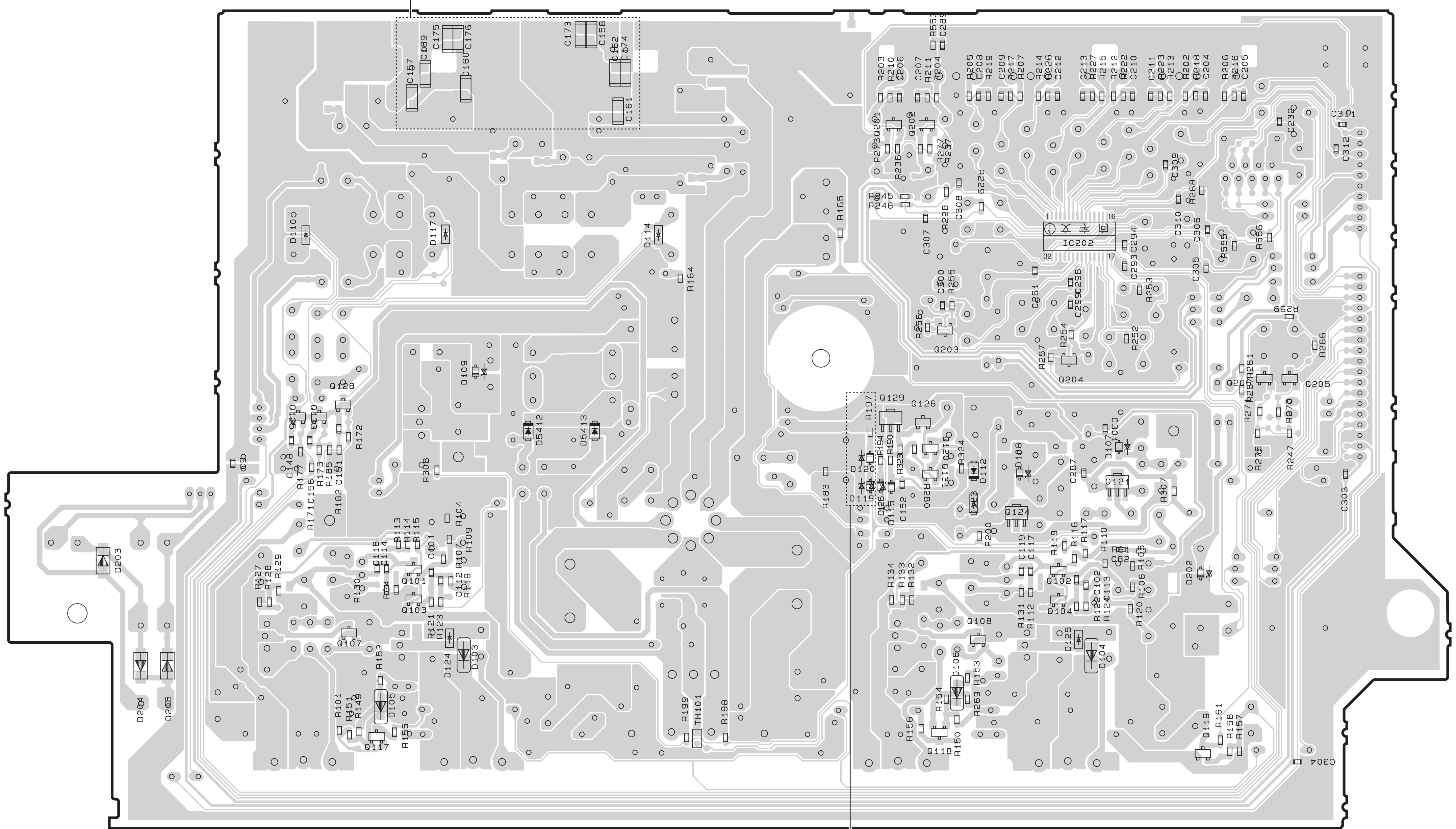
Safety measures

- Some internal parts in this product contain high voltages and are dangerous. Be sure to take safety measures during servicing, such as wearing insulating gloves.
- Note that the capacitors indicated below are dangerous even after the power is turned off because an electric charge remains and a high voltage continues to exist there. Before starting any repair work, connect a discharging resistor (5 k-ohms/10 W) to the terminals of each capacitor indicated below to discharge electricity.
  The time required for discharging is about 30 seconds per each.
  C140, C143 and C149 on MAIN (1) P.C.B.

• Semiconductor Location

Ref no.	Location
D113	E6
Q105	C5
Q106	G5
Q109	C5
Q110	D5
Q111	G5
Q112	F5
Q113	C6
Q114	D6
Q115	G6
Q116	F6
Q123	D4

MAIN (1) (Side B)



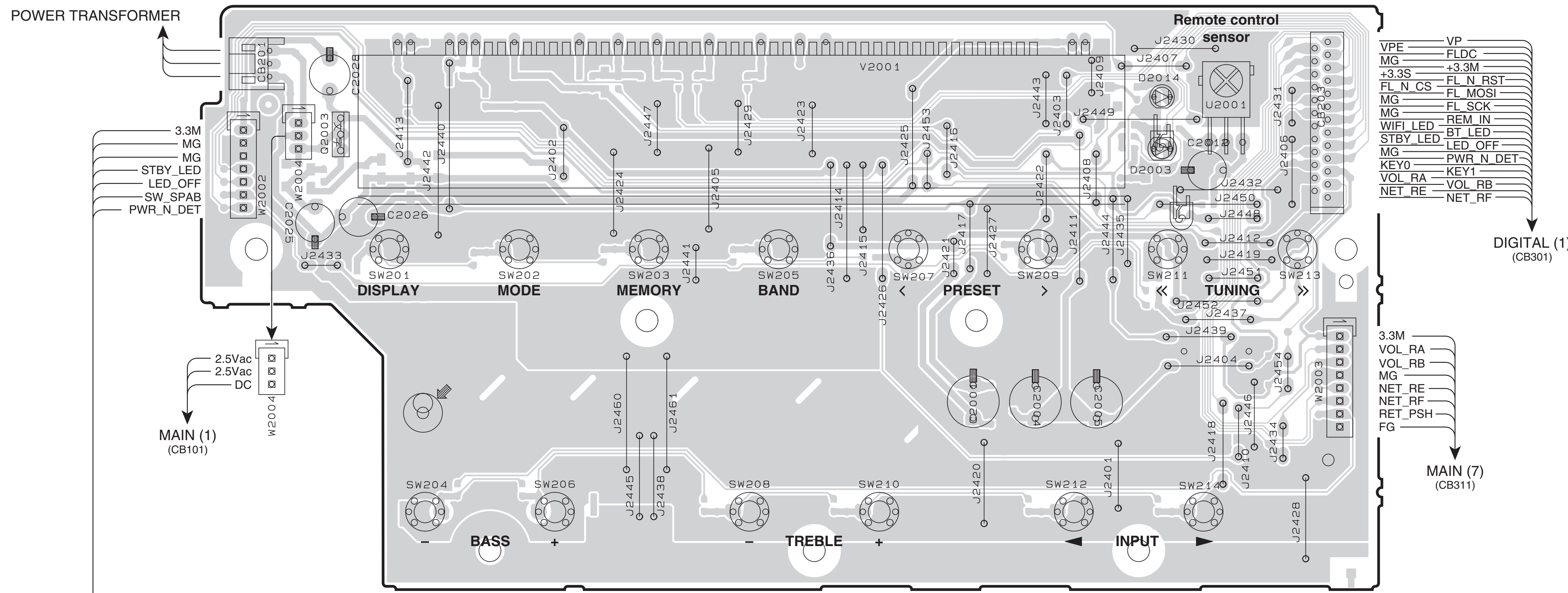
• Semiconductor Location

Ref no.	Location
D103	D5
D104	G5
D105	C6
D106	F6
D107	G4
D108	G5
D109	D4
D110	C3
D112	F5
D114	E3
D115	F5
D117	D3
D119	F5
D120	F4
D123	F5
D124	D5
D125	G5
D126	F4
D202	G5
D203	B5
D204	B5
D205	B5
D5412	D4
D5413	E4
IC202	G3
Q101	D5
Q102	G5
Q103	D5
Q104	G5
Q107	C5
Q108	F5
Q117	C6
Q118	F6
Q119	G6
Q120	F4
Q121	G5
Q124	G5
Q125	C4
Q126	F4
Q128	C4
Q129	F4
Q131	F5
Q133	C4
Q201	F3
Q202	F3
Q203	F4
Q204	G4
Q205	H4
Q206	H4

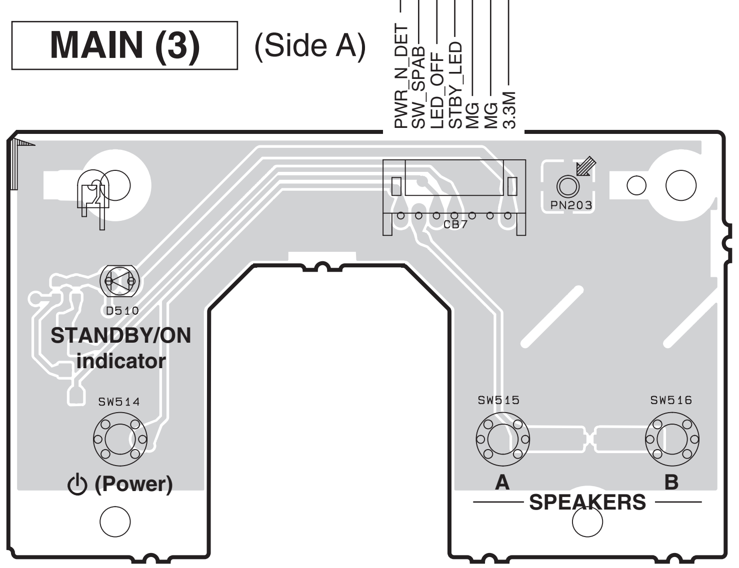
• Semiconductor Location

Ref no.	Location
D2003	G2
D2014	G2
D510	B6
IC351	J6
Q2003	C2

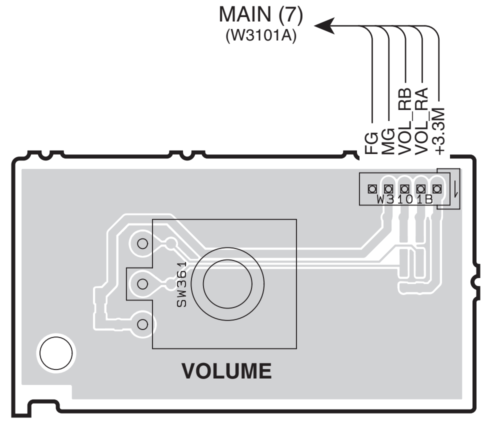
MAIN (2) (Side A)



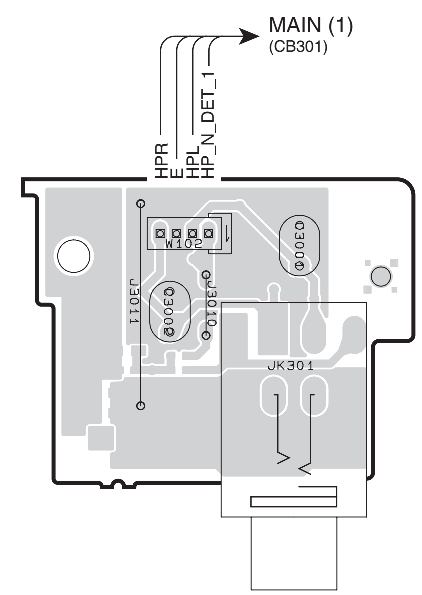
MAIN (3) (Side A)



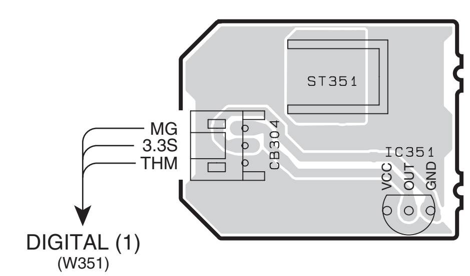
MAIN (4) (Side A)



MAIN (5) (Side A)



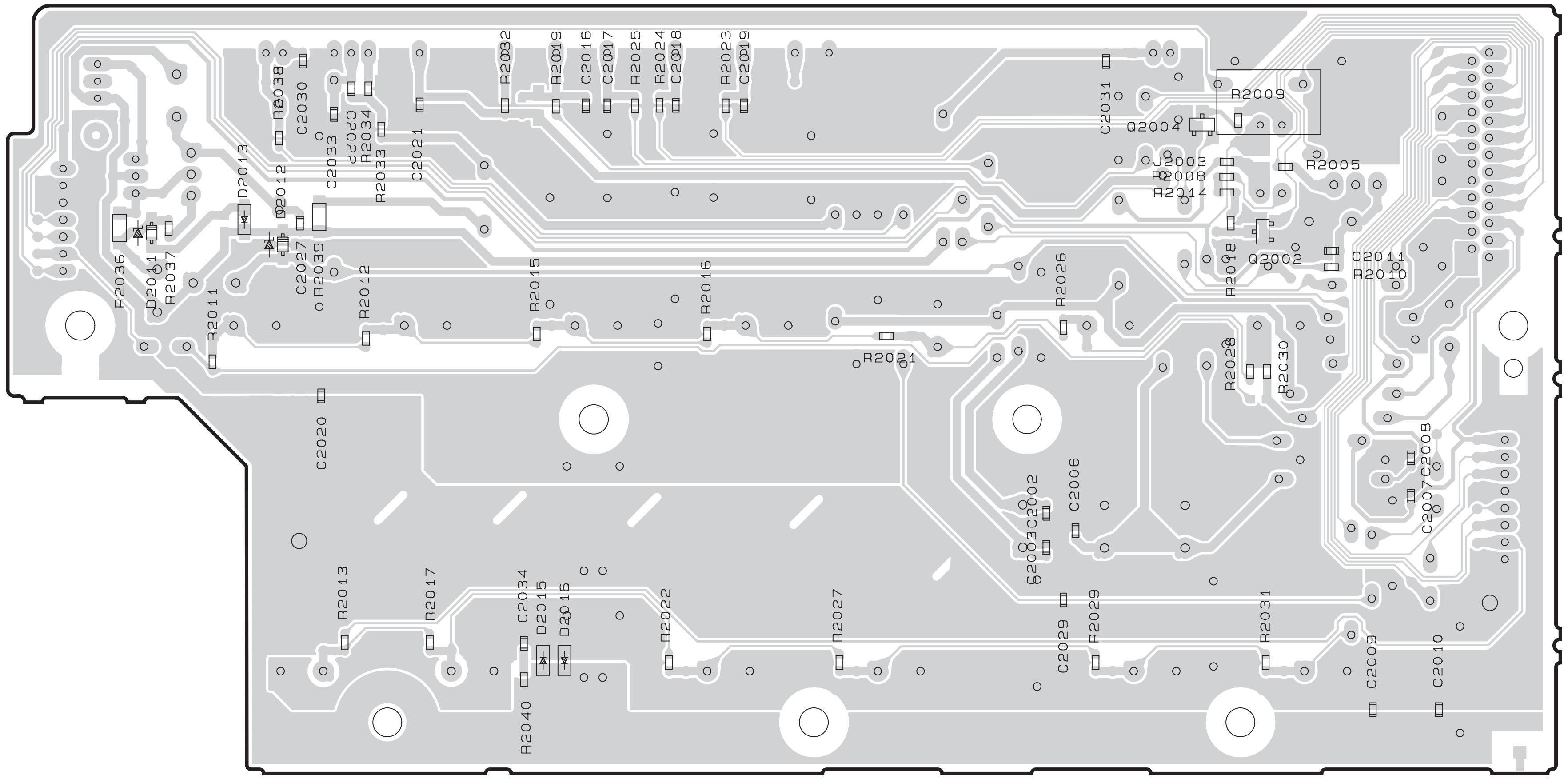
MAIN (6) (Side A)



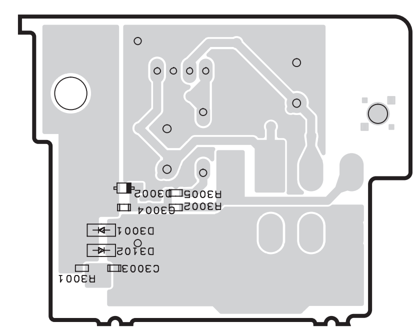
• Semiconductor Location

Ref no.	Location
D2011	C2
D2012	D2
D2013	D2
D2015	E4
D2016	E4
D3001	B6
D3002	G6
D3102	B6
Q2002	G2
Q2004	G2
Q511	B6
Q512	B6

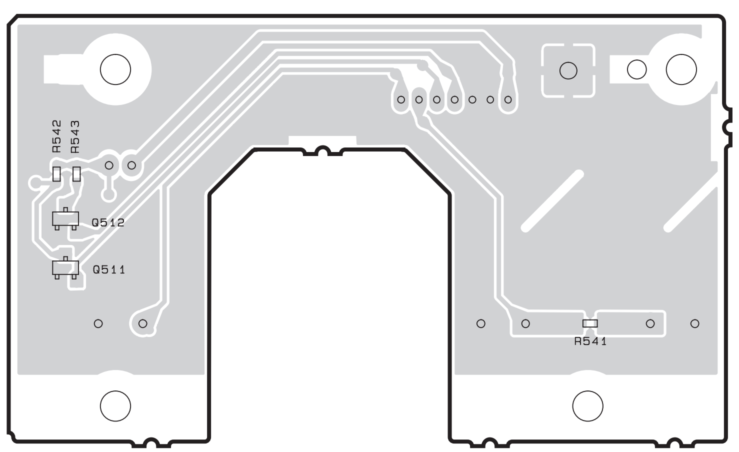
MAIN (2) (Side B)



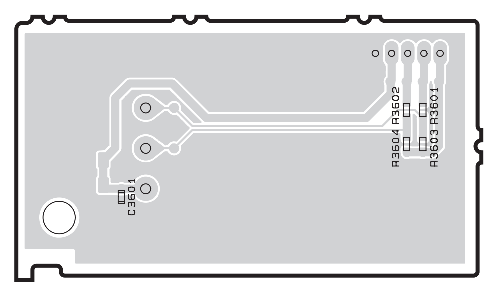
MAIN (5) (Side B)



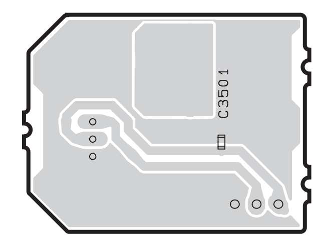
MAIN (3) (Side B)



MAIN (4) (Side B)



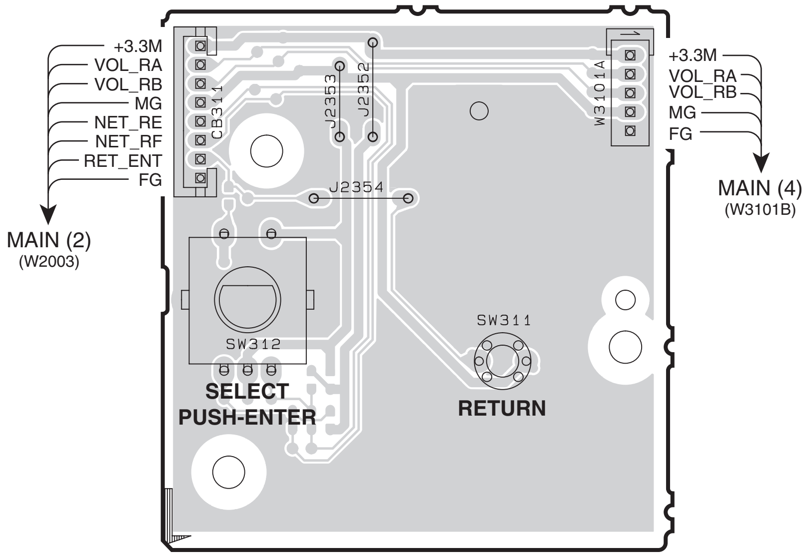
MAIN (6) (Side B)



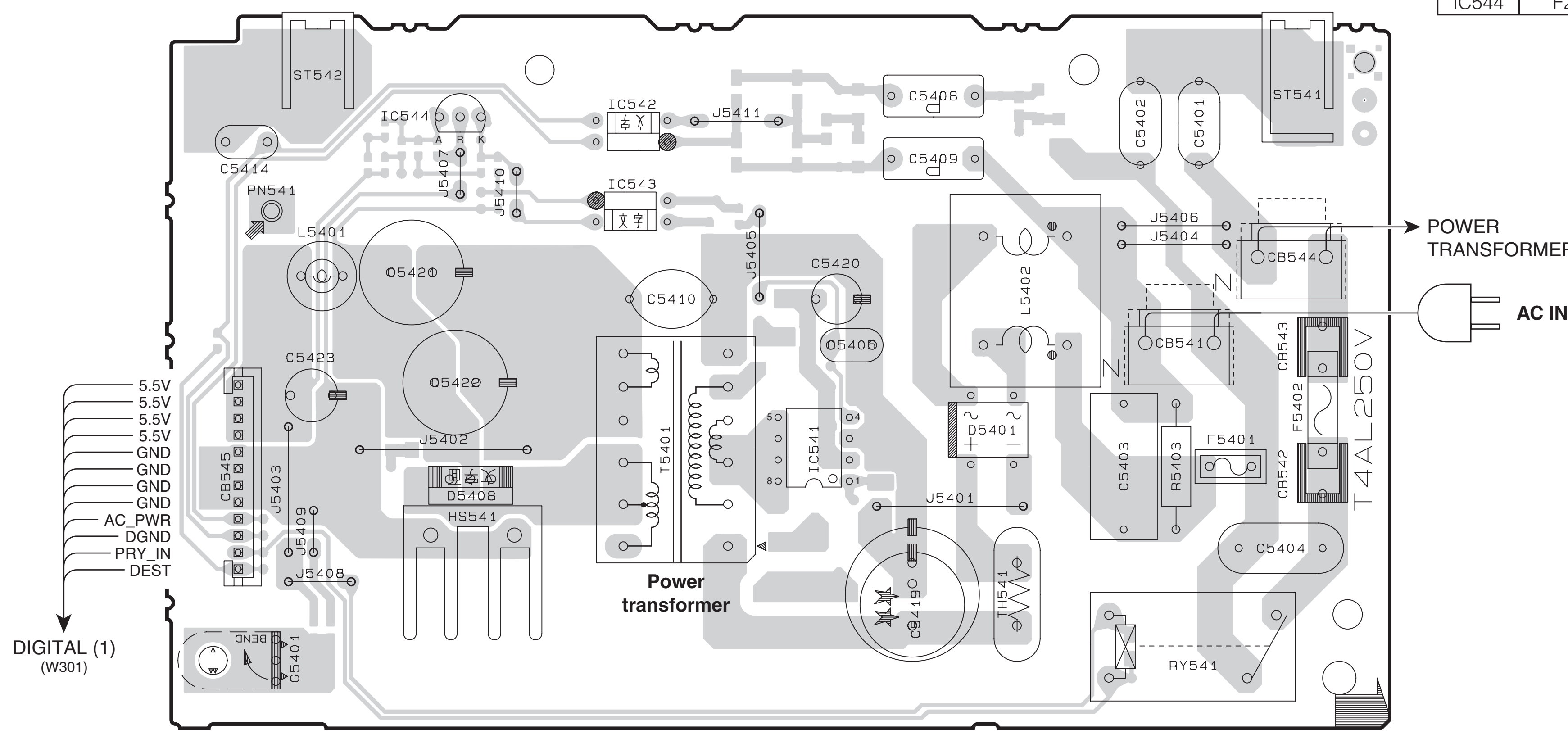
• Semiconductor Location

Ref no.	Location
D5401	H3
D5408	F3
IC541	G3
IC542	F2
IC543	F2
IC544	F2

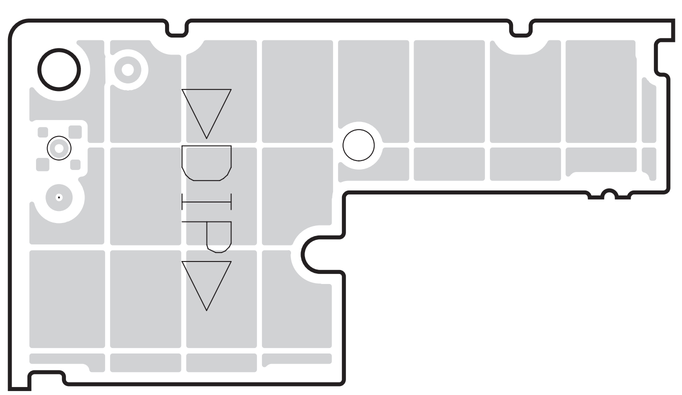
MAIN (7) (Side A)



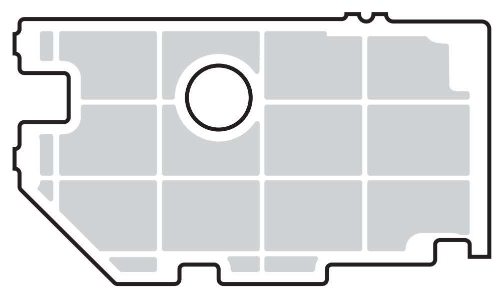
MAIN (8) (Side A)



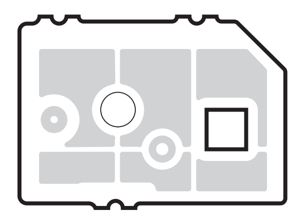
MAIN (9) (Side A)



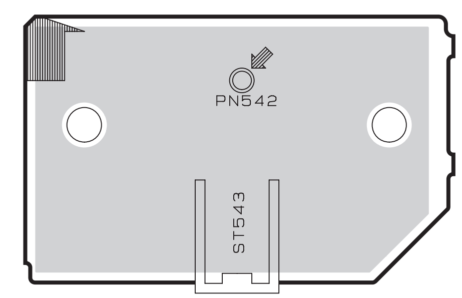
MAIN (10) (Side A)



MAIN (11) (Side A)



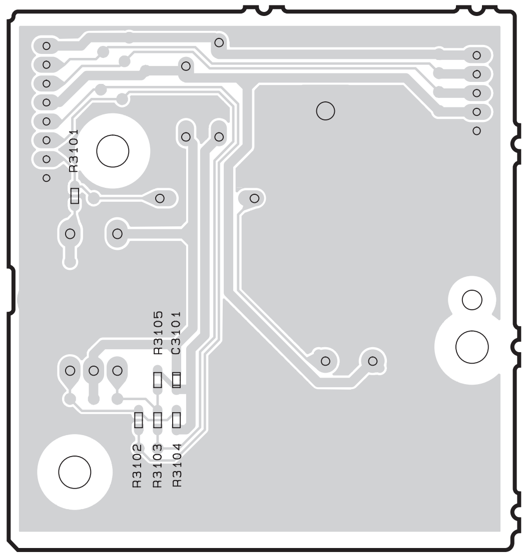
MAIN (12) (Side A)



• Semiconductor Location

Ref no.	Location
D5402	G3
D5403	G4
D5404	G2
D5405	G2
D5406	G2
D5407	G2
D5409	H4

MAIN (7) (Side B)



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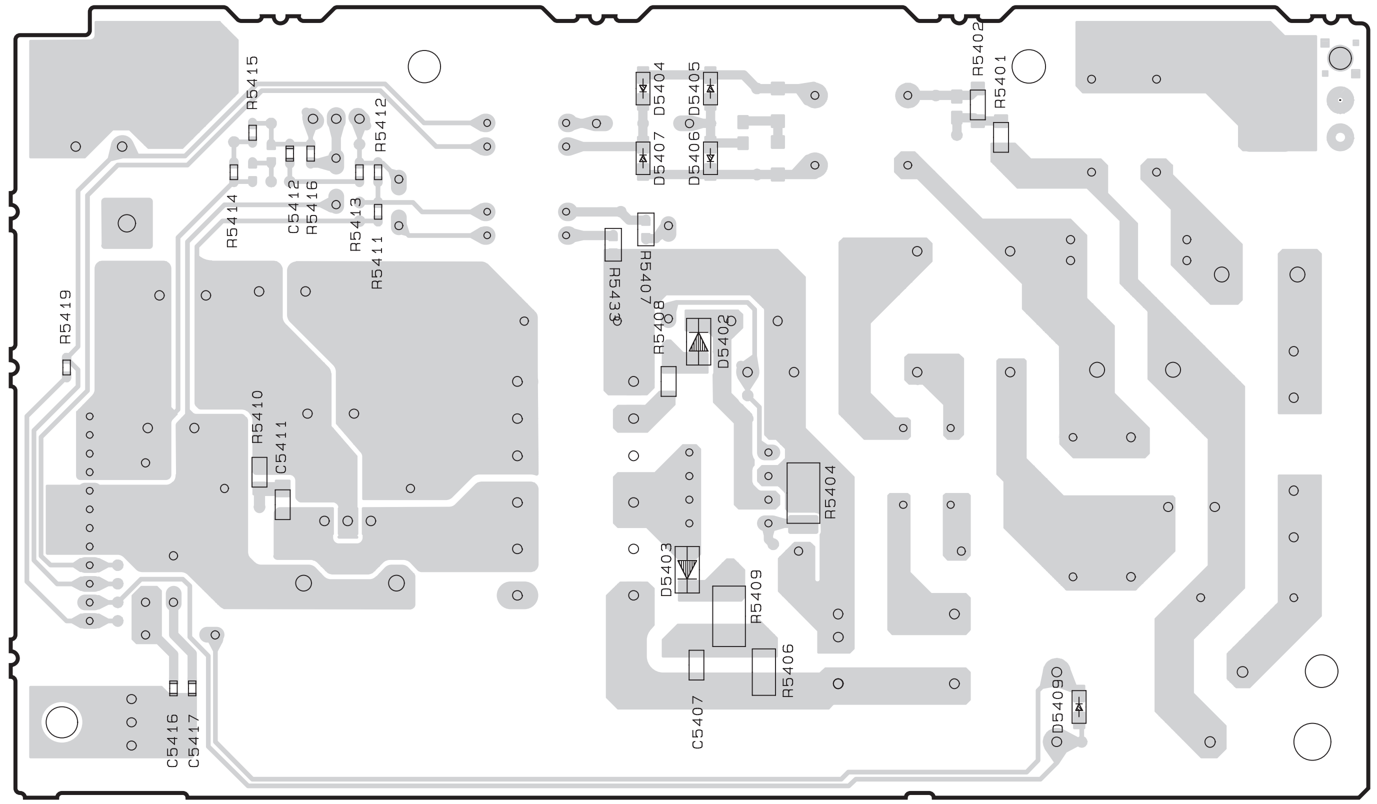

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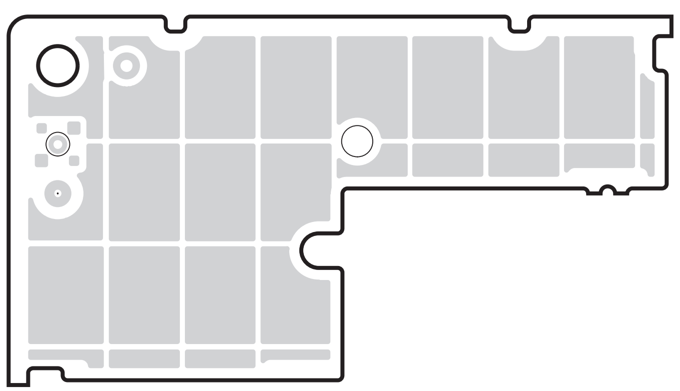

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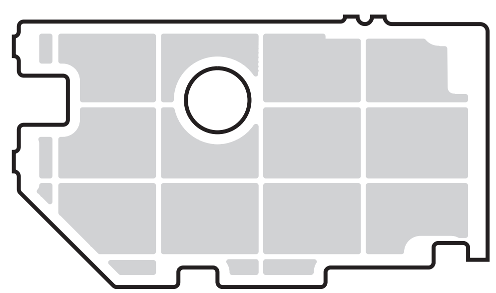

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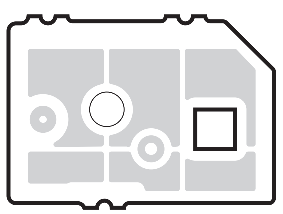

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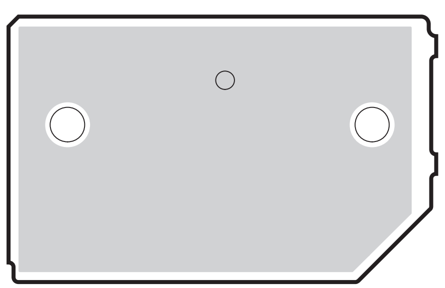

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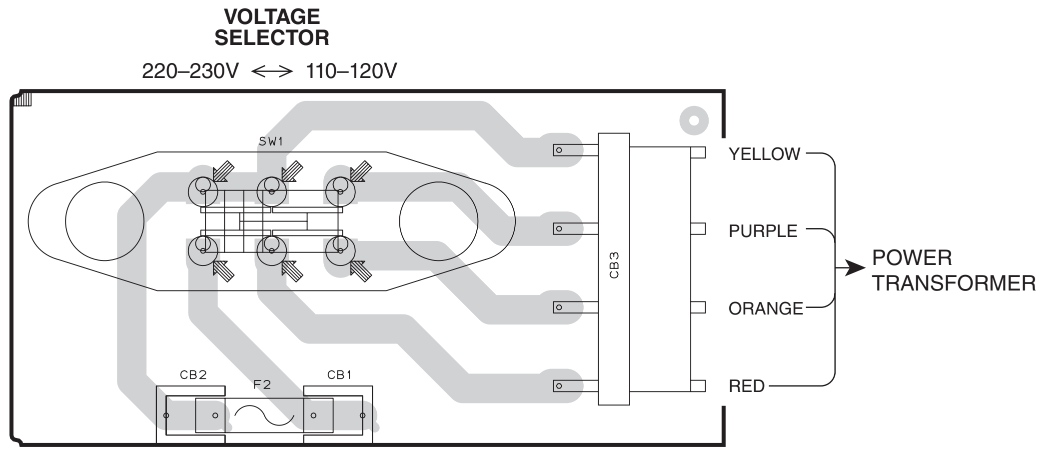
5
6
7

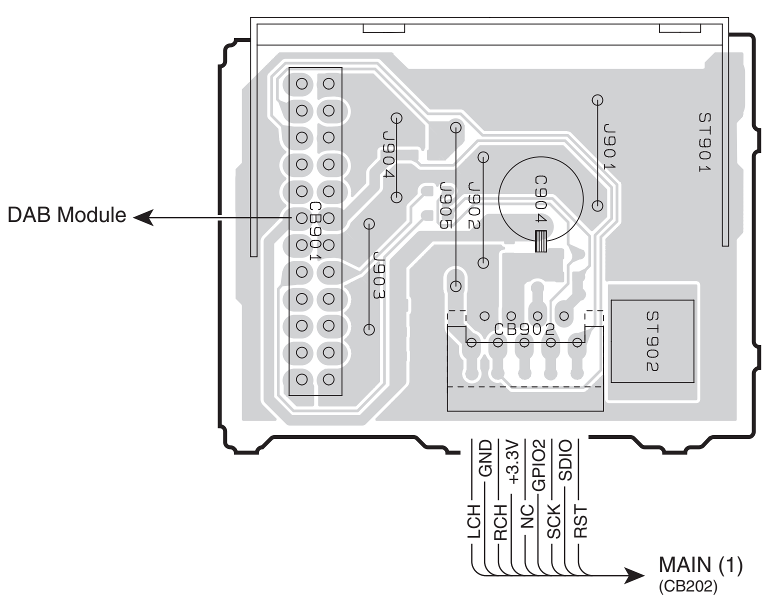
SWITCH (Side A)

R, S models



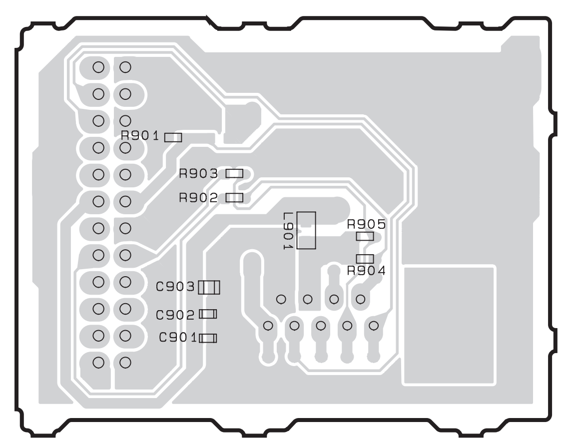
DAB (Side A)

(R-N402D)



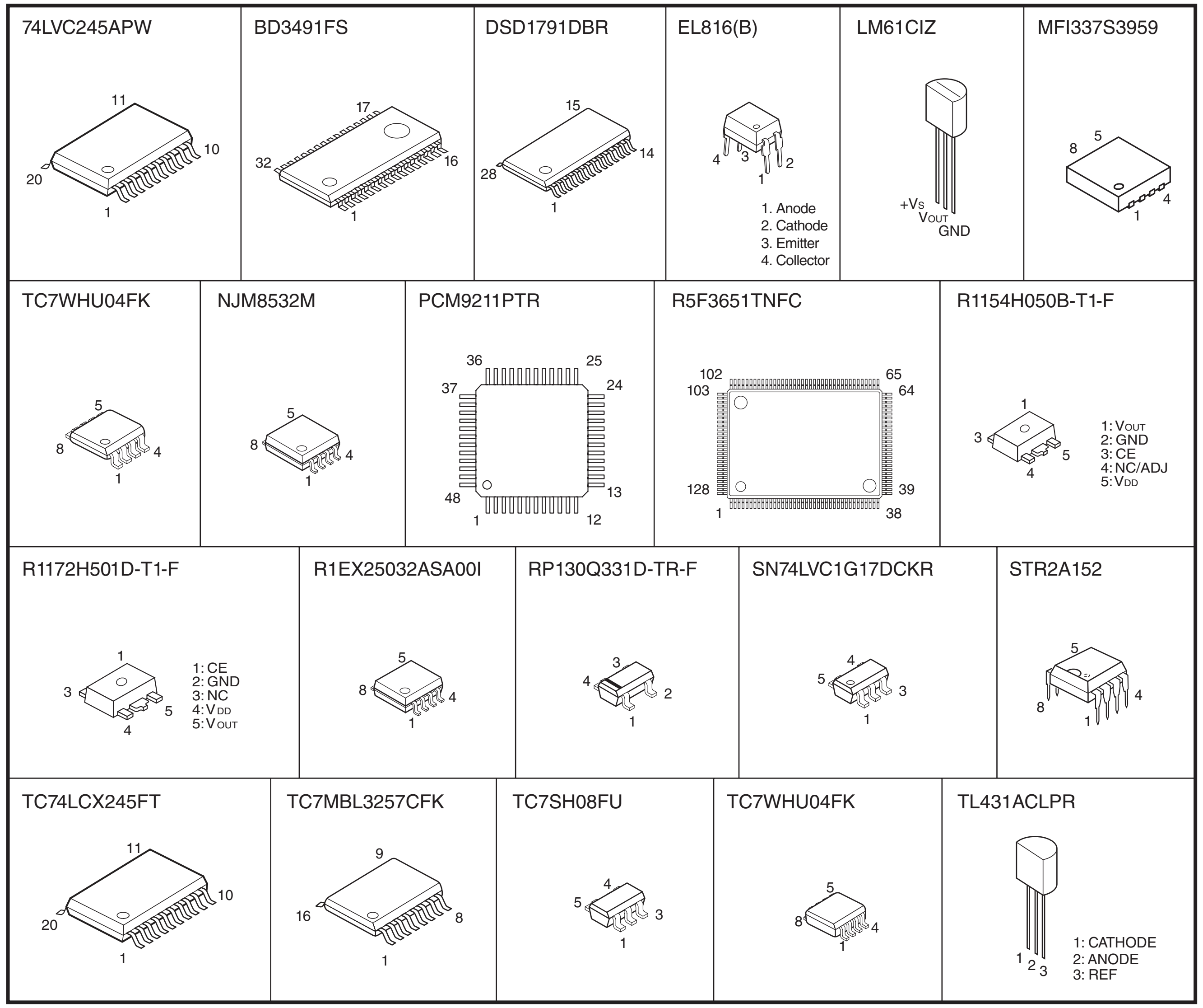
DAB (Side B)

(R-N402D)

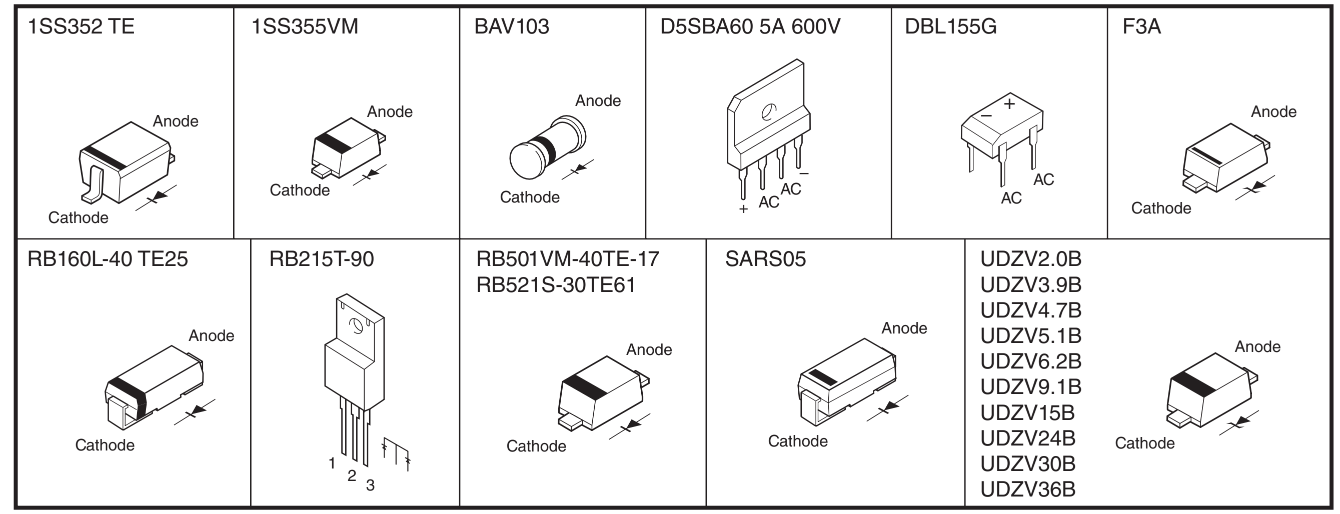


■ PIN CONNECTION DIAGRAMS

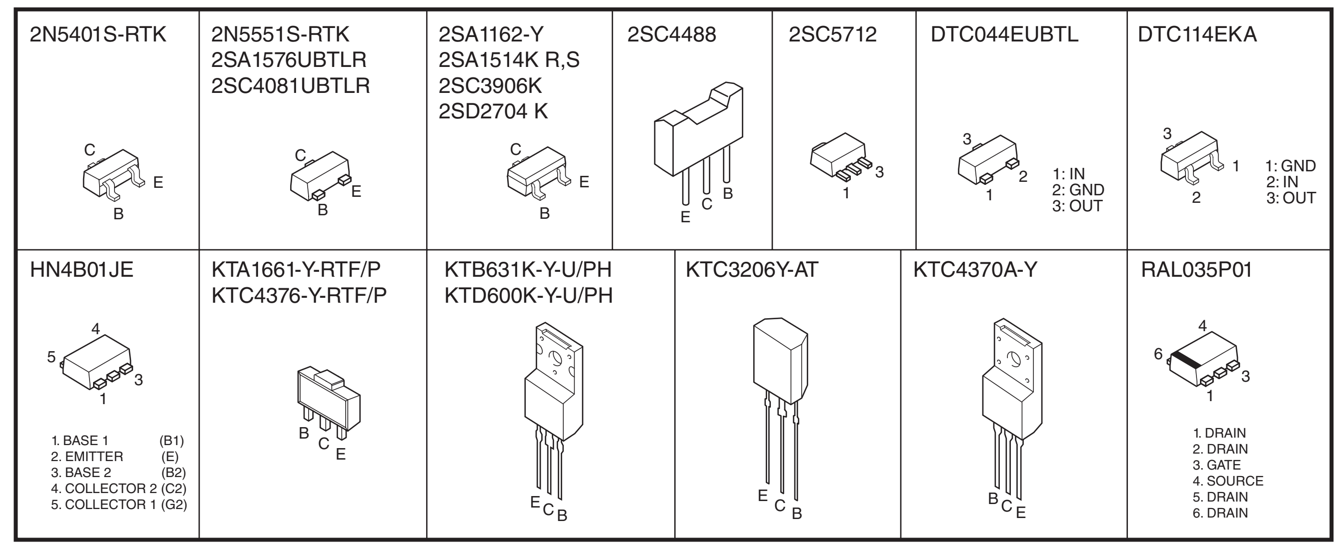
• ICs



• Diodes



•Transistors

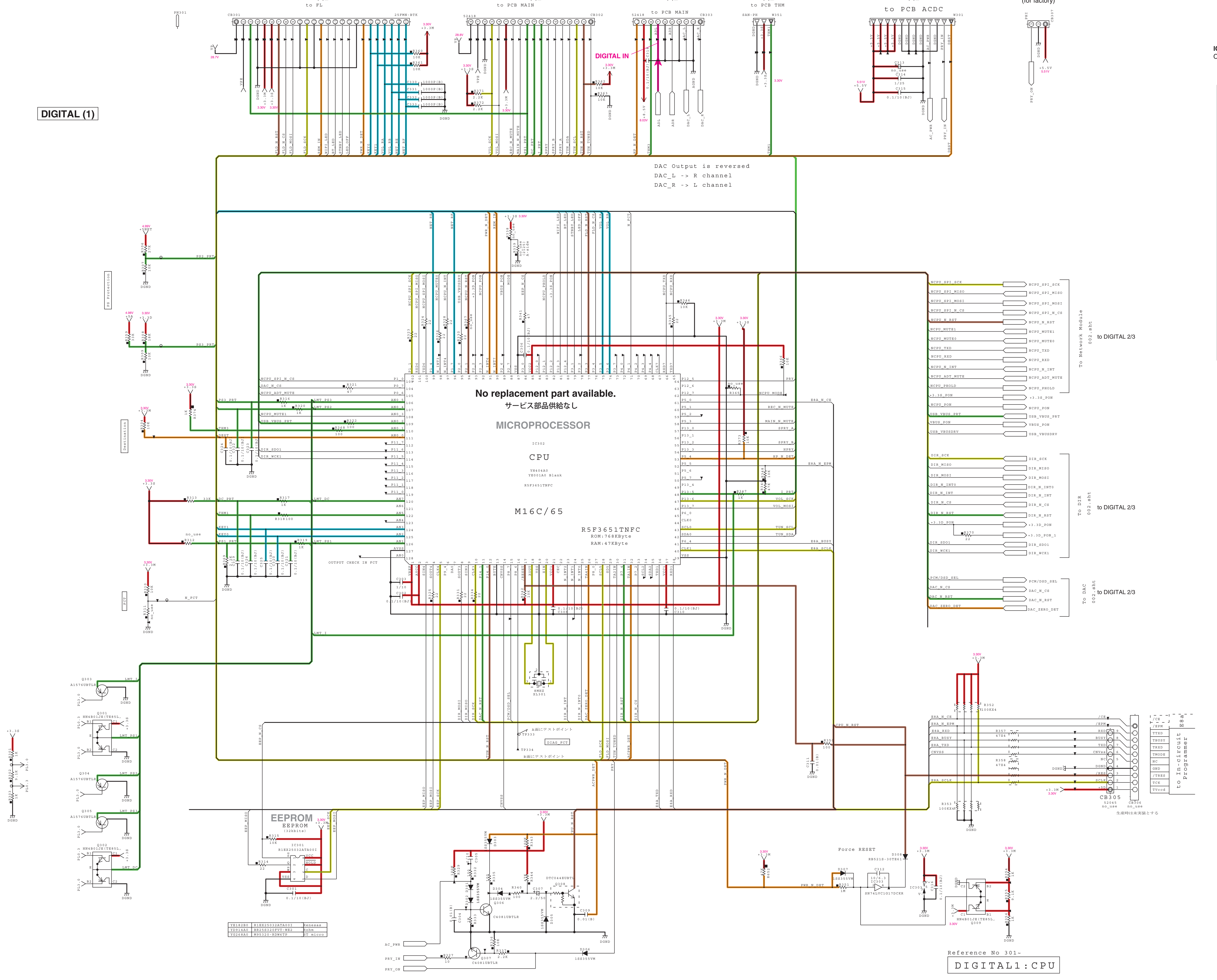


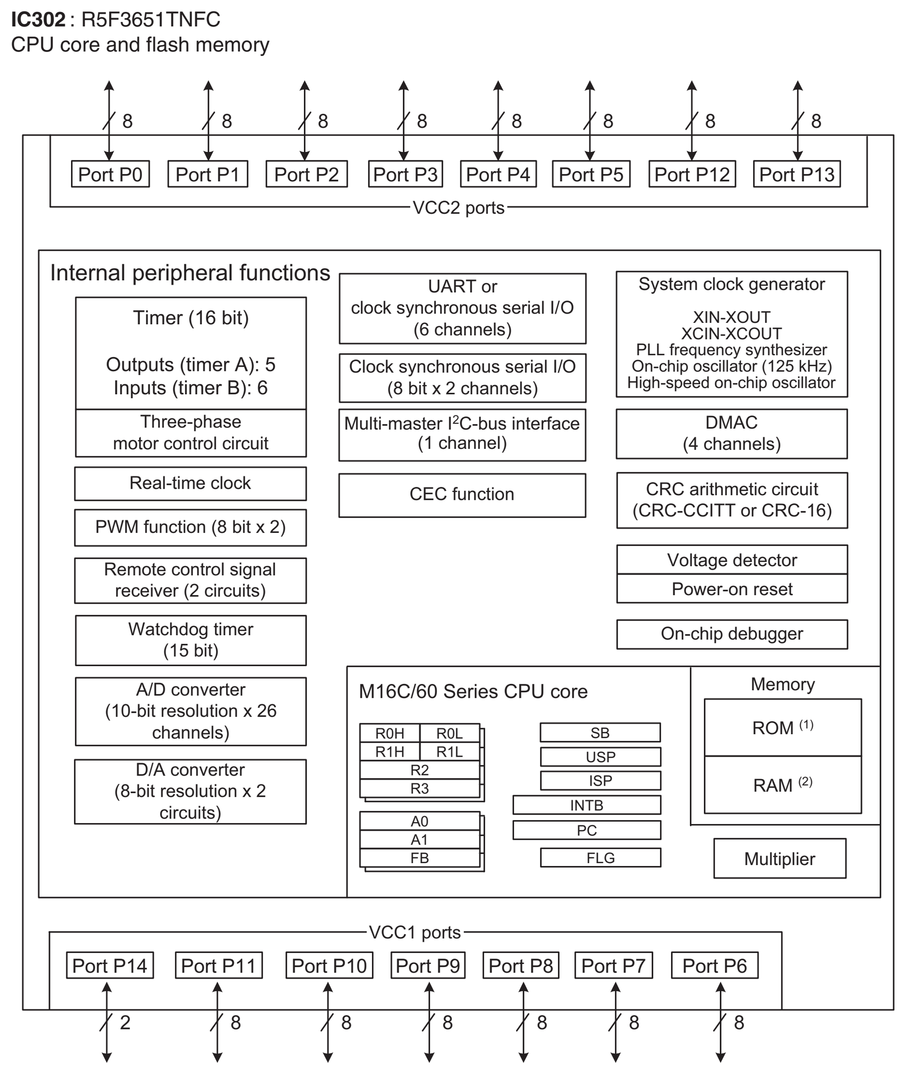
MEMO

MEMO

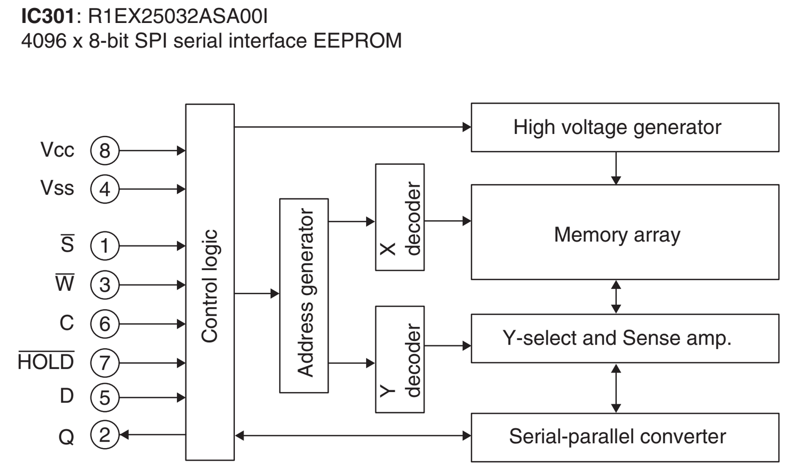
■ SCHEMATIC DIAGRAMS

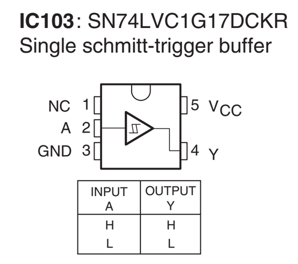
DIGITAL 1/3





Notes:
1. ROM size depends on MCU type.
2. RAM size depends on MCU type.





RESISTOR

REMARKS	PARTS NAME
NO MARK	CARBON FILM RESISTOR (P=5)
	CARBON FILM RESISTOR (P=10)
	METAL OXIDE FILM RESISTOR
	METAL FILM RESISTOR
	METAL PLATE RESISTOR
	FIRE PROOF CARBON FILM RESISTOR
	CEMENT MOLDED RESISTOR
	SEMI VARIABLE RESISTOR
	CHIP RESISTOR

CAPACITOR

REMARKS	PARTS NAME
NO MARK	ELECTROLYTIC CAPACITOR
	SOLID ELECTROLYTIC CAPACITOR
NO MARK	CERAMIC CAPACITOR
	CERAMIC TUBULAR CAPACITOR
	POLYESTER FILM CAPACITOR
	POLYSTYRENE FILM CAPACITOR
	MICA CAPACITOR
	POLYPROPYLENE FILM CAPACITOR
	SEMICONDUCTIVE CERAMIC CAPACITOR

NOTICE (model)

(J)····· JAPAN
(U)····· U.S.A
(C)····· CANADA
(R)····· GENERAL
(T)····· CHINA
(K)····· KOREA
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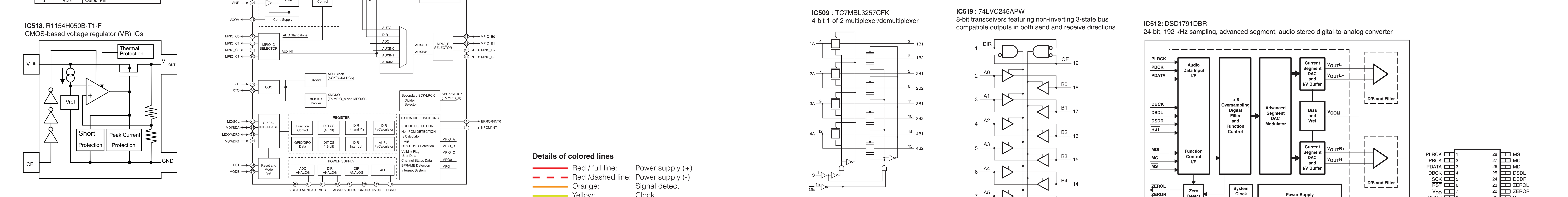
(H)····· THAI

★ All voltages are measured with a 10M Ω /V DC electronic voltmeter.
★ Components having special characteristics are marked ⚠ and must be replaced with parts having specifications equal to those originally installed.
★ Schematic diagram is subject to change without notice.

Details of colored lines

Red / full line:	Power supply (+)
Red /dashed line:	Power supply (-)
Orange:	Signal detect
Yellow:	Clock
Green:	Protection detect
Brown:	Reset signal
Blue:	Panel key input



RESISTOR

REMARKS	PARTS NAME
NO MARK	CARBON FILM RESISTOR (P=5)
◪	CARBON FILM RESISTOR (P=10)
△	METAL OXIDE FILM RESISTOR
△	METAL FILM RESISTOR
☒	METAL PLATE RESISTOR
◪	FIRE PROOF CARBON FILM RESISTOR
□	CEMENT MOLDED RESISTOR
⊘	SEMI VARIABLE RESISTOR
▤	CHIP RESISTOR

CAPACITOR

REMARKS	PARTS NAME
NO MARK	ELECTROLYTIC CAPACITOR
⊗	SOLID ELECTROLYTIC CAPACITOR
NO MARK	CERAMIC CAPACITOR
◉	CERAMIC TUBULAR CAPACITOR
◎	POLYESTER FILM CAPACITOR
○	POLYSTYRENE FILM CAPACITOR
⦶	MICA CAPACITOR
Ⓟ	POLYPROPYLENE FILM CAPACITOR
⊜	SEMICONDUCTIVE CERAMIC CAPACITOR

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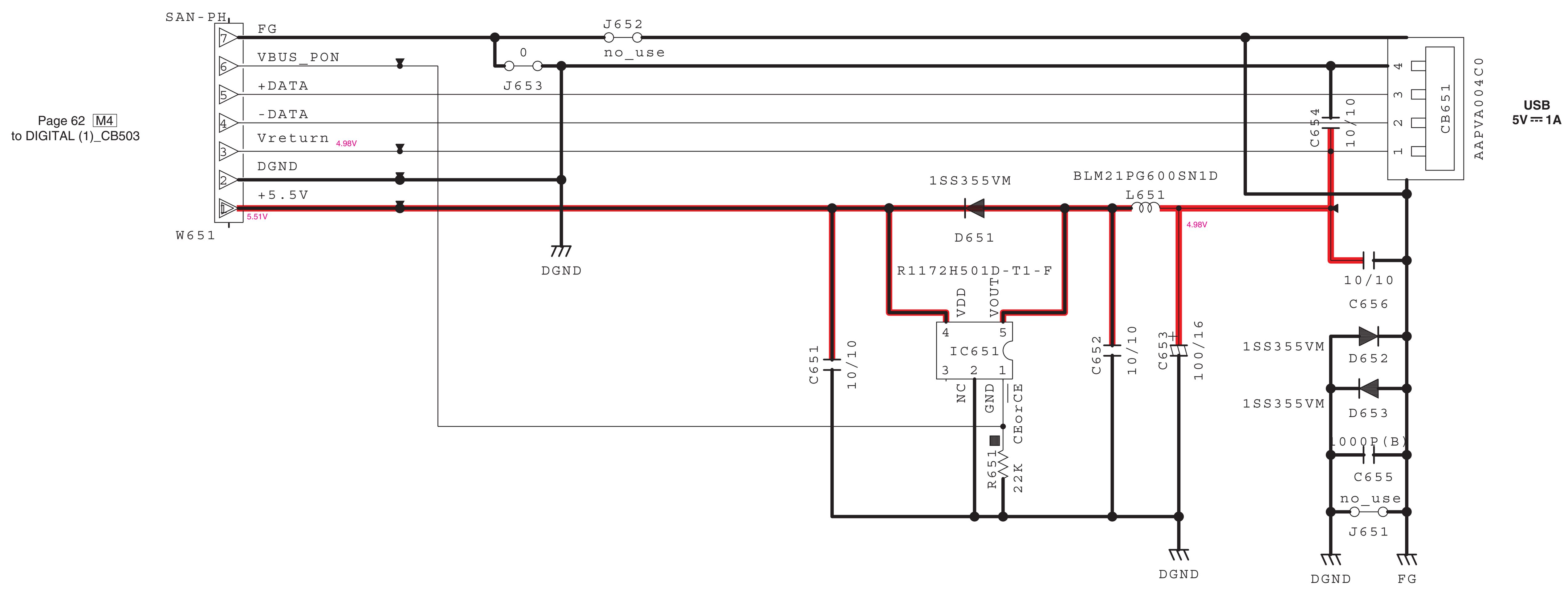
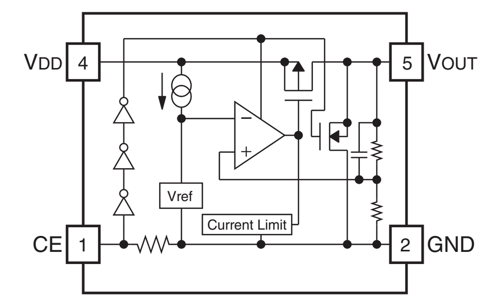
DIGITAL (2)

USB CONNECTOR



IC651: R1172H501D-T1-F
Voltage regulator



Pin No.	Symbol	Description
1	CE	Chip Enable Pin ("H" Active)
2	GND	Ground Pin
4	VDD	Input Pin
5	VOUT	Output Pin

Reference No 651~

USB Connector

RESISTOR

REMARKS	PARTS NAME
NO MARK	CARBON FILM RESISTOR (P=5)
◩	CARBON FILM RESISTOR (P=10)
△	METAL OXIDE FILM RESISTOR
△	METAL FILM RESISTOR
⊠	METAL PLATE RESISTOR
◩	FIRE PROOF CARBON FILM RESISTOR
□	CEMENT MOLDED RESISTOR
⊘	SEMI VARIABLE RESISTOR
▤	CHIP RESISTOR

CAPACITOR

REMARKS	PARTS NAME
NO MARK	ELECTROLYTIC CAPACITOR
⊗	SOLID ELECTROLYTIC CAPACITOR
NO MARK	CERAMIC CAPACITOR
◉	CERAMIC TUBULAR CAPACITOR
◎	POLYESTER FILM CAPACITOR
○	POLYSTYRENE FILM CAPACITOR
⦶	MICA CAPACITOR
Ⓟ	POLYPROPYLENE FILM CAPACITOR
⊜	SEMICONDUCTIVE CERAMIC CAPACITOR

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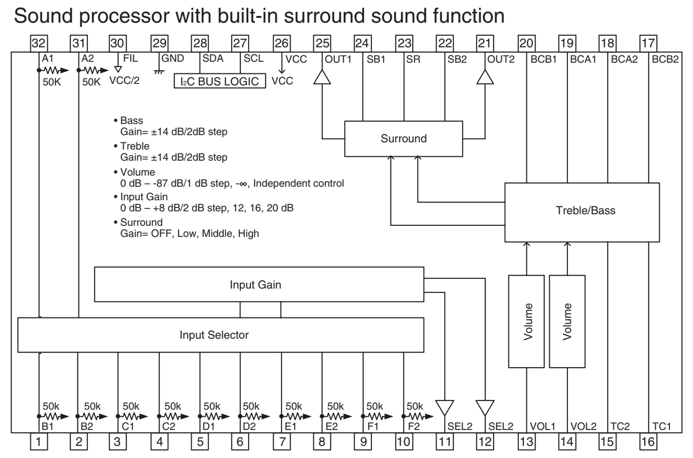
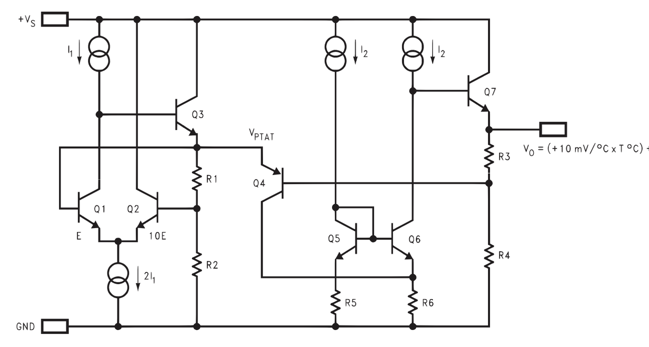
Brown:	Reset signal
Blue:	Panel key input

IC202: BD3491FS
Sound processor with built-in surround sound function



IC351: LM61CIZ
Temperature sensor



RESISTOR

REMARKS	PARTS NAME
NO MARK	CARBON FILM RESISTOR (P=5)
	CARBON FILM RESISTOR (P=10)
	METAL OXIDE FILM RESISTOR
	METAL FILM RESISTOR
	METAL PLATE RESISTOR
	FIRE PROOF CARBON FILM RESISTOR
	CEMENT MOLDED RESISTOR
	SEMI VARIABLE RESISTOR
	CHIP RESISTOR

CAPACITOR

REMARKS	PARTS NAME
NO MARK	ELECTROLYTIC CAPACITOR
	SOLID ELECTROLYTIC CAPACITOR
NO MARK	CERAMIC CAPACITOR
	CERAMIC TUBULAR CAPACITOR
	POLYESTER FILM CAPACITOR
	POLYSTYRENE FILM CAPACITOR
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(A)····· AUSTRALIA	(S)····· BRAZIL
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Safety measures

- Some internal parts in this product contain high voltages and are dangerous. Be sure to take safety measures during servicing, such as wearing insulating gloves.
- Note that the capacitors indicated below are dangerous even after the power is turned off because an electric charge remains and a high voltage continues to exist there. Before starting any repair work, connect a discharging resistor (5 k-ohms/10 W) to the terminals of each capacitor indicated below to discharge electricity.
  The time required for discharging is about 30 seconds per each.
  C140 and C143 on MAIN (1) P.C.B.

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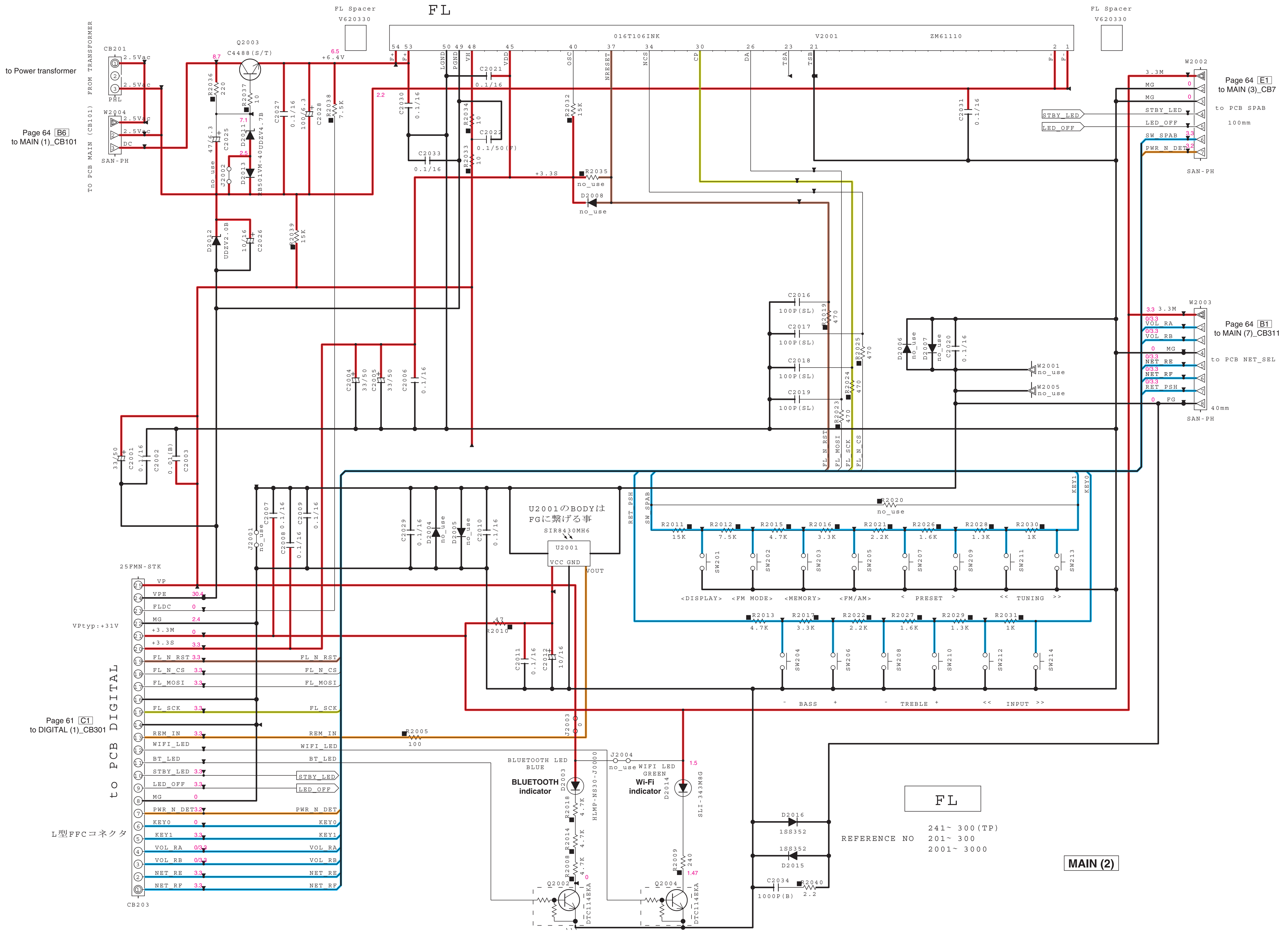
Brown:	Reset signal
Blue:	Panel key input

MAIN 2/3



MAIN (2)

RESISTOR

REMARKS	PARTS NAME
NO MARK	CARBON FILM RESISTOR (P=5)
◪	CARBON FILM RESISTOR (P=10)
△	METAL OXIDE FILM RESISTOR
⧋	METAL FILM RESISTOR
⊠	METAL PLATE RESISTOR
◪	FIRE PROOF CARBON FILM RESISTOR
□	CEMENT MOLDED RESISTOR
⊘	SEMI VARIABLE RESISTOR
▤	CHIP RESISTOR

CAPACITOR

REMARKS	PARTS NAME
NO MARK	ELECTROLYTIC CAPACITOR
⊗	SOLID ELECTROLYTIC CAPACITOR
NO MARK	CERAMIC CAPACITOR
◉	CERAMIC TUBULAR CAPACITOR
◎	POLYESTER FILM CAPACITOR
○	POLYSTYRENE FILM CAPACITOR
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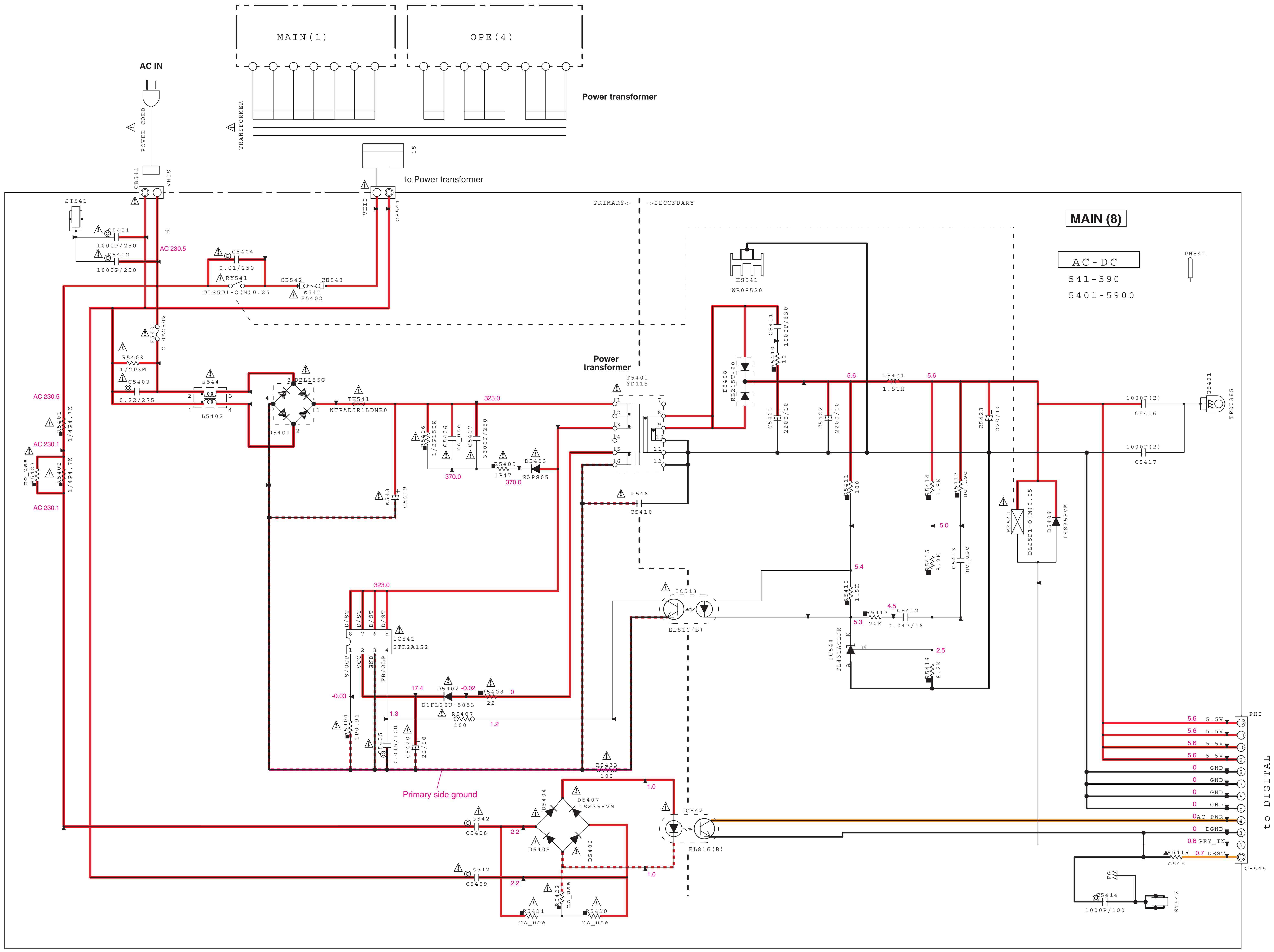
MAIN 3/3



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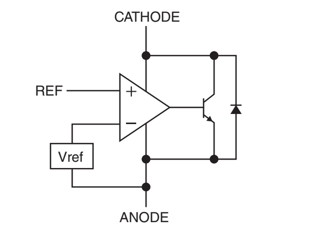
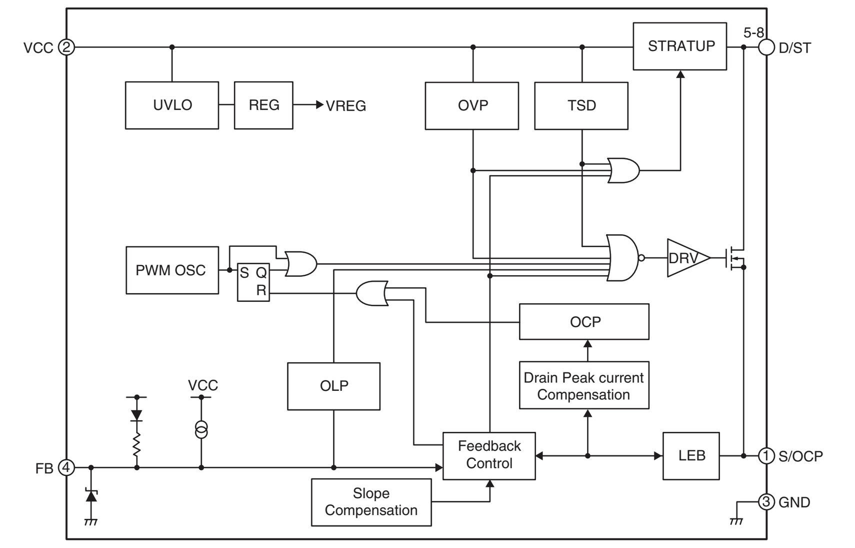
to DIGITAL (1)_W301

IC544: TL431ACLPR
Adjustable precision shunt regulators



IC541: STR2A152
Switching regulator



RESISTOR

REMARKS	PARTS NAME
NO MARK	CARBON FILM RESISTOR (P=5)
	CARBON FILM RESISTOR (P=10)
	METAL OXIDE FILM RESISTOR
	METAL FILM RESISTOR
	METAL PLATE RESISTOR
	FIRE PROOF CARBON FILM RESISTOR
	CEMENT MOLDED RESISTOR
	SEMI VARIABLE RESISTOR
	CHIP RESISTOR

CAPACITOR

REMARKS	PARTS NAME
NO MARK	ELECTROLYTIC CAPACITOR
	SOLID ELECTROLYTIC CAPACITOR
NO MARK	CERAMIC CAPACITOR
	CERAMIC TUBULAR CAPACITOR
	POLYESTER FILM CAPACITOR
	POLYSTYRENE FILM CAPACITOR
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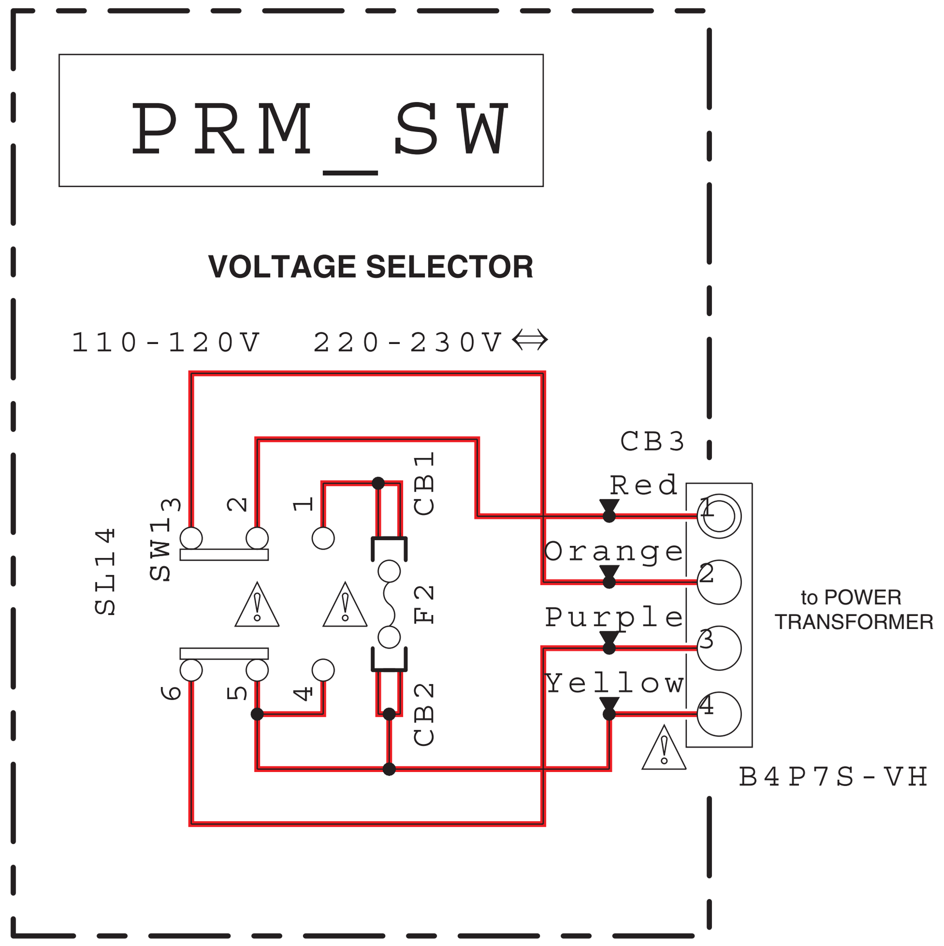
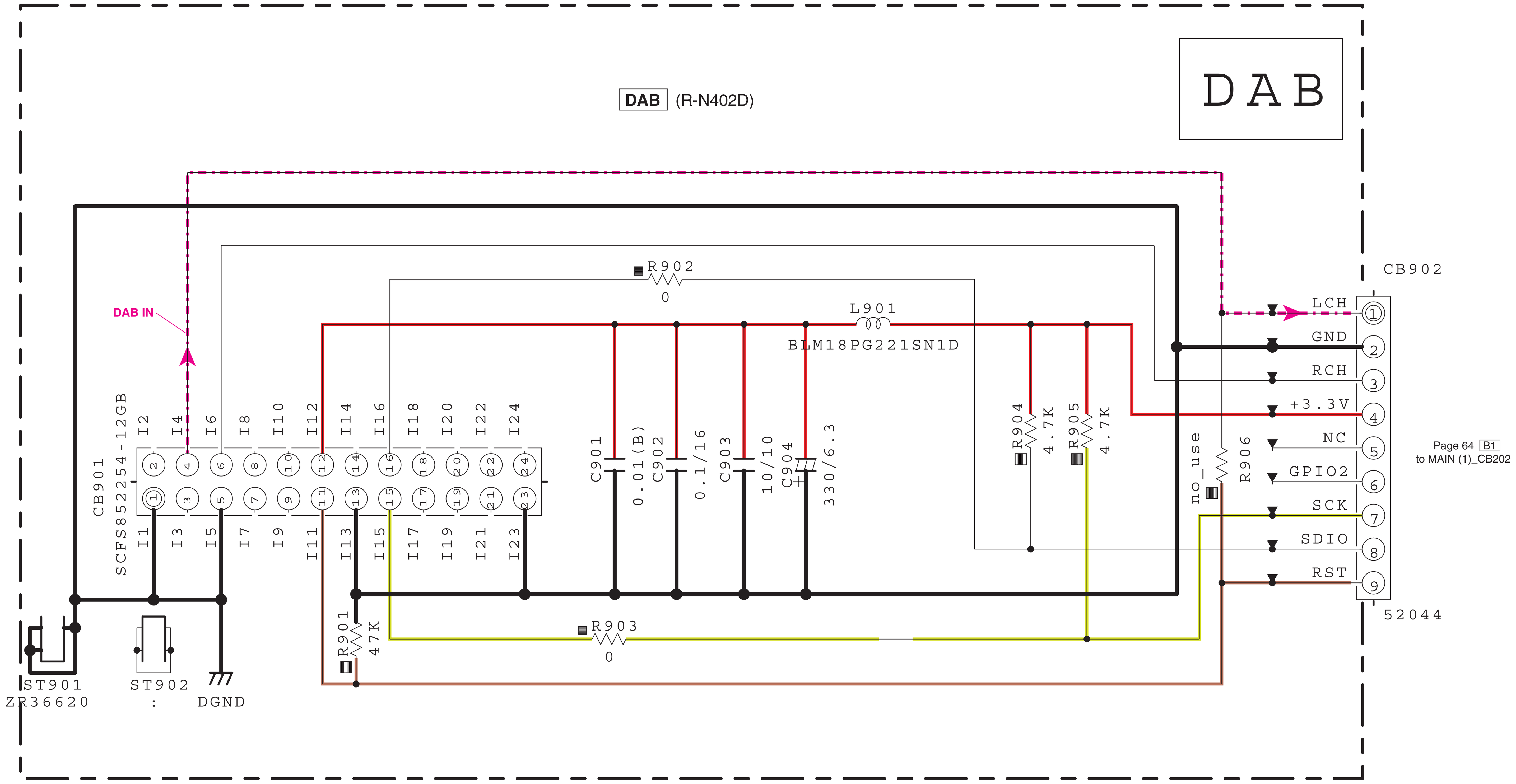
Brown:	Reset signal
Blue:	Panel key input

SWITCH/DAB

SWITCH (R model)





RESISTOR

REMARKS	PARTS NAME
NO MARK	CARBON FILM RESISTOR (P=5)
	CARBON FILM RESISTOR (P=10)
	METAL OXIDE FILM RESISTOR
	METAL FILM RESISTOR
	METAL PLATE RESISTOR
	FIRE PROOF CARBON FILM RESISTOR
	CEMENT MOLDED RESISTOR
	SEMI VARIABLE RESISTOR
	CHIP RESISTOR

CAPACITOR

REMARKS	PARTS NAME
NO MARK	ELECTROLYTIC CAPACITOR
	SOLID ELECTROLYTIC CAPACITOR
NO MARK	CERAMIC CAPACITOR
	CERAMIC TUBULAR CAPACITOR
	POLYESTER FILM CAPACITOR
	POLYSTYRENE FILM CAPACITOR
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■ REPLACEMENT PARTS LIST

• ELECTRICAL COMPONENT PARTS

WARNING

- Components having special characteristics are marked ⚠ and must be replaced with parts having specifications equal to those originally installed.

ABBREVIATIONS IN THIS LIST ARE AS FOLLOWS:

C.A.EL.CHP : CHIP ALUMI.ELECTROLYTIC CAP
C.CE : CERAMIC CAP
C.CE.ARRAY : CERAMIC CAP ARRAY
C.CE.CHP : CHIP CERAMIC CAP
C.CE.M.CHP : CHIP MULTILAYER CERAMIC CAP
C.CE.SAFTY : RECOGNIZED CERAMIC CAP
C.CE.TUBLR : CERAMIC TUBULAR CAP
C.CE.SMI : SEMI CONDUCTIVE CERAMIC CAP
C.EL : ELECTROLYTIC CAP
C.EL.BP : BIPOLAR ELECTROLYTIC CAP
C.EL.CHP : CHIP ELECTROLYTIC CAP
C.MICA : MICA CAP
C.ML.FLM : MULTILAYER FILM CAP
C.MP : METALLIZED POLYESTER FILM CAP
C.MYLAR : MYLAR FILM CAP
C.MYLAR.ML : MULTILAYER MYLAR FILM CAP
C.NIOB.OXD : NIOBIUM OXIDE CAP
C.PAPER : PAPER CAPACITOR
C.PLS : POLYSTYRENE FILM CAP
C.PML : POLYMER MULTI-LAYER CAPACITOR
C.POL : POLYESTER FILM CAP
C.PP : POLYPROPYLENE FILM CAP
C.PP.CHP : CHIP POLYPROPYLENE FILM CAP
C.TNTL : TANTALIUM CAP
C.TNTL.CHP : CHIP TANTALIUM CAP
C.TRIM : TRIMMER CAP
CN : CONNECTOR
CN.BS.PIN : CONNECTOR,BASE PIN
CN.CANNON : CONNECTOR,CANNON
CN.DIN : CONNECTOR,DIN
CN.FLAT : CONNECTOR,FLAT CABLE
CN.FFC : CONNECTOR,FLEXIBLE FLAT CABLE
CN.HDMI : HDMI CONNECTOR
CN.PHOTO.R : PHOTO FIBER SENSOR,RECEIVED
CN.PHOTO.T : PHOTO FIBER SENSOR,TRANSMITTED
D.SCHOTTKY : SCHOTTKY BARRIER DIODE
DIODE.ARRAY : DIODE ARRAY
DIODE.BRG : DIODE BRIDGE
DIODE.CHP : CHIP DIODE
DIODE.VAR : VARACTOR DIODE
DIODE.ZENR : ZENER DIODE
DIODE.Z.CHP : CHIP ZENER DIODE
DIODE.PHOT : PHOTO DIODE
FER.BEAD : FERRITE BEADS
FER.CORE : FERRITE CORE
FET.CHP : CHIP FET
FL.DSPLY : FLUORESCENT DISPLAY
FLTR.CE : CERAMIC FILTER
FLTR.COMB : COMB FILTER MODULE
FLTR.LC.RF : LC FILTER,EMI
FUSE.CHP : CHIP FUSE
GND.MTL : GROUND PLATE
GND.TERM : GROUND TERMINAL
JUMPER.CN : JUMPER CONNECTOR
JUMPER.TST : JUMPER,TEST POINT
L.DTCT : LIGHT DETECTING MODULE
LED.CHP : CHIP LED
LED.DSPLY : LED DISPLAY
LED.INFRD : LED,INFRARED
PHOT.CPL : PHOTO COUPLER
PHOT.INTR : PHOTO INTERRUPTER
PHOT.RFLCT : PHOTO REFLECTOR
PHOT.TR : PHOTO TRANSISTOR
PIN.TEST : PIN,TEST POINT
PTC.THERM : POSITIVE TEMPERATURE COEFFICIENT THERMISTOR
R.ANTI.SURGE : FIXED ANTI SURGE RESISTOR
R.ARRAY : RESISTOR ARRAY
R.CAR. : CARBON RESISTOR
R.CAR.CHP : CHIP RESISTOR
R.CAR.FP : FLAME PROOF CARBON RESISTOR
R.CEMENT : CEMENT RESISTOR
R.CHP : CHIP RESISTOR
R.FUS : FUSIBLE RESISTOR
R.MTL.CHP : CHIP METAL FILM RESISTOR
R.MTL.FLM : METAL FILM RESISTOR
R.MTL.OXD : METAL OXIDE FILM RESISTOR
R.MTL.PLAT : METAL PLATE RESISTOR
RSNR.CE : CERAMIC RESONATOR
RSNR.CRYS : CRYSTAL RESONATOR
SCR.BND.HD : BIND HEAD B-TIGHT SCREW
SCR.TERM : SCREW TERMINAL
SCR.TR : SCREW,TRANSISTOR
SURG.PRTCT : SURGE PROTECTOR
SUPRT.PCB : P.C.B. SUPPORT
SW.LEVER : LEVER SWITCH
SW.MICRO : MICRO SWITCH
SW.LEAF : LEAF SWITCH
SW.PUSH : PUSH SWITCH
SW.RT : ROTARY SWITCH
SW.RT.ENC : ROTARY ENCODER
SW.RT.MTR : ROTARY SWITCH WITH MOTOR
SW.SLIDE : SLIDE SWITCH
SW.TACT : TACT SWITCH
TERM.SP : SPEAKER TERMINAL
TERM.WRAP : WRAPPING TERMINAL
THRMST.CHP : CHIP THERMISTOR
TR : TRANSISTOR
TR.CHP : CHIP TRANSISTOR
TR.DGT : DIGITAL TRANSISTOR
TR.DGT.CHP : CHIP DIGITAL TRANSISTOR
TR.PAIR : PAIR TRANSISTOR
TRANS : TRANSFORMER
TRANS.PULS : PULSE TRANSFORMER
TRANS.PWR : POWER TRANSFORMER
VARISTOR.C : CHIP VARISTOR
VOLT.SELCT : VOLTAGE SELECTOR
VR : ROTARY POTENTIOMETER
VR.MTR : POTENTIOMETER WITH MOTOR
VR.SLIDE : SLIDE POTENTIOMETER
VR.SW : POTENTIOMETER WITH SWITCH
VR.TRIM : TRIMMER POTENTIOMETER

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DIGITAL

	Ref No.	Part No.	Description		Markets
*		ZV095400	P.C.B.	DIGITAL	
	CB301	WE220700	CN.BS.PIN	25P TE	
	CB302	VQ962100	CN.BS.PIN	18P	
	CB303	VQ961300	CN.BS.PIN	10P	
	CB307	VB389900	CN.BS.PIN	3P	
	CB501	ZP457700	CN.SOCKET	60P 134203560W3	
	CB502	ZF286600	CN.LAN	8P HR903125C	
*	CB503	VB390302	CN.BS.PIN	7P	
	CB651	WQ680200	CN.USB	4P TE AAPVA004C0	
	C301-302	US625100	C.CE.CHP	0.1uF 10V	
	C303	US126100	C.CE.CHP	1uF 10V	
	C304	US064100	C.CE.CHP	0.01uF 50V B	
	C305	US126100	C.CE.CHP	1uF 10V	
	C306	US625100	C.CE.CHP	0.1uF 10V	
	C307	WC894700	C.EL.CHP	2.2uF 50V	
	C308	US625100	C.CE.CHP	0.1uF 10V	
	C309	US064100	C.CE.CHP	0.01uF 50V B	
	C310	US625100	C.CE.CHP	0.1uF 10V	
	C311	US064100	C.CE.CHP	0.01uF 50V B	
	C312	WG888300	C.CE.M.CHP	10uF 6.3V	
	C314	US046100	C.CE.CHP	1uF 25V	
	C315-316	US625100	C.CE.CHP	0.1uF 10V	
	C318-326	US625100	C.CE.CHP	0.1uF 10V	
	C330-333	US663100	C.CE.CHP	1000pF 50V	
	C501	US035100	C.CE.CHP	0.1uF 16V B	
	C502	US061220	C.CE.CHP	22pF 50V B	
	C503	US035100	C.CE.CHP	0.1uF 16V B	
	C504	WD758300	C.CE.CHP	10uF 10V	
	C505	US046100	C.CE.CHP	1uF 25V	
	C506	US625100	C.CE.CHP	0.1uF 10V	
	C507	WG251600	C.CE.CHP	4.7uF 6.3V	
	C508	WD758300	C.CE.CHP	10uF 10V	
	C509	US046100	C.CE.CHP	1uF 25V	
	C510	WD758300	C.CE.CHP	10uF 10V	
	C511	WG251600	C.CE.CHP	4.7uF 6.3V	
	C512	US663100	C.CE.CHP	1000pF 50V	
	C513	WD758300	C.CE.CHP	10uF 10V	
	C515	US126100	C.CE.CHP	1uF 10V	
	C516	US634100	C.CE.CHP	0.01uF 16V	
	C517	US661121	C.CE.CHP	12pF 50V	
	C518	WG888300	C.CE.M.CHP	10uF 6.3V	
	C520	US661152	C.CE.CHP	15pF 50V	
	C521	US126100	C.CE.CHP	1uF 10V	
	C523	US634100	C.CE.CHP	0.01uF 16V	
	C524	US126100	C.CE.CHP	1uF 10V	
	C525	US064100	C.CE.CHP	0.01uF 50V B	
	C526	UB214680	C.CE.CHP	0.068uF 25V	
	C527	US063470	C.CE.CHP	4700pF 50V B	
	C528	WD758300	C.CE.CHP	10uF 10V	
	C529-530	US625100	C.CE.CHP	0.1uF 10V	
	C531	WD758300	C.CE.CHP	10uF 10V	
	C532	US634100	C.CE.CHP	0.01uF 16V	
	C533	US135100	C.CE.CHP	0.1uF 16V	
	C534-535	US625100	C.CE.CHP	0.1uF 10V	
	C536	US035100	C.CE.CHP	0.1uF 16V B	
	C537	UU238100	C.EL	100uF 16V	
	C538	US625100	C.CE.CHP	0.1uF 10V	
	C539	UU238100	C.EL	100uF 16V	
	C540-541	US625100	C.CE.CHP	0.1uF 10V	
	C542	US035100	C.CE.CHP	0.1uF 16V B	
	C543	UU267100	C.EL	10uF 50V	
	C544	US625100	C.CE.CHP	0.1uF 10V	
	C545	US064100	C.CE.CHP	0.01uF 50V B	
*	C546-547	ZQ695900	C.PML	0.0033uF 100V	
*	C548-551	ZQ695600	C.PML	0.001uF 100V	
	C552	US035100	C.CE.CHP	0.1uF 16V B	
	C554	UU238100	C.EL	100uF 16V	
	C557-558	US135100	C.CE.CHP	0.1uF 16V	
	C559	US663100	C.CE.CHP	1000pF 50V	
	C561-564	UR867100	C.EL	10uF 50V	
	C570	US046100	C.CE.CHP	1uF 25V	
	C571	US135100	C.CE.CHP	0.1uF 16V	
	C572	US046100	C.CE.CHP	1uF 25V	
	C573	UM397470	C.EL	47uF 16V	
	C574	WD758300	C.CE.CHP	10uF 10V	
	C575	UM397470	C.EL	47uF 16V	
	C576-577	UR867100	C.EL	10uF 50V	
	C578	US135100	C.CE.CHP	0.1uF 16V	
	C580-584	US663100	C.CE.CHP	1000pF 50V	
	C651-652	WD758300	C.CE.CHP	10uF 10V	
	C653	UM398100	C.EL	100uF 16V	
	C654	WD758300	C.CE.CHP	10uF 10V	
	C655	US663100	C.CE.CHP	1000pF 50V	
	C656	WD758300	C.CE.CHP	10uF 10V	
	D301-307	WW783900	DIODE	1SS355VM	
	D308	WR148501	DIODE	RB521S-30TE61	
	D651-655	WW783900	DIODE	1SS355VM	
	IC301	YE182B00	IC.MEMORY	R1EX25032ATA00I	
	IC303	X4453A00	IC	SN74LVC1G17DCKR	
	IC501	YC288A00	IC	RP130Q331D-TR-F	
	IC502	X8385A00	IC	TC7WHU04FK TE85L	
	IC504	YA255A01	IC	R1172H501D-T1-F	
	IC505-506	YC288A00	IC	RP130Q331D-TR-F	
	IC508	YD216A00	IC	PCM9211PTR	
	IC509	YE181A01	IC	TC7MBL3257CFK	
	IC512	X7947A00	IC	DSD1791DBR	
	IC513	XR680A00	IC	TC7SH08FU(TE85L, JF	
*	IC515	YJ125A00	IC	NJM8532M OP AMP	
	IC518	X8944A00	IC	R1154H050B-T1-F	
	IC519	X7787A00	IC	TC74LCX245FT(EL,K)	
	IC651	YA255A01	IC	R1172H501D-T1-F	
	J501	RD350000	R.CHP	0Ω 1/16W J	
	J503	RD350000	R.CHP	0Ω 1/16W J	
	J653	RD350000	R.CHP	0Ω 1/16W J	
	PJ501	V8795700	JACK.PIN	1P	
	Q301-302	WY001400	TR.ARRAY	HN4B01JE	
	Q303-305	WZ461700	TR.CHP	2SA1576UBTLR	
	Q306-307	WZ461800	TR.CHP	2SC4081UBTLR	
	Q308	WW782300	TR.DGT	DTC044EUBTL	
	Q309	WY001400	TR.ARRAY	HN4B01JE	
	Q501	WZ703400	FET	RAL035P01	
	Q502-503	WW782300	TR.DGT	DTC044EUBTL	
	R301	RD456100	R.CHP	1KΩ 1/16W J	
	R302	RD456911	R.CHP	9.1KΩ 1/16W	

* New Parts

* New Parts

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DIGITAL

Ref No.	Part No.	Description			Markets
R303	RD456100	R. CHP	1KΩ	1/16W J	
R304	RD457100	R. CHP	10KΩ	1/16W J	
R306	RD457271	R. CHP	27KΩ	1/16W J	
R307	RD457100	R. CHP	10KΩ	1/16W J	
R308	RD457181	R. CHP	18KΩ	1/16W	
R309-310	RD457100	R. CHP	10KΩ	1/16W J	
R313	RD457331	R. CHP	33KΩ	1/16W	
R314	RD454220	R. CHP	22Ω	1/16W J	
R315	RD457100	R. CHP	10KΩ	1/16W J	
R316-317	RD456100	R. CHP	1KΩ	1/16W J	
R318	RD455100	R. CHP	100Ω	1/16W J	
R319-320	RD456100	R. CHP	1KΩ	1/16W J	
R321	RD454471	R. CHP	47Ω	1/16W	
R322	RD455100	R. CHP	100Ω	1/16W J	
R323-326	RD454220	R. CHP	22Ω	1/16W J	
R327	RD454101	R. CHP	10Ω	1/16W J	
R328	RD454220	R. CHP	22Ω	1/16W J	
R329	RD458100	R. CHP	100KΩ	1/16W	
R330-331	RD454220	R. CHP	22Ω	1/16W J	
R332	RD455100	R. CHP	100Ω	1/16W J	
R333	RD459100	R. CHP	1MΩ	1/16W J	
R334	RD454220	R. CHP	22Ω	1/16W J	
R335	RD455100	R. CHP	100Ω	1/16W J	
R336	RD457100	R. CHP	10KΩ	1/16W J	
R337	RD456220	R. CHP	2. 2KΩ	1/16W J	
R340	RD455100	R. CHP	100Ω	1/16W J	
R341	RD454471	R. CHP	47Ω	1/16W	
R342	RD457100	R. CHP	10KΩ	1/16W J	
R343	RD458100	R. CHP	100KΩ	1/16W	
R344	RD457471	R. CHP	47KΩ	1/16W	
R345	RD454220	R. CHP	22Ω	1/16W J	
R346	RD457100	R. CHP	10KΩ	1/16W J	
R347	RD456100	R. CHP	1KΩ	1/16W J	
R348	RD457100	R. CHP	10KΩ	1/16W J	
R349	RD457471	R. CHP	47KΩ	1/16W	
R350	RD455100	R. CHP	100Ω	1/16W J	
R351	RD459100	R. CHP	1MΩ	1/16W J	
R354	RD456100	R. CHP	1KΩ	1/16W J	
R355	RD456911	R. CHP	9. 1KΩ	1/16W	
R356	RD456100	R. CHP	1KΩ	1/16W J	
R359	RD457331	R. CHP	33KΩ	1/16W	
R360-363	RD457100	R. CHP	10KΩ	1/16W J	
R366	RD457100	R. CHP	10KΩ	1/16W J	
R368	RD455100	R. CHP	100Ω	1/16W J	
R369	RD456220	R. CHP	2. 2KΩ	1/16W J	
R370	RD454220	R. CHP	22Ω	1/16W J	
R371-372	RD456220	R. CHP	2. 2KΩ	1/16W J	
R373	RD457100	R. CHP	10KΩ	1/16W J	
R374	RD356100	R. CHP	1KΩ	1/16W J	
R501	RD455100	R. CHP	100Ω	1/16W J	
R502	RF454820	R. CHP	82Ω	1/16W F	
R503	RD455681	R. CHP	680Ω	1/16W J	
R504	RD456470	R. CHP	4. 7KΩ	1/16W J	
R505	RD454471	R. CHP	47Ω	1/16W	
R506	RD457181	R. CHP	18KΩ	1/16W	
R507	RD455100	R. CHP	100Ω	1/16W J	
R508	RD455181	R. CHP	180Ω	1/16W	

＊ New Parts

Ref No.	Part No.	Description			Markets
R510	RD455181	R. CHP	180Ω	1/16W	
R511-512	RD454220	R. CHP	22Ω	1/16W J	
R513	RD455100	R. CHP	100Ω	1/16W J	
R514	RD457100	R. CHP	10KΩ	1/16W J	
R515	RD454330	R. CHP	33Ω	1/16W J	
R516	RD455681	R. CHP	680Ω	1/16W J	
R517-519	RD454220	R. CHP	22Ω	1/16W J	
R520	RD355680	R. CHP	680Ω	1/16W J	
R521	RD454220	R. CHP	22Ω	1/16W J	
R522	RD454330	R. CHP	33Ω	1/16W J	
R523	RD457100	R. CHP	10KΩ	1/16W J	
R524-526	RD454330	R. CHP	33Ω	1/16W J	
R527	RD454220	R. CHP	22Ω	1/16W J	
R528	RD454330	R. CHP	33Ω	1/16W J	
R529	RD456470	R. CHP	4. 7KΩ	1/16W J	
R530-532	RD454220	R. CHP	22Ω	1/16W J	
R533	RD457100	R. CHP	10KΩ	1/16W J	
R534-538	RD458100	R. CHP	100KΩ	1/16W	
R539	RD457100	R. CHP	10KΩ	1/16W J	
R540-543	RF356200	R. MTL. CHP	2KΩ	1/16W D	
R544-545	RF356330	R. MTL. CHP	3. 3KΩ	1/16W D	
R546-549	RF355390	R. MTL. CHP	390Ω	1/16W D	
R550-551	RF356330	R. MTL. CHP	3. 3KΩ	1/16W D	
R552	RD253220	R. CHP	2. 2Ω	1/10W J	
R553	RD454220	R. CHP	22Ω	1/16W J	
R554	RD458100	R. CHP	100KΩ	1/16W	
R555	RD456470	R. CHP	4. 7KΩ	1/16W J	
R556	RD454330	R. CHP	33Ω	1/16W J	
R557	RD457100	R. CHP	10KΩ	1/16W J	
R558	RD458100	R. CHP	100KΩ	1/16W	
R559-560	RD456470	R. CHP	4. 7KΩ	1/16W J	
R564-566	RD454330	R. CHP	33Ω	1/16W J	
R567-568	RD454220	R. CHP	22Ω	1/16W J	
R570-571	RD357390	R. CHP	39KΩ	1/16W J	
R572-575	RD357330	R. CHP	33KΩ	1/16W J	
R576-577	RD357390	R. CHP	39KΩ	1/16W J	
R578-579	RD457100	R. CHP	10KΩ	1/16W J	
R580-583	RD456470	R. CHP	4. 7KΩ	1/16W J	
R586	RD454330	R. CHP	33Ω	1/16W J	
R589-590	RD457100	R. CHP	10KΩ	1/16W J	
R651	RD357220	R. CHP	22KΩ	1/16W J	
R652	RD357470	R. CHP	47KΩ	1/16W J	
U501	ZF740400	CN. PHOTO. R	1P JSR2165		
XL301	WA782502	RSNR. CE	8MHz		
XL501	WS190000	RSNR. CRYS	24. 576MHz DSX321G		

＊ New Parts

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MAIN

	Ref No.	Part No.	Description			Markets
*		ZV094800	P.C.B.	MAIN		U
*		ZV094900	P.C.B.	MAIN		R
*		ZV095000	P.C.B.	MAIN		T
*		ZV095100	P.C.B.	MAIN		K
*		ZV095200	P.C.B.	MAIN		A
*		ZV095300	P.C.B.	MAIN		BG
*		ZV097200	P.C.B.	MAIN		L
*		ZV097300	P.C.B.	MAIN		S
	CB7	VB858600	CN.BS.PIN	7P		
	CB101	VB389900	CN.BS.PIN	3P		
	CB102	VQ963900	CN.BS.PIN	18P		
	CB103	LB932031	CN.BS.PIN	3P		
	CB104	VQ963100	CN.BS.PIN	10P		
	CB201	VB858200	CN.BS.PIN	3P		
	CB202	VQ047200	CN.BS.PIN	9P		
*	CB203	WE222300	CN.BS.PIN	FMN 25P SE		
	CB301	VB390001	CN.BS.PIN	4P		
	CB304	VB858200	CN.BS.PIN	3P		
	CB311	VB390402	CN.BS.PIN	8P		
⚠	CB541	VG879900	CN.BS.PIN	2P		
	CB542-543	WN103000	CLIP.FUSE	TP00351-31		
⚠	CB544	VG879900	CN.BS.PIN	2P		
	CB545	VB390802	CN.BS.PIN	12P		
	C9	US063100	C.CE.CHP	1000pF	50V B	
	C76	UR867100	C.EL	10uF	50V	
	C82	US062100	C.CE.CHP	100pF	50V B	
	C101-102	US061470	C.CE.CHP	47pF	50V B	
	C103-104	UR867100	C.EL	10uF	50V	
	C108-109	ZD517600	C.MYLAR	100pF	100V	
	C110-111	UR838100	C.EL	100uF	16V	
	C112-113	US060500	C.CE.CHP	5pF	50V B	
	C114	US062150	C.CE.CHP	150pF	50V B	
	C117-119	US062150	C.CE.CHP	150pF	50V B	
	C120	UR897470	C.EL	47uF	100V	
	C121-122	UR897100	C.EL	10uF	100V	
	C123	UR897470	C.EL	47uF	100V	
	C124-125	UR897100	C.EL	10uF	100V	
	C130	UR866220	C.EL	2.2uF	50V	
	C131-132	ZD520400	C.MYLAR	0.022uF	100V	
	C137	UR818100	C.EL	100uF	6.3V	
	C139	UR848330	C.EL	330uF	25V	
*	C140	ZU644800	C.EL	6800uF	71V	
	C141	UR867100	C.EL	10uF	50V	
*	C143	ZU644800	C.EL	6800uF	71V	
	C144	UR829100	C.EL	1000uF	10V	
	C146	UR867470	C.EL	47uF	50V	
	C147	UR867100	C.EL	10uF	50V	
	C148	US064100	C.CE.CHP	0.01uF	50V B	
	C149	UR779100	C.EL	1000uF	63V	
	C151	US064100	C.CE.CHP	0.01uF	50V B	
	C152	US062100	C.CE.CHP	100pF	50V B	
	C154-155	WW194100	C.MYLAR	0.1uF	100V	
	C156	US064100	C.CE.CHP	0.01uF	50V B	
	C157-158	WN200100	C.CE.M.CHP	0.01uF	100V	TKABGL
	C160-162	WN200100	C.CE.M.CHP	0.01uF	100V	TKABGL
	C166-167	WW194100	C.MYLAR	0.1uF	100V	
	C168-169	UR877330	C.EL	33uF	63V	
	C170	ZD517600	C.MYLAR	100pF	100V	
	C172	WW194100	C.MYLAR	0.1uF	100V	
	C173-176	WN200100	C.CE.M.CHP	0.01uF	100V	TKABGL

* New Parts

	Ref No.	Part No.	Description			Markets
	C189	WN200100	C.CE.M.CHP	0.01uF	100V	TKABGL
	C204-205	US062220	C.CE.CHP	220pF	50V B	
	C206-207	US061470	C.CE.CHP	47pF	50V B	
	C208-213	US062220	C.CE.CHP	220pF	50V B	
	C216-217	UR837100	C.EL	10uF	16V	
	C230-231	UR867100	C.EL	10uF	50V	
	C232	US135100	C.CE.CHP	0.1uF	16V	
	C233	UR837330	C.EL	33uF	16V	
	C234-245	UR867100	C.EL	10uF	50V	
	C246-247	ZD519100	C.MYLAR	1800pF	100V	
	C248-251	UR865220	C.EL	0.22uF	50V	
	C253-254	UR867100	C.EL	10uF	50V	
	C259-260	UR866100	C.EL	1uF	50V	
	C261	US135100	C.CE.CHP	0.1uF	16V	
	C262	UR829100	C.EL	1000uF	10V	
	C263	UR867100	C.EL	10uF	50V	
	C266-267	UR866220	C.EL	2.2uF	50V	
	C275	UR867100	C.EL	10uF	50V	
	C287	US135100	C.CE.CHP	0.1uF	16V	
	C289	US163100	C.CE.CHP	1000pF	50V	
	C290-292	WF203300	C.EL	3300uF	16V	
	C293-294	US061680	C.CE.CHP	68pF	50V B	
	C298-299	US062470	C.CE.CHP	470pF	50V B	
	C300-301	US061680	C.CE.CHP	68pF	50V B	
	C302	WF203300	C.EL	3300uF	16V	
	C303-312	US062470	C.CE.CHP	470pF	50V B	
	C2001	UM417330	C.EL	33uF	50V	
	C2002	US035100	C.CE.CHP	0.1uF	16V B	
	C2003	US064100	C.CE.CHP	0.01uF	50V B	
	C2004-2005	UM417330	C.EL	33uF	50V	
	C2006-2011	US035100	C.CE.CHP	0.1uF	16V B	
	C2012	UM397100	C.EL	10uF	16V	
	C2016-2019	US062100	C.CE.CHP	100pF	50V B	
	C2020-2021	US035100	C.CE.CHP	0.1uF	16V B	
	C2022	US065100	C.CE.CHP	0.1uF	50V B	
	C2025	UM387470	C.EL	47uF	16V	
	C2026	UM397100	C.EL	10uF	16V	
	C2027	US035100	C.CE.CHP	0.1uF	16V B	
	C2028	UM388100	C.EL	100uF	6.3V	
	C2029-2031	US035100	C.CE.CHP	0.1uF	16V B	
	C2033	US035100	C.CE.CHP	0.1uF	16V B	
	C2034	US063100	C.CE.CHP	1000pF	50V B	
	C3001-3002	ZK674400	C.MYLAR	3300pF	100V	
	C3003	US064100	C.CE.CHP	0.01uF	50V B	
	C3004	US063100	C.CE.CHP	1000pF	50V B	
	C3101	US135100	C.CE.CHP	0.1uF	16V	
	C3501	US135100	C.CE.CHP	0.1uF	16V	
	C3601	US035100	C.CE.CHP	0.1uF	16V B	
⚠	C5401-5402	WQ902301	C.CE.SAFTY	1000pF	250V	
⚠	C5403	ZM881200	C.CE.SAFTY	0.22 275V	U.C.IEC	
⚠	C5404	WQ939401	C.CE.SAFTY	0.01uF	250V	
⚠	C5405	ZD520200	C.MYLAR	0.015uF	100V	
⚠	C5407	WR246900	C.CE.CHP	3300pF	250V	
⚠	C5408-5409	WJ361200	C.POL.MTL	0.047uF	400V	U
⚠	C5408-5409	WC041600	C.POL.MTL	0.022uF	630V	RABGLS
⚠	C5410	ZH377601	C.CE.SAFTY	3300pF	250V	U
⚠	C5410	WQ902201	C.CE.SAFTY	2200pF	250V	RABGLS
	C5411	WJ322300	C.CE.M.CHP	1000pF	630V	
	C5412	US034471	C.CE.CHP	0.047uF	16V B	
	C5414	ZD518800	C.MYLAR	1000pF	100V	

* New Parts

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	Ref No.	Part No.	Description		Markets
	C5416-5417	US063100	C.CE.CHP	1000pF 50V B	
⚠	C5419	WW766000	C.EL	220uF 220V	U
⚠	C5419	WW766100	C.EL	150uF 400V	RS
⚠	C5419	WQ852500	C.EL	68uF 400V	TKABGL
⚠	C5420	UR867220	C.EL	22uF 50V	
	C5421-5422	WH772400	C.EL	2200uF 10V	
	C5423	WH771600	C.EL	220uF 10V	
	D103-106	ZA984400	DIODE	BAV103	
	D107	WY162900	DIODE.ZENR	UDZV3.9B	
	D108	WY163800	DIODE.ZENR	UDZV9.1B	
	D109	WY165000	DIODE.ZENR	UDZV30B	
	D110	WW783900	DIODE	1SS355VM	
⚠	D112	ZU129100	DIODE.CHP	F3A	
⚠	D113	VN953300	DIODE.BRG	D5SBA60 5A 600V	
	D114	WW783900	DIODE	1SS355VM	
	D115	WY164300	DIODE.ZENR	UDZV15B	
	D117	WW783900	DIODE	1SS355VM	
	D119-120	WY165200	DIODE.ZENR	UDZV36B	RS
⚠	D123-125	ZP358601	DIODE	1SS352 TE	
	D126	WY164800	DIODE.ZENR	UDZV24B	
	D202	WY163400	DIODE.ZENR	UDZV6.2B	
	D203-205	VS597600	DIODE.CHP	RB160L-40 TE25	
	D510	WA467800	LED	SEL6910A-CD	
	D2003	ZR754101	LED	HLMP-NS30 Blue	
	D2011	WY163101	DIODE.ZENR	UDZV4.7B	
	D2012	ZC674600	DIODE.ZENR	UDZV2.0B	
	D2013	ZQ873500	D.SCHOTTKY	RB501VM-40TE-17	
	D2014	WP354200	LED	GREEN	
	D2015-2016	ZP358601	DIODE	1SS352 TE	
	D3001	WW783900	DIODE	1SS355VM	
	D3002	WY163200	DIODE.ZENR	UDZV5.1B	
	D3102	WW783900	DIODE	1SS355VM	
⚠	D5401	WW872000	DIODE.BRG	DBL155G 1.5A 600	
⚠	D5402	VV463000	DIODE.CHP	1.1A 200V D1FL20U	
⚠	D5403	WW170700	DIODE	SARS05	
⚠	D5404-5407	WW783900	DIODE	1SS355VM	
	D5408	WW745500	D.SCHOTTKY	RB215T-90 20A 90V	
	D5409	WW783900	DIODE	1SS355VM	
⚠	D5412-5413	ZU129100	DIODE.CHP	F3A	
⚠	F5401	ZM616500	FUSE	2A 250V	
⚠	F5402	WQ211100	FUSE	8A 125V	URS
⚠	F5402	VV071801	FUSE	4A 250V	TKABGL
	G101	V5995800	PLATE.GND		
	IC202	YF439A00	IC	BD3491FS	
	IC351	X0515B00	IC	LM61CIZ THERMAL	
	IC541	YD359A00	IC	STR2A152	
	IC542-543	WJ688100	PHOT.CPL	EL816(B)	
	IC544	YA276A00	IC	TL431AC 2.5-36V	
	J2003	RD350000	R.CHP	0Ω 1/16W J	
	JK301	WZ975700	JACK.PHONE	MSJ-064-05B-B-RF	
	PJ202	WD195400	JACK.PIN	6P	
	PJ203	WD195200	JACK.PIN	4P	
	Q101-104	WD896300	TR	2SA1514K R,S	
	Q105-106	WC292201	TR	KTC3206Y-AT	
	Q107-108	WC883400	TR	2SD2704 K	
⚠	Q109	ZG214200	TR	KTD600K-Y-U/PH	
⚠	Q110-111	ZG214100	TR	KTB631K Y,GR	
⚠	Q112	ZG214200	TR	KTD600K-Y-U/PH	
⚠ #	Q113-116	ZU111000	TR.POWER	2SC4468 O,P,Y	
	Q117-118	WF549900	TR	2SC3906K T146 R,S	
	Q119-120	WC397600	TR	2N5401S-RTK/P	
⚠	Q121	ZC256500	TR	2SC5712	
	Q123	WZ177900	TR	KTC4370A-Y	
⚠	Q124	ZE933000	TR	KTC4376-Y-RTF/P	
	Q125	WC398300	TR	2N5551S-RTK/P	
	Q126	WC397600	TR	2N5401S-RTK/P	
	Q128	WC398300	TR	2N5551S-RTK/P	
⚠	Q129	ZG245700	TR	KTA1661-Y-RTF/P	
	Q131	WC397600	TR	2N5401S-RTK/P	
	Q133	WC398300	TR	2N5551S-RTK/P	
	Q201-204	WC883400	TR	2SD2704 K	
	Q205-206	VJ927202	TR	2SA1162-Y(TE85R,F)	
	Q511-512	VV655400	TR.DGT	DTC114EKA	
	Q2002	VV655400	TR.DGT	DTC114EKA	
	Q2003	VP872700	TR	2SC4488 S,T	
	Q2004	VV655400	TR.DGT	DTC114EKA	
	R61	RD353220	R.CHP	2.2Ω 1/16W J	
	R64	RD355820	R.CHP	820Ω 1/16W J	
	R101	RD356100	R.CHP	1KΩ 1/16W J	
	R102-103	V8071300	R.MTL.FLM	470Ω 1W	
	R104-105	RD357470	R.CHP	47KΩ 1/16W J	
	R106-107	RD356100	R.CHP	1KΩ 1/16W J	
	R109-110	RD357330	R.CHP	33KΩ 1/16W J	
	R112	RD355820	R.CHP	820Ω 1/16W J	
	R113-118	RD356820	R.CHP	8.2KΩ 1/16W J	
	R119-120	RD356120	R.CHP	1.2KΩ 1/16W J	
	R121-122	RD357330	R.CHP	33KΩ 1/16W J	
	R123-124	RD355470	R.CHP	470Ω 1/16W J	
	R127-129	RD356820	R.CHP	8.2KΩ 1/16W J	
	R130-131	RD355120	R.CHP	120Ω 1/16W J	
	R132-134	RD356820	R.CHP	8.2KΩ 1/16W J	
	R135-136	WW865700	R.CAR.FP	5.6KΩ 1/4W	
	R137-138	WW865102	R.CAR.FP	2.2KΩ 1/4W	
⚠	R139	WW863301	R.CAR.FP	100Ω 1/4W	
	R140-141	WW863901	R.CAR.FP	330Ω 1/4W	
⚠	R142	WW863301	R.CAR.FP	100Ω 1/4W	
⚠	R143-146	WW861300	R.CAR.FP	2.2Ω 1/4W	
⚠	R147-148	ZG861500	R.WW	0.1Ω 2W	
	R149-150	RD355510	R.CHP	510Ω 1/16W J	
	R151	RD358330	R.CHP	330KΩ 1/16W J	
	R152-153	RD357150	R.CHP	15KΩ 1/16W J	
	R154	RD358330	R.CHP	330KΩ 1/16W J	
	R155-157	RD357470	R.CHP	47KΩ 1/16W J	
	R158	RD357100	R.CHP	10KΩ 1/16W J	
⚠	R159-160	V8070300	R.MTL.FLM	10Ω 1W	
	R161	RD357470	R.CHP	47KΩ 1/16W J	
	R162-163	WW862101	R.CAR.FP	10Ω 1/4W	
	R164	RD357470	R.CHP	47KΩ 1/16W J	
	R165	RD357560	R.CHP	56KΩ 1/16W J	
⚠ *	R167	VC762501	R.MTL.OXD	2.7KΩ 2W	
⚠	R168	WW863301	R.CAR.FP	100Ω 1/4W	
*	R169	WW864701	R.CAR.FP	1.2KΩ 1/4W	
	R170	WW863701	R.CAR.FP	220Ω 1/4W	
	R171-173	RD356470	R.CHP	4.7KΩ 1/16W J	
	R174-175	WW863301	R.CAR.FP	100Ω 1/4W	
	R177	RD357470	R.CHP	47KΩ 1/16W J	
⚠	R178	WW864600	R.CAR.FP	1KΩ 1/4W	
	R179	HL007100	R.MTL.OXD	10KΩ 1/2W	
	R182	RD357470	R.CHP	47KΩ 1/16W J	
	R183	RD357100	R.CHP	10KΩ 1/16W J	

* New Parts

* New Parts

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	Ref No.	Part No.	Description			Markets
	R185	RD357470	R.CHP	47KΩ	1/16W J	
	R186	WW863901	R.CAR.FP	330Ω	1/4W	UTKABG
	R186	WW863200	R.CAR.FP	82Ω	1/4W	RLS
	R187	WW865200	R.CAR.FP	2.4KΩ	1/4W	
	R190	RD356330	R.CHP	3.3KΩ	1/16W J	
	R191	WW863901	R.CAR.FP	330Ω	1/4W	UTKABG
	R191	WW863200	R.CAR.FP	82Ω	1/4W	RLS
	R192	WW975500	R.MTL.OXD	39KΩ	1/4w	
	R194	RD357150	R.CHP	15KΩ	1/16W J	
	R196	WW863901	R.CAR.FP	330Ω	1/4W	UTKABG
	R196	WW863200	R.CAR.FP	82Ω	1/4W	RLS
	R197	RD357100	R.CHP	10KΩ	1/16W J	RS
	R198-199	RD358160	R.CHP	160KΩ	1/16W J	
	R200	RD357390	R.CHP	39KΩ	1/16W J	
	R202	RD355470	R.CHP	470Ω	1/16W J	
	R203-204	RD356100	R.CHP	1KΩ	1/16W J	
	R205-207	RD355470	R.CHP	470Ω	1/16W J	
	R210-211	RD358100	R.CHP	100KΩ	1/16W J	
	R212-215	RD355470	R.CHP	470Ω	1/16W J	
	R216-219	RD358470	R.CHP	470KΩ	1/16W J	
	R222-223	RD358470	R.CHP	470KΩ	1/16W J	
	R226-227	RD358470	R.CHP	470KΩ	1/16W J	
	R228-229	RD355220	R.CHP	220Ω	1/16W J	
	R236-237	RD357100	R.CHP	10KΩ	1/16W J	
	R245-246	RD355100	R.CHP	100Ω	1/16W J	
	R247	RD358100	R.CHP	100KΩ	1/16W J	
	R252-253	RD356470	R.CHP	4.7KΩ	1/16W J	
	R254-255	RD356120	R.CHP	1.2KΩ	1/16W J	
	R256-257	RD357100	R.CHP	10KΩ	1/16W J	
	R259	RD358100	R.CHP	100KΩ	1/16W J	
	R261	RD358100	R.CHP	100KΩ	1/16W J	
	R266-267	RD355100	R.CHP	100Ω	1/16W J	
	R269	RD356100	R.CHP	1KΩ	1/16W J	
	R270-271	RD356470	R.CHP	4.7KΩ	1/16W J	
	R272	V8070500	R.MTL.FLM	22Ω	1W	
	R273	RD356120	R.CHP	1.2KΩ	1/16W J	
	R275	RD358100	R.CHP	100KΩ	1/16W J	
	R277	RD356120	R.CHP	1.2KΩ	1/16W J	
⚠	R278-279	V8070700	R.MTL.FLM	47Ω	1W	
	R280	RD357100	R.CHP	10KΩ	1/16W J	
⚠	R281-284	VC756301	R.MTL.OXD	10Ω	2W	
	R288	RD353220	R.CHP	2.2Ω	1/16W J	
	R307	RD357330	R.CHP	33KΩ	1/16W J	
	R308	RD358100	R.CHP	100KΩ	1/16W J	
⚠	R316-317	WW861600	R.CAR.FP	3.9Ω	1/4W	
	R318-320	WW863800	R.CAR.FP	270Ω	1/4W	
⚠ *	R322	VC762501	R.MTL.OXD	2.7KΩ	2W	
	R323	RD357100	R.CHP	10KΩ	1/16W J	
	R324	RD357470	R.CHP	47KΩ	1/16W J	
	R541	RD357470	R.CHP	47KΩ	1/16W J	
	R542-543	RD355620	R.CHP	620Ω	1/16W J	
	R553	RD353220	R.CHP	2.2Ω	1/16W J	
	R555-556	RD358100	R.CHP	100KΩ	1/16W J	
	R2005	RD355100	R.CHP	100Ω	1/16W J	
	R2008	RD356470	R.CHP	4.7KΩ	1/16W J	
	R2009	RD355240	R.CHP	240Ω	1/16W J	
	R2010	RD354470	R.CHP	47Ω	1/16W J	
	R2011	RD357150	R.CHP	15KΩ	1/16W J	
	R2012	RD356750	R.CHP	7.5KΩ	1/16W J	
	R2013-2015	RD356470	R.CHP	4.7KΩ	1/16W J	

* New Parts

	Ref No.	Part No.	Description			Markets
	R2016-2017	RD356330	R.CHP	3.3KΩ	1/16W J	
	R2018	RD356470	R.CHP	4.7KΩ	1/16W J	
	R2019	RD355470	R.CHP	470Ω	1/16W J	
	R2021-2022	RD356220	R.CHP	2.2KΩ	1/16W J	
	R2023-2025	RD355470	R.CHP	470Ω	1/16W J	
	R2026-2027	RD356160	R.CHP	1.6KΩ	1/16W J	
	R2028-2029	RD356130	R.CHP	1.3KΩ	1/16W J	
	R2030-2031	RD356100	R.CHP	1KΩ	1/16W J	
	R2032	RD357150	R.CHP	15KΩ	1/16W J	
	R2033-2034	RD354101	R.CHP	10Ω	1/16W J	
*	R2036	RD155222	R.CHP	220Ω	1/4 J	
	R2037	RD354101	R.CHP	10Ω	1/16W J	
	R2038	RD356750	R.CHP	7.5KΩ	1/16W J	
*	R2039	RD157152	R.CAR.CHP	15KΩ	1/4W	
	R2040	RD353220	R.CHP	2.2Ω	1/16W J	
	R3001	RD353220	R.CHP	2.2Ω	1/16W J	
	R3002	RD356100	R.CHP	1KΩ	1/16W J	
	R3005	RD356100	R.CHP	1KΩ	1/16W J	
	R3101	RD356750	R.CHP	7.5KΩ	1/16W J	
	R3102-3103	RD357100	R.CHP	10KΩ	1/16W J	
	R3104-3105	RD357470	R.CHP	47KΩ	1/16W J	
	R3601-3602	RD357100	R.CHP	10KΩ	1/16W J	
	R3603-3604	RD357470	R.CHP	47KΩ	1/16W J	
	R5401-5402	WR033300	R.CHP	4.7KΩ	1/4W	
⚠	R5403	WU547900	R.ANTI.SUR	3MΩ	1/2W	
⚠	R5404	WY023000	R.CAR.CHP	0.91Ω	1W	
⚠	R5406	WW745400	R.CHP	150KΩ	1/2W	
⚠	R5407	ZN108500	R.FUSE	100Ω	1/16W	
⚠	R5408	RD154220	R.CHP	22Ω	1/4W J	
⚠	R5409	VF167800	R.CHP	47Ω	1W	
	R5410	RD154102	R.CHP	10Ω	1/4W J	
	R5411	RD355181	R.CHP	180Ω	1/16W J	
	R5412	RD356150	R.CHP	1.5KΩ	1/16W J	
	R5413	RD357220	R.CHP	22KΩ	1/16W J	
	R5414	RF456180	R.CHP	1.8KΩ	1/16W F	
	R5415-5416	RF456820	R.CHP	8.2KΩ	1/16W F	
	R5433	ZN108500	R.FUSE	100Ω	1/16W	
⚠	RY101-103	ZJ359600	RELAY	981-2A-48DS-SP7		
⚠	RY541	WQ804101	RELAY	DC DLS5D1-O(M)0.25		
	ST1	V4040500	SCR.TERM	M3		
	ST5	V4040500	SCR.TERM	M3		
	ST102-103	V4040500	SCR.TERM	M3		
	ST351	WA246200	SCR.TERM	3.5		
	ST543	V4040500	SCR.TERM	M3		
	SW201-214	WD483100	SW.TACT	SKRGAAD010		
	SW311	WD483100	SW.TACT	SKRGAAD010		
	SW312	ZE287300	SW.RT	XRE0126 299-9834		
	SW361	WU974000	SW.RT.ENC	REB161(9X7)PVB20FI		
	SW514-516	WD483100	SW.TACT	SKRGAAD010		
⚠	T5401	YD115C01	TRANS.SUB			
	TE102	ZH270100	TERM.SP	8P BANANA		URTAS
	TE102	ZH270200	TERM.SP	8P NON-BANANA		KBGL
	TH101	V9760200	THRMST.CHP	NCP18XH103J03RB		
⚠	TH541	WF544600	PTC.THERM	NTPAD5R1LDNB0 5.1		
	U2001	WW715100	L.DTCT	SIR8430MH6		
	V2001	ZM611100	FL.DSPLY	016ST106INK		
		V6203300	SPACER.FL			
		WE774200	SCR.BND.HD	3x10	MFZN2W3	

* New Parts

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	Ref No.	Part No.	Description			Markets
*		ZV098000	P.C.B.	MAIN		BG
	CB7	VB858600	CN.BS.PIN	7P		
	CB101	VB389900	CN.BS.PIN	3P		
	CB102	VQ963900	CN.BS.PIN	18P		
	CB103	LB932031	CN.BS.PIN	3P		
	CB104	VQ963100	CN.BS.PIN	10P		
	CB201	VB858200	CN.BS.PIN	3P		
	CB202	VQ047200	CN.BS.PIN	9P		
*	CB203	WE222300	CN.BS.PIN	FMN 25P SE		
	CB301	VB390001	CN.BS.PIN	4P		
	CB304	VB858200	CN.BS.PIN	3P		
	CB311	VB390402	CN.BS.PIN	8P		
	CB541	VG879900	CN.BS.PIN	2P		
	CB542-543	WN103000	CLIP.FUSE	TP00351-31		
	CB544	VG879900	CN.BS.PIN	2P		
	CB545	VB390802	CN.BS.PIN	12P		
	C9	US063100	C.CE.CHP	1000pF	50V B	
	C76	UR867100	C.EL	10uF	50V	
	C82	US062100	C.CE.CHP	100pF	50V B	
	C101-102	US061470	C.CE.CHP	47pF	50V B	
	C103-104	UR867100	C.EL	10uF	50V	
	C108-109	ZD517600	C.MYLAR	100pF	100V	
	C110-111	UR838100	C.EL	100uF	16V	
	C112-113	US060500	C.CE.CHP	5pF	50V B	
	C114	US062150	C.CE.CHP	150pF	50V B	
	C117-119	US062150	C.CE.CHP	150pF	50V B	
	C120	UR897470	C.EL	47uF	100V	
	C121-122	UR897100	C.EL	10uF	100V	
	C123	UR897470	C.EL	47uF	100V	
	C124-125	UR897100	C.EL	10uF	100V	
	C130	UR866220	C.EL	2.2uF	50V	
	C131-132	ZD520400	C.MYLAR	0.022uF	100V	
	C137	UR818100	C.EL	100uF	6.3V	
	C139	UR848330	C.EL	330uF	25V	
*	C140	ZU644800	C.EL	6800uF	71V	
	C141	UR867100	C.EL	10uF	50V	
*	C143	ZU644800	C.EL	6800uF	71V	
	C144	UR829100	C.EL	1000uF	10V	
	C146	UR867470	C.EL	47uF	50V	
	C147	UR867100	C.EL	10uF	50V	
	C148	US064100	C.CE.CHP	0.01uF	50V B	
	C149	UR779100	C.EL	1000uF	63V	
	C151	US064100	C.CE.CHP	0.01uF	50V B	
	C152	US062100	C.CE.CHP	100pF	50V B	
	C154-155	WW194100	C.MYLAR	0.1uF	100V	
	C156	US064100	C.CE.CHP	0.01uF	50V B	
	C157-158	WN200100	C.CE.M.CHP	0.01uF	100V	
	C160-162	WN200100	C.CE.M.CHP	0.01uF	100V	
	C166-167	WW194100	C.MYLAR	0.1uF	100V	
	C168-169	UR877330	C.EL	33uF	63V	
	C170	ZD517600	C.MYLAR	100pF	100V	
	C172	WW194100	C.MYLAR	0.1uF	100V	
	C173-176	WN200100	C.CE.M.CHP	0.01uF	100V	
	C189	WN200100	C.CE.M.CHP	0.01uF	100V	
	C204-205	US062220	C.CE.CHP	220pF	50V B	
	C206-207	US061470	C.CE.CHP	47pF	50V B	
	C208-213	US062220	C.CE.CHP	220pF	50V B	

* New Parts

Ref No.	Part No.	Description			Markets
C216-217	UR837100	C.EL	10uF	16V	
C230-231	UR867100	C.EL	10uF	50V	
C232	US135100	C.CE.CHP	0.1uF	16V	
C233	UR837330	C.EL	33uF	16V	
C234-245	UR867100	C.EL	10uF	50V	
C246-247	ZD519100	C.MYLAR	1800pF	100V	
C248-251	UR865220	C.EL	0.22uF	50V	
C253-254	UR867100	C.EL	10uF	50V	
C259-260	UR866100	C.EL	1uF	50V	
C261	US135100	C.CE.CHP	0.1uF	16V	
C262	UR829100	C.EL	1000uF	10V	
C263	UR867100	C.EL	10uF	50V	
C266-267	UR866220	C.EL	2.2uF	50V	
C275	UR867100	C.EL	10uF	50V	
C287	US135100	C.CE.CHP	0.1uF	16V	
C289	US163100	C.CE.CHP	1000pF	50V	
C290-292	WF203300	C.EL	3300uF	16V	
C293-294	US061680	C.CE.CHP	68pF	50V B	
C298-299	US062470	C.CE.CHP	470pF	50V B	
C300-301	US061680	C.CE.CHP	68pF	50V B	
C302	WF203300	C.EL	3300uF	16V	
C303-312	US062470	C.CE.CHP	470pF	50V B	
C2001	UM417330	C.EL	33uF	50V	
C2002	US035100	C.CE.CHP	0.1uF	16V B	
C2003	US064100	C.CE.CHP	0.01uF	50V B	
C2004-2005	UM417330	C.EL	33uF	50V	
C2006-2011	US035100	C.CE.CHP	0.1uF	16V B	
C2012	UM397100	C.EL	10uF	16V	
C2016-2019	US062100	C.CE.CHP	100pF	50V B	
C2020-2021	US035100	C.CE.CHP	0.1uF	16V B	
C2022	US065100	C.CE.CHP	0.1uF	50V B	
C2025	UM387470	C.EL	47uF	16V	
C2026	UM397100	C.EL	10uF	16V	
C2027	US035100	C.CE.CHP	0.1uF	16V B	
C2028	UM388100	C.EL	100uF	6.3V	
C2029-2031	US035100	C.CE.CHP	0.1uF	16V B	
C2033	US035100	C.CE.CHP	0.1uF	16V B	
C2034	US063100	C.CE.CHP	1000pF	50V B	
C3001-3002	ZK674400	C.MYLAR	3300pF	100V	
C3003	US064100	C.CE.CHP	0.01uF	50V B	
C3004	US063100	C.CE.CHP	1000pF	50V B	
C3101	US135100	C.CE.CHP	0.1uF	16V	
C3501	US135100	C.CE.CHP	0.1uF	16V	
C3601	US035100	C.CE.CHP	0.1uF	16V B	
C5401-5402	WQ902301	C.CE.SAFTY	1000pF	250V	
C5403	ZM881200	C.CE.SAFTY	0.22 275V	U.C.IEC	
C5404	WQ939401	C.CE.SAFTY	0.01uF	250V	
C5405	ZD520200	C.MYLAR	0.015uF	100V	
C5407	WR246900	C.CE.CHP	3300pF	250V	
C5408-5409	WC041600	C.POL.MTL	0.022uF	630V	
C5410	WQ902201	C.CE.SAFTY	2200pF	250V	
C5411	WJ322300	C.CE.M.CHP	1000pF	630V	
C5412	US034471	C.CE.CHP	0.047uF	16V B	
C5414	ZD518800	C.MYLAR	1000pF	100V	
C5416-5417	US063100	C.CE.CHP	1000pF	50V B	
C5419	WQ852500	C.EL	68uF	400V	
C5420	UR867220	C.EL	22uF	50V	

* New Parts

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Ref No.	Part No.	Description		Markets
C5421-5422	WH772400	C.EL	2200uF 10V	
C5423	WH771600	C.EL	220uF 10V	
D103-106	ZA984400	DIODE	BAV103	
D107	WY162900	DIODE.ZENR	UDZV3.9B	
D108	WY163800	DIODE.ZENR	UDZV9.1B	
D109	WY165000	DIODE.ZENR	UDZV30B	
D110	WW783900	DIODE	1SS355VM	
D112	ZU129100	DIODE.CHP	F3A	
D113	VN953300	DIODE.BRG	D5SBA60 5A 600V	
D114	WW783900	DIODE	1SS355VM	
D115	WY164300	DIODE.ZENR	UDZV15B	
D117	WW783900	DIODE	1SS355VM	
D123-125	ZP358601	DIODE	1SS352 TE	
D126	WY164800	DIODE.ZENR	UDZV24B	
D202	WY163400	DIODE.ZENR	UDZV6.2B	
D203-205	VS597600	DIODE.CHP	RB160L-40 TE25	
D510	WA467800	LED	SEL6910A-CD	
D2003	ZR754101	LED	HLMP-NS30 Blue	
D2011	WY163101	DIODE.ZENR	UDZV4.7B	
D2012	ZC674600	DIODE.ZENR	UDZV2.0B	
D2013	ZQ873500	D.SCHOTTKY	RB501VM-40TE-17	
D2014	WP354200	LED	GREEN	
D2015-2016	ZP358601	DIODE	1SS352 TE	
D3001	WW783900	DIODE	1SS355VM	
D3002	WY163200	DIODE.ZENR	UDZV5.1B	
D3102	WW783900	DIODE	1SS355VM	
D5401	WW872000	DIODE.BRG	DBL155G 1.5A 600	
D5402	VV463000	DIODE.CHP	1.1A 200V D1FL20U	
D5403	WW170700	DIODE	SARS05	
D5404-5407	WW783900	DIODE	1SS355VM	
D5408	WW745500	D.SCHOTTKY	RB215T-90 20A 90V	
D5409	WW783900	DIODE	1SS355VM	
D5412-5413	ZU129100	DIODE.CHP	F3A	
F5401	ZM616500	FUSE	2A 250V	
F5402	VV071801	FUSE	4A 250V	
G101	V5995800	PLATE.GND		
IC202	YF439A00	IC	BD3491FS	
IC351	X0515B00	IC	LM61CIZ THERMAL	
IC541	YD359A00	IC	STR2A152	
IC542-543	WJ688100	PHOT.CPL	EL816(B)	
IC544	YA276A00	IC	TL431AC 2.5-36V	
J2003	RD350000	R.CHP	0Ω 1/16W J	
JK301	WZ975700	JACK.PHONE	MSJ-064-05B-B-RF	
PJ202	WD195400	JACK.PIN	6P	
PJ203	WD195200	JACK.PIN	4P	
Q101-104	WD896300	TR	2SA1514K R,S	
Q105-106	WC292201	TR	KTC3206Y-AT	
Q107-108	WC883400	TR	2SD2704 K	
Q109	ZG214200	TR	KTD600K-Y-U/PH	
Q110-111	ZG214100	TR	KTB631K Y,GR	
Q112	ZG214200	TR	KTD600K-Y-U/PH	
# Q113-116	ZU111000	TR.POWER	2SC4468 O,P,Y	
Q117-118	WF549900	TR	2SC3906K T146 R,S	
Q119-120	WC397600	TR	2N5401S-RTK/P	
Q121	ZC256500	TR	2SC5712	
Q123	WZ177900	TR	KTC4370A-Y	
Q124	ZE933000	TR	KTC4376-Y-RTF/P	
Q125	WC398300	TR	2N5551S-RTK/P	
Q126	WC397600	TR	2N5401S-RTK/P	
Q128	WC398300	TR	2N5551S-RTK/P	
Q129	ZG245700	TR	KTA1661-Y-RTF/P	
Q131	WC397600	TR	2N5401S-RTK/P	
Q133	WC398300	TR	2N5551S-RTK/P	
Q201-204	WC883400	TR	2SD2704 K	
Q205-206	VJ927202	TR	2SA1162-Y(TE85R,F)	
Q511-512	VV655400	TR.DGT	DTC114EKA	
Q2002	VV655400	TR.DGT	DTC114EKA	
Q2003	VP872700	TR	2SC4488 S,T	
Q2004	VV655400	TR.DGT	DTC114EKA	
R61	RD353220	R.CHP	2.2Ω 1/16W J	
R64	RD355820	R.CHP	820Ω 1/16W J	
R101	RD356100	R.CHP	1KΩ 1/16W J	
R102-103	V8071300	R.MTL.FLM	470Ω 1W	
R104-105	RD357470	R.CHP	47KΩ 1/16W J	
R106-107	RD356100	R.CHP	1KΩ 1/16W J	
R109-110	RD357330	R.CHP	33KΩ 1/16W J	
R112	RD355820	R.CHP	820Ω 1/16W J	
R113-118	RD356820	R.CHP	8.2KΩ 1/16W J	
R119-120	RD356120	R.CHP	1.2KΩ 1/16W J	
R121-122	RD357330	R.CHP	33KΩ 1/16W J	
R123-124	RD355470	R.CHP	470Ω 1/16W J	
R127-129	RD356820	R.CHP	8.2KΩ 1/16W J	
R130-131	RD355120	R.CHP	120Ω 1/16W J	
R132-134	RD356820	R.CHP	8.2KΩ 1/16W J	
R135-136	WW865700	R.CAR.FP	5.6KΩ 1/4W	
R137-138	WW865102	R.CAR.FP	2.2KΩ 1/4W	
R139	WW863301	R.CAR.FP	100Ω 1/4W	
R140-141	WW863901	R.CAR.FP	330Ω 1/4W	
R142	WW863301	R.CAR.FP	100Ω 1/4W	
R143-146	WW861300	R.CAR.FP	2.2Ω 1/4W	
R147-148	ZG861500	R.WW	0.1Ω 2W	
R149-150	RD355510	R.CHP	510Ω 1/16W J	
R151	RD358330	R.CHP	330KΩ 1/16W J	
R152-153	RD357150	R.CHP	15KΩ 1/16W J	
R154	RD358330	R.CHP	330KΩ 1/16W J	
R155-157	RD357470	R.CHP	47KΩ 1/16W J	
R158	RD357100	R.CHP	10KΩ 1/16W J	
R159-160	V8070300	R.MTL.FLM	10Ω 1W	
R161	RD357470	R.CHP	47KΩ 1/16W J	
R162-163	WW862101	R.CAR.FP	10Ω 1/4W	
R164	RD357470	R.CHP	47KΩ 1/16W J	
R165	RD357560	R.CHP	56KΩ 1/16W J	
* R167	VC762501	R.MTL.OXD	2.7KΩ 2W	
R168	WW863301	R.CAR.FP	100Ω 1/4W	
* R169	WW864701	R.CAR.FP	1.2KΩ 1/4W	
R170	WW863701	R.CAR.FP	220Ω 1/4W	
R171-173	RD356470	R.CHP	4.7KΩ 1/16W J	
R174-175	WW863301	R.CAR.FP	100Ω 1/4W	
R177	RD357470	R.CHP	47KΩ 1/16W J	
R178	WW864600	R.CAR.FP	1KΩ 1/4W	
R179	HL007100	R.MTL.OXD	10KΩ 1/2W	
R182	RD357470	R.CHP	47KΩ 1/16W J	
R183	RD357100	R.CHP	10KΩ 1/16W J	
R185	RD357470	R.CHP	47KΩ 1/16W J	

* New Parts

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Ref No.	Part No.	Description			Markets
R186	WW863901	R.CAR.FP	330Ω	1/4W	
R187	WW865200	R.CAR.FP	2.4KΩ	1/4W	
R190	RD356330	R.CHP	3.3KΩ	1/16W J	
R191	WW863901	R.CAR.FP	330Ω	1/4W	
R192	WW975500	R.MTL.OXD	39KΩ	1/4w	
R194	RD357150	R.CHP	15KΩ	1/16W J	
R196	WW863901	R.CAR.FP	330Ω	1/4W	
R198-199	RD358160	R.CHP	160KΩ	1/16W J	
R200	RD357390	R.CHP	39KΩ	1/16W J	
R202	RD355470	R.CHP	470Ω	1/16W J	
R203-204	RD356100	R.CHP	1KΩ	1/16W J	
R205-207	RD355470	R.CHP	470Ω	1/16W J	
R210-211	RD358100	R.CHP	100KΩ	1/16W J	
R212-215	RD355470	R.CHP	470Ω	1/16W J	
R216-219	RD358470	R.CHP	470KΩ	1/16W J	
R222-223	RD358470	R.CHP	470KΩ	1/16W J	
R226-227	RD358470	R.CHP	470KΩ	1/16W J	
R228-229	RD355220	R.CHP	220Ω	1/16W J	
R236-237	RD357100	R.CHP	10KΩ	1/16W J	
R245-246	RD355100	R.CHP	100Ω	1/16W J	
R247	RD358100	R.CHP	100KΩ	1/16W J	
R252-253	RD356470	R.CHP	4.7KΩ	1/16W J	
R254-255	RD356120	R.CHP	1.2KΩ	1/16W J	
R256-257	RD357100	R.CHP	10KΩ	1/16W J	
R259	RD358100	R.CHP	100KΩ	1/16W J	
R261	RD358100	R.CHP	100KΩ	1/16W J	
R266-267	RD355100	R.CHP	100Ω	1/16W J	
R269	RD356100	R.CHP	1KΩ	1/16W J	
R270-271	RD356470	R.CHP	4.7KΩ	1/16W J	
R272	V8070500	R.MTL.FLM	22Ω	1W	
R273	RD356120	R.CHP	1.2KΩ	1/16W J	
R275	RD358100	R.CHP	100KΩ	1/16W J	
R277	RD356120	R.CHP	1.2KΩ	1/16W J	
R278-279	V8070700	R.MTL.FLM	47Ω	1W	
R280	RD357100	R.CHP	10KΩ	1/16W J	
R281-284	VC756301	R.MTL.OXD	10Ω	2W	
R288	RD353220	R.CHP	2.2Ω	1/16W J	
R307	RD357330	R.CHP	33KΩ	1/16W J	
R308	RD358100	R.CHP	100KΩ	1/16W J	
R316-317	WW861600	R.CAR.FP	3.9Ω	1/4W	
R318-320	WW863800	R.CAR.FP	270Ω	1/4W	
*R322	VC762501	R.MTL.OXD	2.7KΩ	2W	
R323	RD357100	R.CHP	10KΩ	1/16W J	
R324	RD357470	R.CHP	47KΩ	1/16W J	
R541	RD357470	R.CHP	47KΩ	1/16W J	
R542-543	RD355620	R.CHP	620Ω	1/16W J	
R553	RD353220	R.CHP	2.2Ω	1/16W J	
R555-556	RD358100	R.CHP	100KΩ	1/16W J	
R2005	RD355100	R.CHP	100Ω	1/16W J	
R2008	RD356470	R.CHP	4.7KΩ	1/16W J	
R2009	RD355240	R.CHP	240Ω	1/16W J	
R2010	RD354470	R.CHP	47Ω	1/16W J	
R2011	RD357150	R.CHP	15KΩ	1/16W J	
R2012	RD356750	R.CHP	7.5KΩ	1/16W J	
R2013-2015	RD356470	R.CHP	4.7KΩ	1/16W J	
R2016-2017	RD356330	R.CHP	3.3KΩ	1/16W J	
R2018	RD356470	R.CHP	4.7KΩ	1/16W J	

Ref No.	Part No.	Description			Markets
R2019	RD355470	R.CHP	470Ω	1/16W J	
R2021-2022	RD356220	R.CHP	2.2KΩ	1/16W J	
R2023-2025	RD355470	R.CHP	470Ω	1/16W J	
R2026-2027	RD356160	R.CHP	1.6KΩ	1/16W J	
R2028-2029	RD356130	R.CHP	1.3KΩ	1/16W J	
R2030-2031	RD356100	R.CHP	1KΩ	1/16W J	
R2032	RD357150	R.CHP	15KΩ	1/16W J	
R2033-2034	RD354101	R.CHP	10Ω	1/16W J	
*R2036	RD155222	R.CHP	220Ω	1/4 J	
R2037	RD354101	R.CHP	10Ω	1/16W J	
R2038	RD356750	R.CHP	7.5KΩ	1/16W J	
*R2039	RD157152	R.CAR.CHP	15KΩ	1/4W	
R2040	RD353220	R.CHP	2.2Ω	1/16W J	
R3001	RD353220	R.CHP	2.2Ω	1/16W J	
R3002	RD356100	R.CHP	1KΩ	1/16W J	
R3005	RD356100	R.CHP	1KΩ	1/16W J	
R3101	RD356750	R.CHP	7.5KΩ	1/16W J	
R3102-3103	RD357100	R.CHP	10KΩ	1/16W J	
R3104-3105	RD357470	R.CHP	47KΩ	1/16W J	
R3601-3602	RD357100	R.CHP	10KΩ	1/16W J	
R3603-3604	RD357470	R.CHP	47KΩ	1/16W J	
R5401-5402	WR033300	R.CHP	4.7KΩ	1/4W	
R5403	WU547900	R.ANTI.SUR	3MΩ	1/2W	
R5404	WY023000	R.CAR.CHP	0.91Ω	1W	
R5406	WW745400	R.CHP	150KΩ	1/2W	
R5407	ZN108500	R.FUSE	100Ω	1/16W	
R5408	RD154220	R.CHP	22Ω	1/4W J	
R5409	VF167800	R.CHP	47Ω	1W	
R5410	RD154102	R.CHP	10Ω	1/4W J	
R5411	RD355181	R.CHP	180Ω	1/16W J	
R5412	RD356150	R.CHP	1.5KΩ	1/16W J	
R5413	RD357220	R.CHP	22KΩ	1/16W J	
R5414	RF456180	R.CHP	1.8KΩ	1/16W F	
R5415-5416	RF456820	R.CHP	8.2KΩ	1/16W F	
R5419	RD357680	R.CHP	68KΩ	1/16W J	
R5433	ZN108500	R.FUSE	100Ω	1/16W	
RY101-103	ZJ359600	RELAY	981-2A-48DS-SP7		
RY541	WQ804101	RELAY	DC DLS5D1-O(M)0.25		
ST1	V4040500	SCR.TERM	M3		
ST5	V4040500	SCR.TERM	M3		
ST102-103	V4040500	SCR.TERM	M3		
ST351	WA246200	SCR.TERM	3.5		
ST543	V4040500	SCR.TERM	M3		
SW201-214	WD483100	SW.TACT	SKRGAAD010		
SW311	WD483100	SW.TACT	SKRGAAD010		
SW312	ZE287300	SW.RT	XRE0126 299-9834		
SW361	WU974000	SW.RT.ENC	REB161(9X7)PVB20FI		
SW514-516	WD483100	SW.TACT	SKRGAAD010		
T5401	YD115C01	TRANS.SUB			
TE102	ZH270200	TERM.SP	8P NON-BANANA		
TH101	V9760200	THRMST.CHP	NCP18XH103J03RB		
TH541	WF544600	PTC.THERM	NTPAD5R1LDNB0 5.1		
U2001	WW715100	L.DTCT	SIR8430MH6		
V2001	ZM611100	FL.DSPLY	016ST106INK		
	V6203300	SPACER.FL			
	WE774200	SCR.BND.HD	3x10	MFZN2W3	

R-N402

SWITCH

Ref No.	Part No.	Description		Markets
	ZN201600	P.C.B.	SWITCH	RS
CB1-2	WN103000	CLIP.FUSE	TP00351-31	
CB3	V9377900	CN.BS.PIN	4P	
F2	VV071801	FUSE	4A 250V	
SW1	WV382901	SW.SLIDE	SL14	

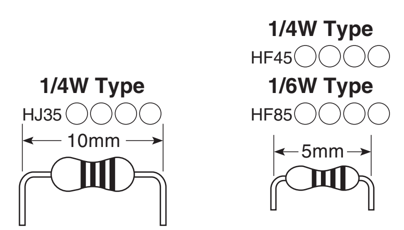
R-N402D

DAB

	Ref No.	Part No.	Description		Markets
*		ZV098100	P.C.B.	DAB	BG
	CB901	ZS226200	CN	24P TE SCFS852254	
	CB902	VQ044400	CN.BS.PIN	9P	
	C901	US064100	C.CE.CHP	0.01uF 50V B	
	C902	US035100	C.CE.CHP	0.1uF 16V B	
	C903	WD758300	C.CE.CHP	10uF 10V	
	C904	UM388330	C.EL	330uF 6.3V	
	R901	RD357470	R.CHP	47KΩ 1/16W J	
	R902-903	RD350000	R.CHP	0Ω 1/16W J	
	R904-905	RD356470	R.CHP	4.7KΩ 1/16W J	
	R906	RD357470	R.CHP	47KΩ 1/16W J	
	ST901	ZR366200	SUPPORT.DAB		

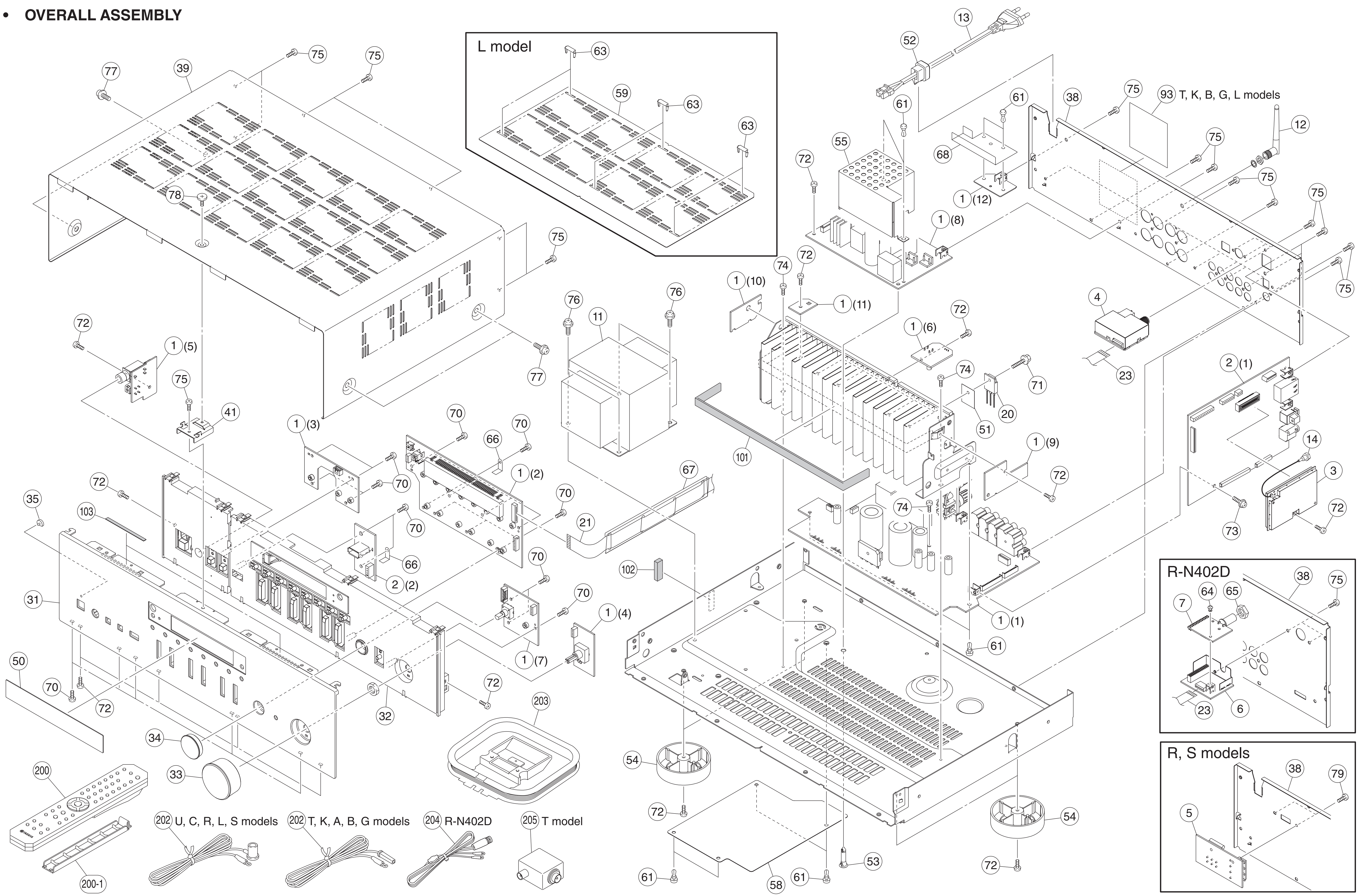
Carbon Resistors

Value	1/4W Type Part No.	1/6W Type Part No.	Value	1/4W Type Part No.	1/6W Type Part No.
1.0 Ω	HJ35 3100	HF85 3100	11 kΩ	HF45 7110	HF45 7110
1.8 Ω	HJ35 3180	✻	12 kΩ	HJ35 7120	HF85 7120
2.2 Ω	HJ35 3220	HF85 3220	13 kΩ	HF45 7130	HF45 7130
3.3 Ω	HJ35 3330	HF85 3330	15 kΩ	HF45 7150	HF45 7150
4.7 Ω	HJ35 3470	HF85 3470	18 kΩ	HF45 7180	HF45 7180
5.6 Ω	HJ35 3560	HF85 3560	22 kΩ	HF45 7220	HF45 7220
10 Ω	HF45 4100	HF45 4100	24 kΩ	HF45 7240	HF45 7240
15 Ω	HJ35 4150	HF85 4150	27 kΩ	HJ35 7270	HF85 7270
22 Ω	HF45 4220	HF45 4220	30 kΩ	HF45 7300	HF45 7300
27 Ω	HJ35 4270	HF85 4270	33 kΩ	HF45 7330	HF45 7330
33 Ω	HF45 4330	HF45 4330	36 kΩ	HF45 7360	HF45 7360
39 Ω	HJ35 4470	HF85 4390	39 kΩ	HF45 7390	HF45 7390
47 Ω	HF45 4470	HF45 4470	47 kΩ	HF45 7470	HF45 7470
56 Ω	HF45 4560	HF45 4560	51 kΩ	HF45 7510	HF45 7510
68 Ω	HF45 4680	HF45 4680	56 kΩ	HF45 7560	HF45 7560
75 Ω	HF45 4750	HF45 4750	62 kΩ	HF45 7620	HF45 7620
82 Ω	HF45 4820	HF45 4820	68 kΩ	HF45 7680	HF45 7680
91 Ω	HF45 4910	HF45 4910	82 kΩ	HF45 7820	HF45 7820
100 Ω	HF45 5100	HF45 5100	91 kΩ	HF45 7910	HF45 7910
110 Ω	HJ35 5110	HF85 5110	100 kΩ	HF45 8100	HF45 8100
120 Ω	HF45 5120	HF45 5120	110 kΩ	HF45 8110	HF45 8110
150 Ω	HF45 5150	HF45 5150	120 kΩ	HF45 8120	HF45 8120
160 Ω	HJ35 5160	✻	130 kΩ	HF45 8130	✻
180 Ω	HF45 5180	HF45 5180	150 kΩ	HF45 8150	HF45 8150
200 Ω	HF45 5200	HF45 5200	180 kΩ	HF45 8180	HF45 8180
220 Ω	HF45 5220	HF45 5220	220 kΩ	HJ35 8220	HF85 8220
270 Ω	HF45 5270	HF45 5270	270 kΩ	HF45 8270	HF45 8270
330 Ω	HF45 5330	HF45 5330	300 kΩ	HF45 8300	HF45 8300
390 Ω	HF45 5390	HF45 5390	330 kΩ	HF45 8330	HF45 8330
430 Ω	HF45 5430	HF45 5430	390 kΩ	HJ35 8390	HF85 8390
470 Ω	HF45 5470	HF45 5470	470 kΩ	HF45 8470	HF45 8470
510 Ω	HF45 5510	HF45 5510	560 kΩ	HJ35 8560	HF85 8560
560 Ω	HF45 5560	HF45 5560	680 kΩ	HJ35 8680	HF85 8680
680 Ω	HF45 5680	HF45 5680	820 kΩ	HJ35 8820	HF85 8820
820 Ω	HF45 5820	HF45 5820	1.0 MΩ	HF45 9100	HF45 9100
910 Ω	HF45 5910	HF45 5910	1.2 MΩ	HJ35 9120	✻
1.0 kΩ	HF45 6100	HF45 6100	1.5 MΩ	HJ35 9150	HF85 9150
1.2 kΩ	HF45 6120	HF45 6120	1.8 MΩ	HJ35 9180	HF85 9180
1.5 kΩ	HF45 6150	HF45 6150	2.2 MΩ	HJ35 9220	HF85 9220
1.8 kΩ	HF45 6180	HF45 6180	3.3 MΩ	HJ35 9330	HF85 9330
2.0 kΩ	HJ35 6200	HF85 6200	3.9 MΩ	HJ35 9390	✻
2.2 kΩ	HF45 6220	HF45 6220	4.7 MΩ	HJ35 9470	HF85 9470
2.4 kΩ	HJ35 6240	HF85 6240			
2.7 kΩ	HF45 6270	HF45 6270			
3.0 kΩ	HF45 6300	HF45 6300			
3.3 kΩ	HF45 6330	HF45 6330			
3.6 kΩ	HJ35 6360	HF85 6360			
3.9 kΩ	HF45 6390	HF45 6390			
4.7 kΩ	HF45 6470	HF45 6470			
5.1 kΩ	HF45 6510	HF45 6510			
5.6 kΩ	HF45 6560	HF45 6560			
6.8 kΩ	HF45 6680	HF45 6680			
8.2 kΩ	HF45 6820	HF45 6820			
9.1 kΩ	HF45 6910	HF45 6910			
10 kΩ	HF45 7100	HF45 7100			



✻ : Not available

- **OVERALL ASSEMBLY**



	Ref No.	Part No.	Description		Remarks	Markets
*	1	ZV094800	P.C.B. ASSEMBLY	MAIN		U
*	1	ZV094900	P.C.B. ASSEMBLY	MAIN		R
*	1	ZV095000	P.C.B. ASSEMBLY	MAIN		T
*	1	ZV095100	P.C.B. ASSEMBLY	MAIN		K
*	1	ZV095200	P.C.B. ASSEMBLY	MAIN		A
*	1	ZV095300	P.C.B. ASSEMBLY	MAIN	R-N402	BG
*	1	ZV098000	P.C.B. ASSEMBLY	MAIN	R-N402D	BG
*	1	ZV097200	P.C.B. ASSEMBLY	MAIN		L
*	1	ZV097300	P.C.B. ASSEMBLY	MAIN		S
*	2	ZV095400	P.C.B. ASSEMBLY	DIGITAL		
	3	YH110H00	NETWORK MODULE	NW-01	written	
	4	ZP339600	TUNER MODULE	TUNER TU-04		URLS
	4	ZP339700	TUNER MODULE	TUNER TU-04		TKABG
	5	ZN201600	P.C.B. ASSEMBLY	SWITCH		RS
*	6	ZV098100	P.C.B. ASSEMBLY	DAB	R-N402D	
*	7	ZV868900	DAB MODULE	VERONA2 FS2445..28	R-N402D	
⚠	11	YF699A00	POWER TRANSFORMER			U
⚠	11	YF700A00	POWER TRANSFORMER			RS
⚠	11	YF702A00	POWER TRANSFORMER			TK
⚠	11	YF703A00	POWER TRANSFORMER			A
⚠	11	YF705A00	POWER TRANSFORMER			BG
⚠	11	YF704A00	POWER TRANSFORMER			L
	12	ZQ099000	DIPOLE ANTENNA	EXTERNAL ANTENNA	L108/82.5,D7.9	
⚠	13	WB120500	POWER CABLE	2m		U
⚠	13	WY042500	POWER CABLE	1.8m		R
⚠	13	WY042600	POWER CABLE	1.8m		T
⚠	13	WY042400	POWER CABLE	1.8m		K
⚠	13	WY042100	POWER CABLE	1.8m		A
⚠	13	WY041100	POWER CABLE	1.8m		B
⚠	13	WY041700	POWER CABLE	1.8m		GL
⚠	13	ZC898501	POWER CABLE	1.8m		S
	14	ZQ099300	ANTENNA CABLE	RF CABLE 200mm	with NTU/WASHER	
#	20	ZU111000	TRANSISTOR	2SC4468 O,P,Y	Q113-Q116	
	23	MF109101	FLEXIBLE FLAT CABLE	9P 100mm P=1.25		
*	31	ZU856400	FRONT PANEL		R-N402 BL	
*	31	ZU856300	FRONT PANEL		R-N402 SI	
*	31	ZU886400	FRONT PANEL		R-N402D BL	
*	31	ZU886300	FRONT PANEL		R-N402D SI	
*	32	ZU872800	SUB PANEL		BL	
*	32	ZU872900	SUB PANEL		SI	
	33	ZH629500	KNOB	VOLUME	BL	
	33	ZH629600	KNOB	VOLUME	SI	
	34	ZG193000	KNOB	SELECT	SI	
	34	ZG193100	KNOB	SELECT	BL	
	35	WP080600	LENS LED			
*	38	ZV162900	REAR PANEL			U
*	38	ZV163000	REAR PANEL			RS
*	38	ZV163100	REAR PANEL		R-N402	TKABGL
*	38	ZV163400	REAR PANEL		R-N402D	BG
*	39	ZV829700	TOP COVER		BL	
*	39	ZV829800	TOP COVER		SI	
*	41	ZU882800	SSUPPORT	TOP COVER		

* New Parts

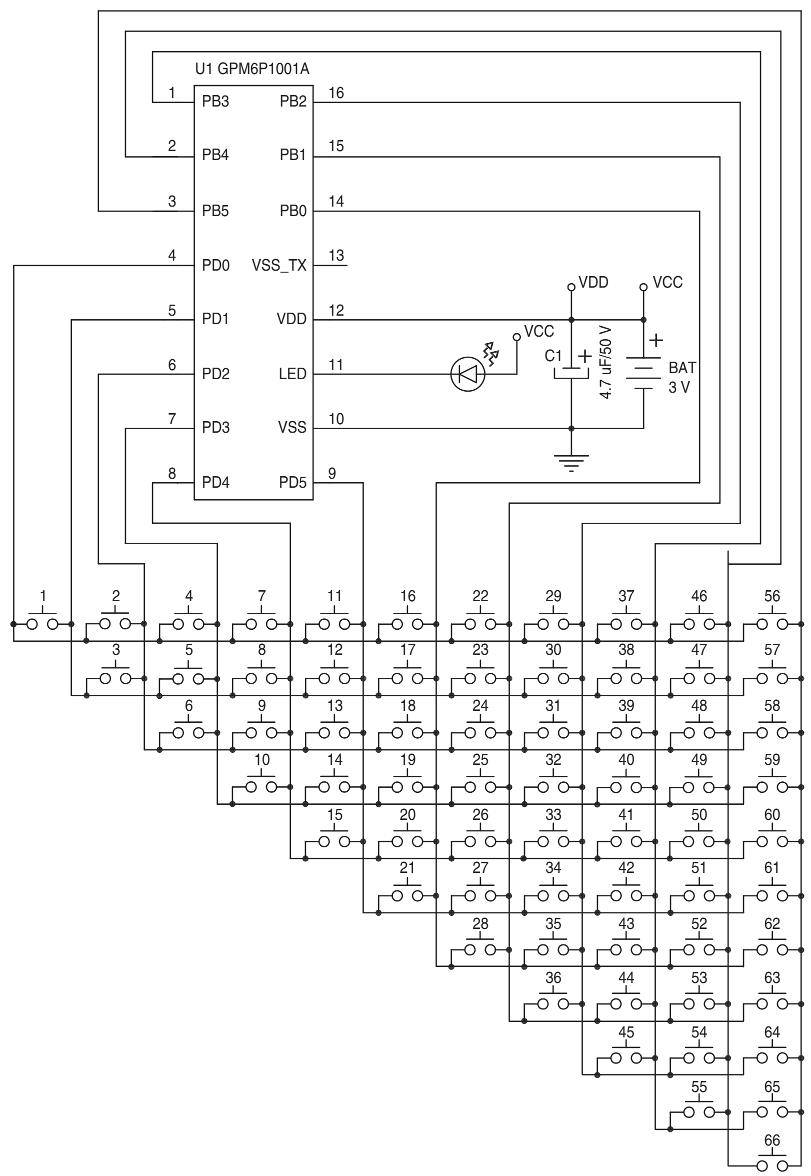
Finish .... BL: Black color, SI: Silver color

	Ref No.	Part No.	Description		Remarks	Markets
*	50	ZV167400	WINDOW SHEET		R-N402	
*	50	ZV167500	WINDOW SHEET		R-N402D	
	51	VV849300	RADIATION SHEET	19x24		
	52	V2438700	CORD STOPPER	10P1		
	53	WC172700	LOCKING CARD SPACER	KGLS-18RT KITAGAWA		
	54	ZC1818i0	LEG	D60/H21 Black	20x20x2 R2	
	55	ZP287200	BARRIER SHEET	87x81.5x53		
	58	ZP549600	BARRIER SHEET	165x121.5x0.5		
	59	ZT935100	TOP COVER SHEET	TOP COVER, Clear	SI	L
*	60	ZV996400	BARRIER SHEET	ACDC		BG
	61	VQ368600	PUSH RIVET	P3555-B		
	63	WJ053800	RIVET	TOP COVER, Clear	SI	L
	64	ZS601900	PUSH RIVET	P2648(B)	R-N402D	
	65	WG205000	NUT	3/8 UNEF-32	R-N402D	
*	66	ZV724800	EARTH PLATE			
*	67	ZV662900	BARRIER SHEET	FFC		
*	68	ZV663000	BARRIER SHEET	CABLE 35.5x70x10.5		
	70	WE774800	BIND HEAD P-TIGHT SCREW	3x8 MFZN2W3		
	71	VK173200	SCREW FOR TRANSISTOR	3x15 SP MFC2		
	72	WE774300	BIND HEAD B-TIGHT SCREW	3x8 MFZN2W3		
	73	WG959600	PW HEAD TAPPING B-T. SCREW	3x6-8 MFZN2W3		
	74	ZK590400	BIND B-TIGHT SCREW	3x8 MFZN2W3		
	75	WE774100	BIND HEAD BONDING B-T. SCREW	3x8 MFZN2B3		
	76	ZK590300	BIND S-TIGHT SCREW	4x10 SP MFZN2W3		
	77	VH313200	PW HEAD S-TIGHT SCREW	4x8-10 MFNI3BL	BL	
	77	VD069600	PW HEAD S-TIGHT SCREW	4x8-10 MFNI33	SI	
	78	WE200400	DISH HEAD B-TIGHT SCREW	3x6 MFNI33		
	78	WE200500	DISH HEAD B-TIGHT SCREW	3x6 MFNI3BL		
	79	WE877900	BIND HEAD S-TIGHT SCREW	3x6 MFZN2W3		R
	93	ZV167600	LABEL T			T
*	93	ZV167700	LABEL K			K
*	93	ZV167800	LABEL BG		R-N402	BG
*	93	ZV167900	LABEL L			L
	101	WQ621800	DAMPER	2x10x310		
	102	V5881100	CUSHION	5x8x25		
	103	ZK306200	INSULATOR	4x60x0.25		
			ACCESSORIES			
*	200	ZU738900	REMOTE CONTROL	RAX34		
	200-1	ZV285900	BATTERY COVER	Black	6005A001	
	202	V6267000	FM ANTENNA	1.4m 1pc	R-N402	URLS
	202	VQ147100	FM ANTENNA	1.4m 1pc	R-N402	TKABG
	203	VQ307400	AM ANTENNA	1.2m 1pc	R-N402	
	204	WG233200	DAB/FM ANTENNA	1.6m 1pc	R-N402D	
	210	ZQ643000	ANTENNA ISOLATOR	11D05Z59 1pc		T
			BATTERY	R6,AA,UM-3 2pcs		
			SERVICE TOOL			
		ZK708100	MHF CONNECTOR REMOVER	HIM-13002	90224-001	

* New Parts

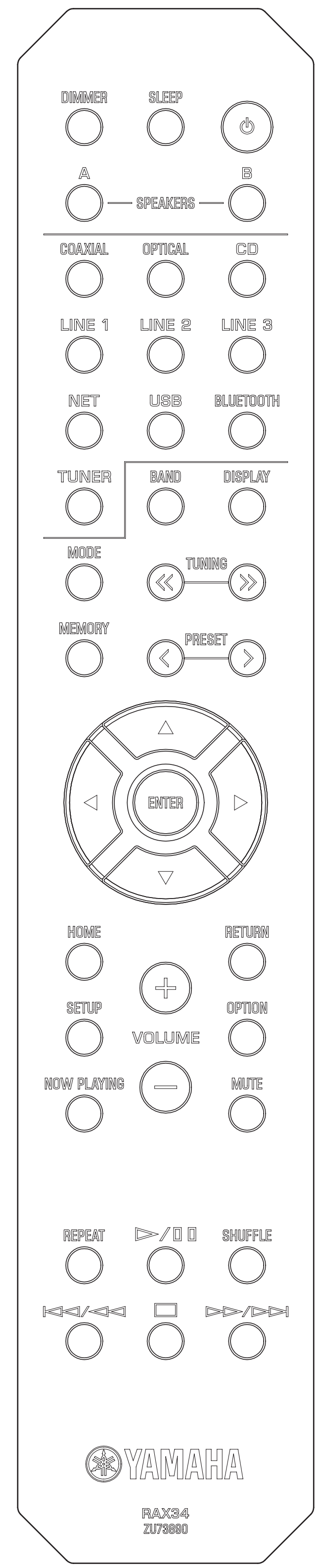
■ REMOTE CONTROL

SCHEMATIC DIAGRAM

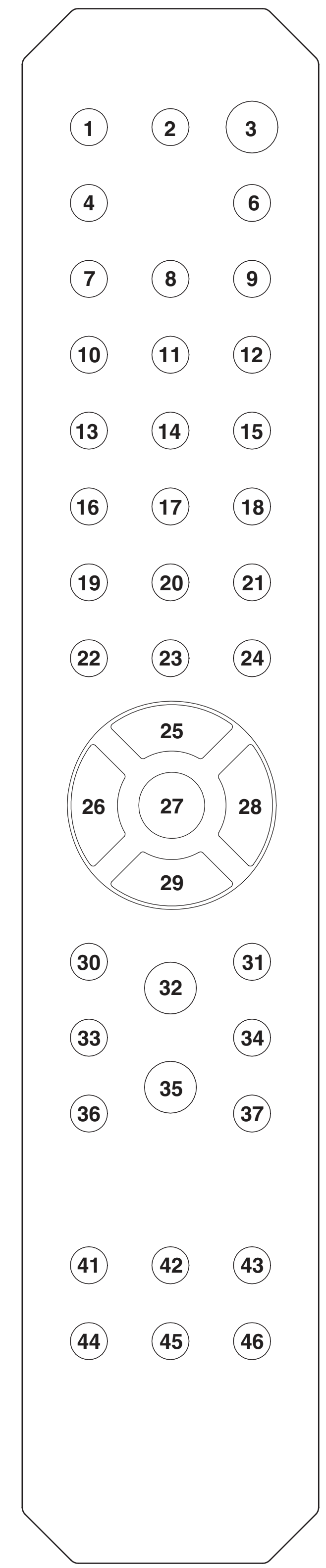


PANEL

RAX34



KEY NO. LAYOUT



KEY CODE

Key No	Function	Code
1	DIMMER	7A-82FD
2	SLEEP	7A-30
3	⏻ POWER	7E-2A
4	SPEAKERS A	7A-9A
5	–	–
6	SPEAKERS B	7A-9B
7	COAXIAL	7A-18
8	OPTICAL	7A-532C
9	CD	7A-15
10	LINE 1	7A-19
11	LINE 2	7A-C1
12	LINE 3	7A-C0
13	NET	7F01-3F
14	USB	7F01-720D
15	BLUETOOTH	7A-BEC1
16	TUNER	7A-16
17	BAND	7A-AE
18	DISPLAY	7F01-60
19	MODE	7F01-66
20	TUNING «	7F01-641B
21	TUNING »	7F01-611E
22	MEMORY	7F01-6718
23	PRESET <	7F01-5E21
24	PRESET >	7F01-5B24
25	▲ (up)	7A-9D
26	◀ (left)	7A-9F
27	ENTER	7A-DE
28	▶ (right)	7A-9E
29	▼ (down)	7A-9C
30	HOME	7A-C2
31	RETURN	7A-AA
32	VOLUME +	7A-1A
33	SETUP	7A-84
34	OPTION	7A-6B14
35	VOLUME –	7A-1B
36	NOW PLAYING	7A-433C
37	MUTE	7A-1C
41	REPEAT	7F01-45
42	►/▮▮ (play/pause)	7F01-99E6
43	SHUFFLE	7F01-46
44	▮◄◄/◄◄ (skip –)	7F01-6C
45	■ (stop)	7F01-69
46	►►/►►▮ (skip +)	7F01-6D

MEMO

R-N402/R-N402D